PHYS4071

Big Bang Cosmology & Inflation

Course Notes

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Introduction

About These Notes

These notes accompany the PHYS4071 Big Bang Cosmology & Inflation course in Hong Kong University of Science and Technology (HKUST). They are intended as a self-contained supplement to the lectures, providing both conceptual grounding and mathematical rigour at a level appropriate for readers with a background in special relativity, basic linear algebra, and basic calculus.

The material covers different cosmological topics, including but not limited to the Friedmann equations, and the energy budget of the ΛCDM model. Each topic is structured to serve different learning objectives—from building physical intuition to solving examination-style problems and exploring advanced derivations.

These notes do not replace lecture slides or the recommended textbooks. They are best used alongside them as a structured study companion.

If you find any errors or have suggestions for improvement, please feel free to contact me at `yhip@connect.ust.hk`.

Document Conventions

Block Components

Each topic is presented using a consistent **5-component block structure** designed to support multiple learning styles:

Physics Intuition

Purpose: Builds conceptual understanding through plain language explanations.
This section provides the *motivation* behind the physics concept, explores its *historical development*, and offers *intuitive explanations* that help you grasp what the theory is really about before diving into equations.

Use this when: You're first encountering a new concept or need to refresh the "big picture" before tackling mathematics.

Theory

Purpose: Provides rigorous mathematical treatment of the concept. Here you'll find detailed *derivations*, *fundamental assumptions*, key *equations*, and connections between *mathematics and physical intuition*. This is the technical foundation required for exams and deeper problem-solving.

Use this when: You need to understand how to apply formulas, prepare for exams, or work through problems rigorously.

Question Examples

Purpose: Demonstrates practical application through worked problems.
Each question example mimics exam-style problems and includes complete *solution walkthroughs*. These help you bridge the gap between theory and problem-solving, and expose common pitfalls.

Use this when: Practicing problem-solving, preparing for assessments, or checking if you can apply the formulas correctly.

Foundation

Purpose: Supplies background knowledge and detailed proof steps that underpin the main content.
This section presents *prerequisite mathematics or physics*, *step-by-step derivations* that are too detailed to include in the Theory block, and *intermediate results* needed for a thorough understanding. Reading this is optional for the exam but essential for genuine mastery.

Use this when: A derivation in Theory feels too abrupt, you want to verify an intermediate step, or you need to solidify prerequisite concepts.

Extra

Purpose: Provides supplementary material that extends *beyond* the course syllabus.
This section offers *interesting asides*, *historical context*, *connections to current research*, and citations to deeper references. The content here is intentionally out-of-scope for the course examination but enriches the broader picture of each topic.

Use this when: You're curious to explore beyond the syllabus, or when a topic connects to your personal research interests.

Suggested Reading Paths:

1. **Exam Preparation:** *Physics Intuition* → *Theory* → *Question Examples*
2. **Deep Learning:** *Physics Intuition* → *Foundation* → *Theory* → *Question Examples* → *Extra*
3. **Quick Review:** *Theory* → *Question Examples*

Special Annotations

In addition to the five block components above, two inline annotations appear throughout the text to flag specific types of information:

IMPORTANT NOTE

Purpose: Alerts you to a result, condition, or sign convention that is easy to get wrong or is frequently tested.
An Important Note signals a *common misconception*, a *critical sign*, a *non-obvious condition* (e.g. a constraint that must hold for an equation to be valid), or an *exam trap* worth memorising explicitly.

Treat these as mandatory reading regardless of which reading path you follow.

REMARK

Purpose: Highlights a useful observation, connection, or naming convention.
A Remark draws attention to a *conceptual link* between results, an *elegant reformulation*, a *naming convention* (e.g. why a quantity carries its particular name), or a *minor but clarifying point* that does not rise to the level of an Important Note.

Use this when: Looking for a deeper understanding of why an equation takes the form it does, or for useful mnemonics.

Mathematical Conventions

The following conventions are used throughout these notes:

Metric signature. We adopt the *mostly-plus* (or "$(-, +, +, +)$") signature:

$$g_{\mu\nu} = \mathrm{diag}(-1, +1, +1, +1).$$

IMPORTANT NOTE

Many general-relativity textbooks use the opposite, *mostly-minus* $(+, -, -, -)$ signature. The two conventions differ by an overall sign in the Ricci scalar and stress-energy trace. Always check the signature convention when comparing formulae across sources.

Natural units. In sections on particle physics or early-universe thermodynamics, we may set $c = \hbar = k_B = 1$. Quantities are then expressed in powers of eV (or GeV). Explicit c and $\hbar$ factors are restored when numerical values are quoted in SI units.

Key symbols.

Symbol	Meaning	Typical units
H_0	Hubble constant today	$\mathrm{km\,s^{-1}\,Mpc^{-1}}$
h	Dimensionless Hubble: $H_0 = 100h$	—
Ω_i	Density parameter for species i	—
$a(t)$	Scale factor ($a_0 = 1$ today)	—
z	Cosmological redshift	—
ρ_{crit}	Critical density	$\mathrm{kg\,m^{-3}}$

REMARK

The convention $a_0 \equiv a(t_0) = 1$ is standard and simplifies many formulae, e.g. $1 + z = 1/a(t_e)$. Some older texts normalise a differently; always check the definition when reading the literature.

differential notation. We use $\dot{x} \equiv \frac{dx}{dt}$ for time derivatives of some variables $x(t)$.

CHAPTER 1
Fundamentals (lecture 1-2)

Before we can start to build up our universe model, we need to first clarify some basic concepts and notations first for convenience.

1.1 Natural Units

Physics Intuition

In SI units, we treat time, length, mass, and energy as distinct physical quantities with different dimensions and units.

To make calculations more convenient, we introduce natural units to connect these quantities via fundamental constants, so that we can express everything in terms of a single unit and same type of dimension (i.e. energy).

Theory

In natural units, we set $c = \hbar = k_B = 1$.
When $c = 1$, mass and energy are measured in the same units and dimension in terms of E^1 by $E = mc^2 = m$.

Question Example 1.1: Dimension of Gravitational Constant

What is the dimension of the gravitational constant G in natural units?

Solution. We know that $F = G\frac{m_1 m_2}{r^2}$ and $F = ma$.
We get dimension of a is E^1 by $a = v/t = x/t^2 = \frac{E^{-1}}{E^{-1} \cdot E^{-1}} = E^1$.
Then we get dimension of F is E^2 by $F = ma = E^1 \cdot E^1 = E^2$.
Finally, we get dimension of G is E^{-2} by $G = \frac{Fr^2}{m_1 m_2} = E^2 \cdot E^{-2}/(E^1 \cdot E^1) = E^{-2}$.

In general, we define the dimension of a quantity directly in terms of n if its dimension is E^n for simplicity. Below are some useful conversion factors between SI and natural units:

Quantity	**Related equation**	**SI dimension**	**Natural unit**
Energy	$E = mc^2$	$\mathrm{kg\,m^2\,s^{-2}}$	1
Mass	$m = E/c^2$	kg	1
Frequency	$\nu = E/h$	Hz	1
Time	$t = 1/\nu$	s	-1
Space	$x = ct$	m	-1
Temperature	$T = E/k_B$	K	1
Velocity	$v = x/t$	$\mathrm{m\,s^{-1}}$	0
Momentum	$p = E/c$	$\mathrm{kg\,m\,s^{-1}}$	1
Angular Velocity	$\omega = rv$	$\mathrm{rad\,s^{-1}}$	-1
Angular Momentum	$L = \frac{1}{2}mv\omega$	$\mathrm{kg\,m^2\,s^{-1}}$	0
Luminosity	$\mathcal{L} = E/t$	W	2
Force	$F = p/t$	N	2
Pressure	$P = F/A$	Pa	4
Mass Density	$\rho = m/V$	$\mathrm{kg\,m^{-3}}$	4
Energy Density	$\rho = E/V$	$\mathrm{J\,m^{-3}}$	4
Gravitational Constant	$G = Fr^2/m_1m_2$	$\mathrm{m^3\,kg^{-1}\,s^{-2}}$	-2
Redshift	$z = \lambda_0/\lambda_e - 1$	—	0
Cosmological Scale Factor	$a(t) = 1/(1+z)$	—	0
Hubble Parameter	$H = \dot{a}/a$	$\mathrm{s^{-1}}$	1

Remark

In the Heavide-Lorentz system of natural units, we also set $\epsilon_0 = \mu_0 = 1$. Which one can hence get the dimension of charge q is 0.

1.2 4-Vectors and Tensors

1.2.1 4-Vectors

FOUNDATION

In special relativity, we treat the time and space coordinates on equal footing as a 4-vector $x^\mu = (t, x, y, z) = (x^0, x^1, x^2, x^3)$. or in other words, we used the 4-vector to combine the time and space coordinates together.

THEORY

By using the 4-vector, we can write the Lorentz transformation in-terms of the Lorentz transformation matrix and the 4-vector:
WLOG, we can consider a boost in the x-direction with velocity v,

$$\begin{pmatrix} t' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} t \\ x \\ y \\ z \end{pmatrix},$$

where $\gamma = 1/\sqrt{1 - v^2}$ is the Lorentz factor, and $\beta = v/c$ is the velocity in units of c ($\beta = v$ in natural units).

1.2.2 Einstein Summation Convention

PHYSICS INTUITION

Each time when we are doing the 4-vectors operations like the Lorentz transformation, we are doing the summation of the products of the components of the 4-vectors and the Lorentz transformation matrix.

It can be very tedious to write down the summation every time, so we want a more compact notation to represent the summation.

THEORY

Using the example of the Lorentz transformation above, if we want to do the transformation in traditional way, we have to write down the summation explicitly:

$$x^{\mu'} = \Sigma_{\nu=0}^{3} \Lambda^\mu{}_\nu x^\nu.$$

By using the Einstein summation convention, we can simply write down the transformation as:

$$x^{\mu'} = \Lambda^\mu{}_\nu x^\nu.$$

1.2.3 Metric Tensor

Physics Intuition

In a non-Euclidean space, such as the Minkowski spacetime in special relativity, the distance between two points is not simply what we seen in our eyes.
The spacetime interval between two events(or points) is no longer measured by the Pythagorean theorem, so we need to use another ruler to measure it.

In general, we used a metric tensor as the ruler to measure the distance of 2 point in a specific spacetime.

Theory

In traditional Euclidean space, the metric tensor is simply the identity matrix, so the distance between two points is given by the Pythagorean theorem:

$$ds^2 = \begin{pmatrix} dx & dy & dz \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} dx \\ dy \\ dz \end{pmatrix} = dx^2 + dy^2 + dz^2.$$

In Minkowski spacetime, the metric tensor is given by:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

so the spacetime interval is given by:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -dt^2 + dx^2 + dy^2 + dz^2$$

or in other words, the spacetime interval is calculated by the dot product of the 4-vector with itself by using the metric tensor as the dot product operator:

Theory

$$A \cdot B = g_{\mu\nu} A^\mu B^\nu = \begin{pmatrix} A^0 & A^1 & A^2 & A^3 \end{pmatrix} \begin{pmatrix} g_{00} & g_{01} & g_{02} & g_{03} \\ g_{10} & g_{11} & g_{12} & g_{13} \\ g_{20} & g_{21} & g_{22} & g_{23} \\ g_{30} & g_{31} & g_{32} & g_{33} \end{pmatrix} \begin{pmatrix} B^0 \\ B^1 \\ B^2 \\ B^3 \end{pmatrix}$$

1.2.4 Tensors

PHYSICS INTUITION

Now you may wonder, why sometimes the index is up, and sometimes it is down? Is there any difference between them?
The answer is yes, they are different.

The upper index represents a **contravariant** component, while the lower index represents a **covariant** component.
By the combination of the **contravariant** and **convariant** components, we can construct a more general mathematical object called a **tensor**.

FOUNDATION

In tensor calculus and differential geometry, the position of an index (upper or lower) indicates how the component of the tensor transforms when you change the coordinate system.

Direct Mapping:

- **Upper Index** (μ in V^μ): Represents a **Contravariant** component.
- **Lower Index** (μ in V_μ): Represents a **Covariant** component.

Transformation Rules:
The terms *contravariant* and *covariant* describe how components change relative to **basis vectors** when switching coordinates (e.g., Cartesian to Polar, or meters to centimeters).

Contravariant (Upper Index):

- **Symbol:** V^μ
- **Transformation:** Transforms **inversely** (against) the basis vectors.
- **Law:** $V'^\mu = \dfrac{\partial x'^\mu}{\partial x^\nu} V^\nu$
- **Intuition:** If your ruler shrinks (smaller basis vectors), the numerical value must grow to describe the same physical length. Example: $1 \text{ m} = 100 \text{ cm}$.

Covariant (Lower Index):

- **Symbol:** V_μ
- **Transformation:** Transforms **in the same way** (with) the basis vectors.

- **Law:** $V'_\mu = \dfrac{\partial x^\nu}{\partial x'^\mu} V_\nu$
- **Intuition:** Like a gradient (slope). If the ruler shrinks, the slope per unit distance also shrinks.

Geometric Interpretation:

- **Contravariant Vectors** (V^μ): Standard "arrows" tangent to a curve, living in the **Tangent Space**.
- **Covariant Vectors** (V_μ): Called **Dual Vectors** or **One-Forms**. They are functions that map vectors to scalars and live in the **Cotangent Space**. Geometrically visualized as stacked parallel surfaces pierced by vectors.

Einstein Summation Convention:
The distinction is critical, repeated indices (one upper, one lower) indicate summation. Only these are coordinate-invariant scalars.

- **Valid:** $S = A^\mu B_\mu \quad \Rightarrow$ scalar invariant
- **Invalid:** $A^\mu B^\mu$ or $A_\mu B_\mu \quad \Rightarrow$ not coordinate-invariant (not summation)

Summary Table

Property	Contravariant	Covariant
Index Position	Upper ($^\mu$)	Lower ($_\mu$)
Notation	V^μ	V_μ
Also Called	Vector	Dual Vector / One-Form
Transformation	Opposite to basis	Same as basis
Example	Displacement, Velocity	Gradient
Metric Operation	Lowered by $g_{\mu\nu}$	Raised by $g^{\mu\nu}$

THEORY

Raising and Lowering Indices:
With the **Metric Tensor** ($g_{\mu\nu}$), you can convert between contravariant and covariant components:

- **Lowering:** $V_\mu = g_{\mu\nu} V^\nu$ (contravariant $\to$ covariant)
- **Raising:** $V^\mu = g^{\mu\nu} V_\nu$ (covariant $\to$ contravariant)

And the contravariant(inverse) metric $g^{\mu\nu}$ is the inverse of the covariant metric $g_{\mu\nu}$, so we have:

THEORY

$$g^{\mu\nu} g_{\nu\sigma} = \delta^{\mu}{}_{\sigma},$$

where $\delta^{\mu}{}_{\sigma}$ is the Kronecker delta, which is 1 if $\mu = \sigma$ and 0 otherwise.

As a results, we can further simplified the operations like the dot product and trace:

THEORY

$$A \cdot B = g_{\mu\nu} A^{\mu} B^{\nu} = A_{\nu} B^{\nu} = A^{\mu} B_{\mu}$$
$$T = g^{\mu\nu} T_{\mu\nu} = g_{\mu\nu} T^{\mu\nu} = T^{\nu}{}_{\nu} = T_{\mu}{}^{\mu}$$

Question Example 1.2: Trace of Spherical Minkowski Metric

Calculate the trace of the spherical Minkowski metric $g_{\mu\nu} = \mathrm{diag}(-1, 1, r^2, r^2 \sin^2\theta)$.

Solution. We need to first calculate the inverse metric $g^{\mu\nu}$, which is given by:

$$g^{\mu\nu} = g_{\mu\nu}^{-1} = \mathrm{diag}(-1, 1, r^{-2}, r^{-2} \sin^{-2}\theta).$$

Then we can calculate the trace:

$$Trace = g^{\mu\nu} g_{\mu\nu} = (-1)(-1) + (1)(1) + (r^{-2})(r^2) + (r^{-2}\sin^{-2}\theta)(r^2\sin^2\theta) = 4.$$

IMPORTANT NOTE

In flat Euclidean space with Cartesian coordinates, $g_{\mu\nu} = \delta_{\mu\nu}$ (identity), so $V^{\mu} = V_{\mu}$ numerically.

However, for other spacetimes (e.g., Minkowski, curved), the metric is not the identity, and V^{μ} and V_{μ} can differ significantly. The distinction becomes crucial in General Relativity and curvilinear coordinates. To be safe, treat them as different objects and always check the metric when raising/lowering indices.

EXTRA

The tensor T is said to be of type (p,q) if it has:

- p: The number of contravariant indices(upper indices)
- q: The number of convariant indices(lower indices)

Tensor with different types cannot be added together or equal to each other, in such a meaning, you can think of the type of a tensor as its “dimension” in the space of tensors, and only tensors with the same type can be compared or combined.

Tensor Rank: The total rank of a Tensor T is p + q (let said p + q = n), and we will said it is a rank-n-Tensor

Note that the rank of a tensor is not the same as linear algebra rank of a matrix. The rank of a tensor is the total number of indices, while the linear algebra rank is the maximum number of linearly independent rows or columns in a matrix. Thats mean a rank-2 tensor can have a linear algebra rank of 0, 1, or 2 depending on its components.

Tensor Type	**p**	**q**	**Rank**	**Example**
(0,0)	0	0	0	Scalar (e.g., Temperature)
(1,0)	1	0	1	Contravariant Vector (e.g., Displacement V^μ)
(0,1)	0	1	1	Covariant Vector (e.g., Gradient V_μ)
(1,1)	1	1	2	Mixed Tensor (e.g., Linear Transformation $\Lambda^\mu{}_\nu$)
(2,0)	2	0	2	Contravariant Tensor (e.g., Inverse Metric $g^{\mu\nu}$)
(0,2)	0	2	2	Covariant Tensor (e.g., Metric Tensor $g_{\mu\nu}$)
(1,3)	1	3	4	Riemann Curvature ($R^\rho_{\sigma\mu\nu}$)

1.2.5 Energy-Momentum Tensor

PHYSICS INTUITION

In Newtonian mechanics, we thought mass is the source of gravity, but in relativity, we found that mass is a form of energy($E = mc^2$), so energy must also be a source of gravity.
Further, momentum is also a form of energy($E^2 = p^2c^2 + m^2c^4$), so momentum must also be a source of gravity.

The energy-momentum tensor expand the concept of a scalar energy density to a 4x4 matrix that track of all "components" in a region of spacetime that contributes to gravity.

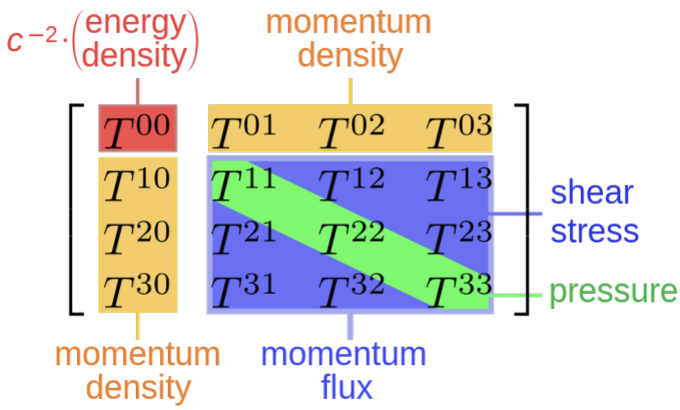

Figure 1.1: The components of the energy-momentum tensor $T^{\mu\nu}$ and their physical interpretations.

FOUNDATION

Interpretation of Components:
For a component $T^{\mu\nu}$ of the energy-momentum tensor, it represents the flux of the μ-th component of the 4-momentum across a volume of constant ν-th coordinate.

(1) Time-Energy Component: T^{00} (Top-Left Corner)

- **What it is:** Energy density — the amount of energy per unit volume.
- **Includes:** Mass (via $E = mc^2$), heat, light, and chemical energy.
- **Gravitational role:** Primary source of gravity for everyday objects (e.g., the Earth).

(2) Energy Flux Components: T^{01}, T^{02}, T^{03} (Top Row)

- **What they measure:** Energy flowing through surfaces in the x-, y-, and z-directions.
- **Physical picture:** A beam of light passing through the volume carries energy in a direction.
- **Key insight:** In relativity, energy flux is equivalent to momentum density. Moving energy carries momentum!

(3) Momentum Density Components: T^{10}, T^{20}, T^{30} (Left Column)

- **What they measure:** The amount of momentum stored inside the volume.
- **Surprising fact:** These values are typically equal to the energy flux components above!
- **Implication:** Energy flow and momentum density are intimately related.

(4) Stress and Pressure Components: T^{11} to T^{33} (Bottom-Right 3×3 Grid)

- **Physical meaning:** Forces and stresses acting within the material.
- **Diagonal terms** (T^{11}, T^{22}, T^{33}): **Pressure**
  - Gas molecules bouncing off the walls of the volume exert pressure.
  - **Remarkable fact:** Pressure creates gravity!
  - A box of hot gas (high pressure) has slightly more gravitational mass than cold gas with the same rest mass, because pressure contributes to the stress-energy tensor.
- **Off-diagonal terms** $(T^{12}, T^{23}$, etc.): **Shear Stress**
  - Internal friction between different parts of the material.
  - Analogy: Sliding the top half of a deck of cards sideways relative to the bottom.

Foundation

The energy-momentum tensor has 2 important properties:

- **(1) Symmetry:** $T^{\mu\nu} = T^{\nu\mu}$
  - Reason 1: Energy flux and momentum density are equal, so $T^{0i} = T^{i0}$.
  - Reason 2: The stress tensor (bottom-right 3×3) is symmetric due to the conservation of angular momentum. If there were a torque, it would cause the material to rotate infinitely fast, which would violate the conservation of angular momentum.
- **(2) Conservation:** $\partial_\mu T^{\mu\nu} = 0$
  - imagine a small box in space-time. The conservation law states that the amount of energy-momentum flowing into the box along the μ-th direction must equal the amount flowing out along the ν-th direction. This ensures that energy and momentum are conserved locally.

Important Note

The energy-momentum tensor is a symmetric so there are only 10 independent components at a single point in spacetime.

If we further consider the conservation law, we can reduce the number of degree of freedom to 6, as it provides 4 additional constraints on the components of the energy-momentum tensor.

(i.e. 10 independent components, with 4 constraints from conservation, leaves 6 degrees of freedom.)

Remark

In fact, for different type of fluids, the can have different number of degree of freedom, 6 is just the upper bound for a general fluid. For example, a perfect fluid has only 5 degree of freedom, which are the energy density(1) and pressure(1) and velocity(3), as it is isotropic and has no shear stress.

THEORY

For frictionless perfect fluid, the energy-momentum tensor can be written as:

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu},$$

where ρ is the energy density, P is the pressure, and u^μ is the 4-velocity of the fluid.
WLOG, we can consider a comoving frame where the fluid is at rest, so $u^\mu = (1, 0, 0, 0)$, then we can simplify the energy-momentum tensor to:

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & -P & 0 & 0 \\ 0 & 0 & -P & 0 \\ 0 & 0 & 0 & -P \end{pmatrix}$$

The perfect fluid is a good approximation for many astrophysical and cosmological systems, such as stars, galaxies, and the large-scale structure of the universe. As it is isotropic and has no shear stress, it can be used to model the behavior of matter in these systems without having to consider the complexities of viscosity and heat conduction.

Question Example 1.3: Trace of Energy-Momentum Tensor for Perfect Fluid

Calculate the trace of the energy-momentum tensor for a perfect fluid.

Solution. We can calculate the trace by contracting the indices of the energy-momentum tensor with the metric tensor:

$$T = g_{\mu\nu}T^{\mu\nu} = g_{00}T^{00} + g_{11}T^{11} + g_{22}T^{22} + g_{33}T^{33}.$$

Substituting the values from the perfect fluid energy-momentum tensor, we get:

$$T = (-1)(\rho) + (1)(-P) + (1)(-P) + (1)(-P) = -\rho + 3P.$$

1.3 Einstein Field Equations

In this course, the Einstein field equations (EFE) are not main focus, but it is still important to understand the basic form of the EFE and how it relates to the energy-momentum tensor, as it is the fundamental equation that describes how matter and energy influence the curvature of spacetime, which is the basis for our understanding of gravity and cosmology.

Theory

The Einstein field equations (EFE) are the fundamental equations of General Relativity that describe how matter and energy influence the curvature of spacetime. They can be written as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu},$$

where $G_{\mu\nu}$ is the Einstein tensor, which encodes the curvature of spacetime, $T_{\mu\nu}$ is the energy-momentum tensor, which describes the distribution of matter and energy, and G is the gravitational constant.

The EFE can be thought of as a set of ten coupled, nonlinear partial differential equations that relate the geometry of spacetime (through $G_{\mu\nu}$) to the energy and momentum content (through $T_{\mu\nu}$).

We can also write the EFE in terms of trace-reversed form:

Theory

$$R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2}Tg_{\mu\nu} \right),$$

where $T = g^{\mu\nu}T_{\mu\nu}$ is the trace of the energy-momentum tensor. This form is often more convenient for calculations, as it directly relates the Ricci curvature tensor $R_{\mu\nu}$ to the energy-momentum tensor without the need to compute the scalar curvature R separately.

1.4 Cosmological Principle

The cosmological principle is a fundamental assumption in cosmology that states that the universe is homogeneous and isotropic on large scales. This means that the universe looks the same in all directions (isotropic) and has the same properties at every point (homogeneous) when viewed on sufficiently large scales.

Extra

The cosmic Microwave Background (CMB) radiation provides strong evidence for the cosmological principle, as it is observed to be nearly uniform in all directions, with only tiny fluctuations at the level of one part in 100,000.

CHAPTER 2

Hubble's Law and Cosmological Redshift (lecture 3-4)

With the fundamental knowledge in last chapter, we can now start to build up our universe model.

2.1 Robertson-Walker Metric

2.1.1 Metric for a Homogeneous and Isotropic Universe

Physics Intuition

In general relativity, the geometry of spacetime can be curved.
Therefore, we need another metric instead of the Minkowski metric to describe the geometry of the universe.
The Robertson-Walker metric was hence introduced to describe a homogeneous and isotropic universe in large scales.

Theory

By the cosmological principle, the universe is homogeneous and isotropic on large scales.
In mathematics, there are only three possible spatial geometries that satisfy these conditions, defined by the constant curvature k:

- $k = 0$: flat geometry (Euclidean space)
- $k > 0$: closed geometry (spherical space)
- $k < 0$: open geometry (hyperbolic space)

And we need to write the line element dl^2 for such a spacetime.
Note that the cosmological principle only states that the universe is homogeneous and isotropic, but it does not forbid the universe to expand or contract.
Therefore, we need to introduce a time-dependent scale factor $a(t)$ to describe the expansion of the universe.

$$ds^2 = -dt^2 + a^2(t)dl^2$$

To fullfill the isotropy condition, we will use spherical coordinates to write the spatial line element dl^2:

$$dl^2 = g_{rr}dr^2 + g_{\theta\theta}d\theta^2 + g_{\phi\phi}d\phi^2$$

Foundation

you may ask why we dont need to consider the cross terms like $g_{r\theta}drd\theta$ in the line element.

The reason is that the cross terms would break the isotropy of the universe. For example, if we have a cross term like $g_{r\theta}drd\theta$, it would mean that the radial distance between two points in the universe would depend on the direction we are looking at, which would violate the isotropy condition.

Therefore, we can safely ignore the cross terms in the line element when we are considering a homogeneous and isotropic universe.

Theory

Flat geometry ($k = 0$):
It is trivial to get the line element for flat geometry, which is just the Euclidean space:

$$dl^2 = dr^2 + r^2 d\Omega^2 = dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

which $d\Omega^2$ is the line element of a unit sphere(r = 1).
Hence, the metric for a flat universe is:

$$ds^2 = -dt^2 + a^2(t)[dr^2 + r^2 d\Omega^2] = -dt^2 + a^2(t)\left[dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

However, for the closed and open geometry, the coefficient of $d\Omega^2$ is not r^2, but a function of r.

Closed geometry ($k > 0$):
To get the line element for closed geometry like spherical space, we first consider a 3-sphere embedded in a 4-dimensional Euclidean space, which can be described by the equation:

$$x^2 + y^2 + z^2 + w^2 = R^2$$

where R is the radius of the 3-sphere.
We can then use spherical coordinates to parameterize the 3-sphere:

$$\begin{cases} x = R\sin\chi\sin\theta\cos\phi \\ y = R\sin\chi\sin\theta\sin\phi \\ z = R\sin\chi\cos\theta \\ w = R\cos\chi \end{cases}$$

where χ is the radial coordinate that ranges from 0 to π.
And we can get the metric components:

$$\begin{cases} g_{\chi\chi} = \frac{\partial}{\partial\chi}(x,y,z,w) \cdot \frac{\partial}{\partial\chi}(x,y,z,w) = R^2 \\ g_{\theta\theta} = \frac{\partial}{\partial\theta}(x,y,z,w) \cdot \frac{\partial}{\partial\theta}(x,y,z,w) = R^2\sin^2\chi \\ g_{\phi\phi} = \frac{\partial}{\partial\phi}(x,y,z,w) \cdot \frac{\partial}{\partial\phi}(x,y,z,w) = R^2\sin^2\chi\sin^2\theta \end{cases}$$

The line element for the 3-sphere can then be derived as:

$$dl^2 = R^2\left[d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

Hence, the metric for a closed universe is:

$$ds^2 = -dt^2 + a^2(t)R^2\left[d\chi^2 + \sin^2\chi(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

Open geometry ($k < 0$):
To get the line element for open geometry like hyperbolic space, we can consider a 3-hyperboloid embedded in a 4-dimensional Minkowski space, which can be described by the equation:

$$-x^2 - y^2 - z^2 + w^2 = R^2$$

where R is the radius of curvature of the hyperbolic space.
We can then use hyperbolic coordinates to parameterize the 3-hyperboloid:

$$\begin{cases} x = R\sinh\chi\sin\theta\cos\phi \\ y = R\sinh\chi\sin\theta\sin\phi \\ z = R\sinh\chi\cos\theta \\ w = R\cosh\chi \end{cases}$$

where χ is the radial coordinate that ranges from 0 to ∞.
And we can get the metric components:

$$\begin{cases} g_{\chi\chi} = \frac{\partial}{\partial\chi}(x,y,z,w) \cdot \frac{\partial}{\partial\chi}(x,y,z,w) = R^2 \\ g_{\theta\theta} = \frac{\partial}{\partial\theta}(x,y,z,w) \cdot \frac{\partial}{\partial\theta}(x,y,z,w) = R^2\sinh^2\chi \\ g_{\phi\phi} = \frac{\partial}{\partial\phi}(x,y,z,w) \cdot \frac{\partial}{\partial\phi}(x,y,z,w) = R^2\sinh^2\chi\sin^2\theta \end{cases}$$

The line element for the 3-hyperboloid can then be derived as:

$$dl^2 = R^2\left[d\chi^2 + \sinh^2\chi(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

Hence, the metric for an open universe is:

$$ds^2 = -dt^2 + a^2(t)R^2\left[d\chi^2 + \sinh^2\chi(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

2.1.2 Robertson-Walker metric

Physics Intuition

In order to be simplified, we want to make a unified form of the metric for all three types of geometry.

Theory

Unified form:

$$\begin{pmatrix} g_{tt} & 0 & 0 & 0 \\ 0 & g_{rr} & 0 & 0 \\ 0 & 0 & g_{\theta\theta} & 0 \\ 0 & 0 & 0 & g_{\phi\phi} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a^2(t)}{1-kr^2} & 0 & 0 \\ 0 & 0 & a^2(t)r^2 & 0 \\ 0 & 0 & 0 & a^2(t)r^2\sin^2\theta \end{pmatrix}$$

where k is the curvature constant that can take values of 0, 1, or -1 for flat, closed, and open geometries, respectively.
And we defined the r:

$$r = \begin{cases} \chi & ,k = 0 \\ \sin\chi & ,k > 0 \\ \sinh\chi & ,k < 0 \end{cases}$$

Foundation

By the substitution of r into the unified metric and letting the unit $R = 1$, you can easily to get the $d\theta^2$ and $d\phi^2$ terms of 3 types of geometries, and get the corresponding dr^2 term by using the chain rule:

$$dr^2 = \begin{cases} d\chi^2 & ,k = 0 \\ \cos^2\chi d\chi^2 & ,k > 0 \\ \cosh^2\chi d\chi^2 & ,k < 0 \end{cases}$$

$$\Rightarrow \frac{dr^2}{1-kr^2} = \begin{cases} d\chi^2 & ,k = 0 \\ d\chi^2 & ,k > 0 \\ d\chi^2 & ,k < 0 \end{cases}$$

And this unified metric is called the **Robertson-Walker metric**, which can be written as:

$$ds^2 = -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right]$$

Remark

In modern cosmology, we will use Friedmann-Lemaître-Robertson-Walker (FLRW) metric, but they are essentially the same, the FLRW metric derived from the Einstein field equations instead of geometry approach like the Robertson-Walker metric, and is used for explaining the dynamics of the universe, while the Robertson-Walker metric is used for describing the geometry of the universe.

2.1.3 the Scale Factor $a(t)$

Physics Intuition

Now we get the Robertson-Walker metric, and we know that through "time" the scale factor $a(t)$ can change, and the metric can be different at different time, which means the proper distance between two points in the universe can change with time.

But what the "time" here means? Isn't the special relativity tells us that there is no absolute time?
The "time" here is the proper time of comoving observers, which are observers that are at rest with respect to the expansion of the universe.
For comoving observers, the proper time τ is the same as the coordinate time t.

Physics Intuition

As the scale factor $a(t)$ applied on the metric instead of the coordinates difference dr, $d\theta$, and $d\phi$, we can think the measurement of coordinate spatial distance is fixed, but the measurement unit (the ruler itself) is changing with time. Given:

$$\begin{aligned} ds^2 &= -dt^2 + a^2(t)\left[\frac{dr^2}{1-kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right] \\ &= -dt^2 + a^2(t)d\chi^2 \end{aligned}$$

and we can get:

$$\Delta D_{proper} = a(t)\Delta\chi$$

which D_{proper} is the ds term, and χ is the comoving distance coordinate.

For example, if we have 2 points on balloon, we draw the marks on the balloon to measure the distance between the two points, when the balloon is expanding, the number of marks between the two points is fixed, but the distance between each mark is increasing with time, which means the proper distance between the two points is increasing with time.

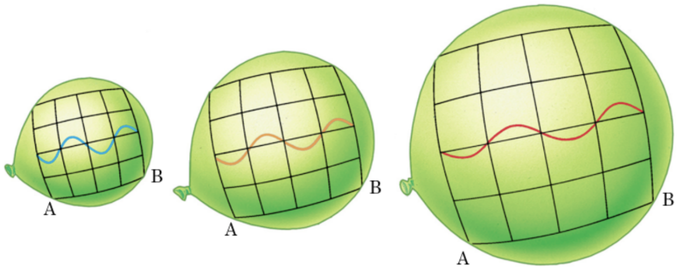

Figure 2.1: The balloon analogy for the expansion of the universe.

Remark

Although scale factor $a(t)$ can change with time, we assume that the curvature constant k does not change with time, which means the geometry of the universe is fixed.

Foundation

If we consider the distance light travel from source to observer, that means $ds^2 = 0$ as light travels along null geodesics, and we can get:

$$\begin{aligned} 0 &= -dt^2 + a^2(t)d\chi^2 \\ dt &= \pm a(t)d\chi \end{aligned}$$

Note when time increase $(dt > 0)$, the light is traveling towards the observer $(d\chi < 0)$, so we need to choose the -ve sign to get the correct direction of light travel.

Hence, we can get the relation between the comoving distance χ of 2 points and the time of travel t:

$$\begin{aligned} dt &= -a(t)d\chi \\ \int_{\chi_0}^{\chi_{source}} d\chi &= -\int_{t_{obs}}^{t_{emit}} \frac{dt}{a(t)} \\ -\chi_{source} &= -\int_{t_{obs}}^{t_{emit}} \frac{dt}{a(t)} \\ \chi_{source} &= \int_{t_{emit}}^{t_{obs}} \frac{dt}{a(t)} \end{aligned}$$

where χ_{source} is the comoving coordinate of the source, χ_0 is the comoving coordinate of the observer, and we can set $\chi_0 = 0$ for simplicity.

Theory

To calculate the **time-independent** comoving distance χ between the source and the observer, we can use the following formula:

$$\chi = \int_{t_{emit}}^{t_{obs}} \frac{dt}{a(t)}$$

which is the integral of the inverse of the scale factor over the time of travel.

Note that the comoving distance χ is fixed once we defined a specific source and observer, and it does not change with time or changing scale factor, which quites counter-intuitive, but it is because the comoving distance is defined in a way that it is not affected by the expansion of the universe, and it is a measure of the distance between two points in the universe that is independent of the expansion.

2.2 Hubble's Law

2.2.1 Redshift and Recession Velocity

Recall the redshift z can occur when the source is moving away from the observer.

Foundation

To derive the Doppler redshift formula in special relativity, we define:

$$\beta \equiv \frac{v}{c}, \qquad \gamma \equiv \frac{1}{\sqrt{1-\beta^2}}$$

where $v > 0$ means the source is receding from the observer.

Let two successive wave crests be emitted with proper time interval $\Delta\tau_{emit}$ in the source rest frame:

$$\Delta\tau_{emit} = \frac{1}{\nu_{emit}}$$

In the observer frame, this emission interval is time-dilated:

$$\Delta t_{emit} = \gamma\,\Delta\tau_{emit}$$

Why do we need the extra travel time?

The key insight is that between the emission of the first crest at time t_1 and the second crest at time $t_1 + \Delta t_{emit}$, the source is *moving away* from the observer. Therefore, the second crest is emitted from a point that is farther from the observer by a distance $\Delta d = v\Delta t_{emit}$. Since the second crest must travel this extra distance at the speed of light c, it requires an additional time to reach the observer.

During this interval, the source moves farther by $v\Delta t_{emit}$, so the second crest has extra travel time:

$$\Delta t_{extra} = \frac{v\Delta t_{emit}}{c} = \frac{v}{c}\Delta t_{emit} = \beta\,\Delta t_{emit}$$

This is the additional travel time caused by the source's recession — the source "runs away" from its own light. Hence the observed arrival interval is:

$$\Delta t_{obs} = \Delta t_{emit} + \Delta t_{extra} = \Delta t_{emit}(1+\beta) = \gamma\Delta\tau_{emit}(1+\beta)$$

Therefore,

$$\nu_{obs} = \frac{1}{\Delta t_{obs}} = \frac{1}{\gamma\Delta\tau_{emit}(1+\beta)} = \frac{\nu_{emit}}{\gamma(1+\beta)} = \nu_{emit}\sqrt{\frac{1-\beta}{1+\beta}}$$

Using $\lambda = c/\nu$, we get:

$$\frac{\lambda_{obs}}{\lambda_{emit}} = \frac{\nu_{emit}}{\nu_{obs}} = \sqrt{\frac{1+\beta}{1-\beta}}$$

By definition of redshift,

$$z \equiv \frac{\lambda_{obs}-\lambda_{emit}}{\lambda_{emit}} = \frac{\lambda_{obs}}{\lambda_{emit}} - 1 = \sqrt{\frac{1+\beta}{1-\beta}} - 1 = \sqrt{\frac{1+v/c}{1-v/c}} - 1$$

Note ν and v are 2 different things, ν is the frequency of the light wave, while v is the velocity of the source. Don't mix them even they look like same!

THEORY

calculation on **Doppler redshift**, based on wavelength:

$$z = \frac{\lambda_{obs}-\lambda_{emit}}{\lambda_{emit}} = \frac{\lambda_{obs}}{\lambda_{emit}} - 1$$

and based on frequency:

$$z = \frac{\nu_{emit}-\nu_{obs}}{\nu_{obs}} = \frac{\nu_{emit}}{\nu_{obs}} - 1$$

and based on the peculiar velocity of the source:

$$z = \sqrt{\frac{1 + v/c}{1 - v/c}} - 1$$

where +ve v means the source is moving away from the observer, and -ve v means the source is moving towards the observer.

2.2.2 Cosmological Redshift

PHYSICS INTUITION

However, in 1929, Edwin Hubble found that most of the galaxies are moving away from us, which does not make sense, if the doppler effect is the only reason for the redshift, then we should expect around half of the galaxies are moving towards us, and the other half are moving away from us, which is not what we observed.

It suggests that there must be another reason for the redshift, and later we know that it is the expansion of the universe that causes the redshift, which dominates over the doppler effect for distant galaxies, and thats why most of the galaxies are moving away from us.

FOUNDATION

Start from the FLRW metric and consider radial light propagation ($d\Omega = 0$):

$$ds^2 = -dt^2 + a^2(t)d\chi^2$$

For photons, $ds^2 = 0$, so:

$$d\chi = \frac{dt}{a(t)}$$

Let a source at fixed comoving coordinate χ_s emit two successive crests. For the first crest (emitted at t_{emit}, observed at t_{obs}):

$$\int_{t_{emit}}^{t_{obs}} \frac{dt}{a(t)} = \chi_s$$

For the second crest (emitted at $t_{emit} + \Delta t_{emit}$, observed at $t_{obs} + \Delta t_{obs}$):

$$\int_{t_{emit}+\Delta t_{emit}}^{t_{obs}+\Delta t_{obs}} \frac{dt}{a(t)} = \chi_s$$

Because both crests travel between the same source and observer, the comoving distance is the same. Subtract the two equations:

$$\int_{t_{emit}}^{t_{emit}+\Delta t_{emit}} \frac{dt}{a(t)} = \int_{t_{obs}}^{t_{obs}+\Delta t_{obs}} \frac{dt}{a(t)}$$

For two successive crests, Δt is very small, so $a(t)$ is nearly constant during each small interval:

$$\frac{\Delta t_{emit}}{a(t_{emit})} \approx \frac{\Delta t_{obs}}{a(t_{obs})} \quad \Rightarrow \quad \frac{\Delta t_{obs}}{\Delta t_{emit}} = \frac{a(t_{obs})}{a(t_{emit})}$$

Now convert to wavelength. For a locally measured light wave, $\lambda = c\,\Delta t$ (or $\lambda \propto \Delta t$ in $c = 1$ units), therefore:

$$\frac{\lambda_{obs}}{\lambda_{emit}} = \frac{\Delta t_{obs}}{\Delta t_{emit}} = \frac{a(t_{obs})}{a(t_{emit})}$$

By the definition of redshift:

$$z \equiv \frac{\lambda_{obs} - \lambda_{emit}}{\lambda_{emit}} = \frac{\lambda_{obs}}{\lambda_{emit}} - 1 = \frac{a(t_{obs})}{a(t_{emit})} - 1$$

THEORY

Calculation on **cosmological redshift**, based on the expansion of the universe:

$$z = \frac{a(t_{obs})}{a(t_{emit})} - 1 = \frac{\lambda_{obs}}{\lambda_{emit}} - 1 = \frac{\nu_{emit}}{\nu_{obs}} - 1$$

and based on the recession velocity of the source:

$$z = \sqrt{\frac{1 + v/c}{1 - v/c}} - 1$$

where $a(t_{obs})$ is the scale factor at the time of observation, and $a(t_{emit})$ is the scale factor at the time of emission.

If we treat the scale factor $a(t)$ will change with time, and let the time of observation be t_0, so that $a(t_0) = a_0 = 1$, then we can rewrite the cosmological redshift as:

$$z = \frac{1}{a(t_{emit})} - 1$$

Remark

The redshift z of galaxies can be contributed by both recession motion(**cosmological redshift**) and peculiar motion(**doppler redshift**), but for distant galaxies $z > 0.01$, the cosmological redshift dominates over the doppler redshift.

2.2.3 Hubble's Law

Physics Intuition

In 1823, Olber's paradox was proposed, which states that if the universe is infinite and static, then the night sky should be bright because there are infinite number of stars in the universe. However, the night sky is dark, which means the universe cannot be infinite and static.

The paradox can not be resolved until the discovery by Edwin Hubble in 1929, who found that the recession velocity of galaxies is proportional to their distance from us, which is now known as Hubble's law. The Hubble's law then become one of the core evidences for the expansion of the universe, and suggest the Big Bang theory as the most plausible model for the origin of the universe, which solves the Olber's paradox by proposing that the universe is neither infinite nor static, but it is expanding from a hot and dense initial state.

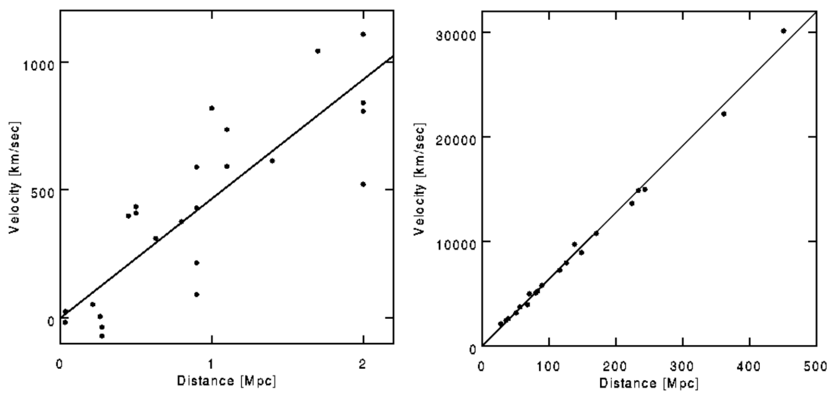

Figure 2.2: Hubble diagram showing the relationship between the recession velocity of galaxies and their distance from us.

The left plot show the original data from Hubble's 1929 paper, and the right plot show the modern data from more distinct galaxies, which is more accurate and has more convincing straight line to fit.

Theory

Hubble's Law states that the recession velocity of a galaxy is proportional to its distance from us:

$$v = HD$$

where H is the Hubble constant, D is the proper distance to the galaxy(true distance between 2 points), and can be changed with time unlike the co-moving distance(the number of marks).

Theory

Let D_{proper} be the proper distance to a galaxy, and χ be the co-moving distance coordinate to the galaxy, then we have:

$$D_{proper}(t) = a(t) \cdot \chi$$

Note χ is independent to time.

The recession velocity v can also be defined as:

$$v \equiv \frac{dD_{proper}}{dt} = \dot{a}(t) \cdot \chi$$

And sub the Hubble's law into the above equation, we can get:

$$\begin{aligned} v &= HD_{proper} \\ \dot{a}(t) \cdot \chi &= Ha(t) \cdot \chi \\ \dot{a}(t) &= Ha(t) \end{aligned}$$

And we know that the Hubble parameter is equal to:

$$H(t) = \frac{\dot{a}(t)}{a(t)} = -\frac{\dot{z}(t)}{1 + z(t)}$$

Remark

For $t > 0$ (after Big Bang), $a(t) > 0$, except at the initial singularity$(t = 0, a(0) = 0)$;

while the observation shows that $H_0 > 0$, which indicates $\dot{a}(t_0) > 0$, which means the universe is expanding at the current time t_0.

Now we can finding the relationship between the redshift z and the distance D of a galaxy, by using the Hubble's law and the definition of cosmological redshift:

Theory

we have cosmological redshift:

$$z = \frac{1}{a(t_{emit})} - 1$$

Important Note

Hubble's constant H_0 is not actually a constant, but it can change with time.

The value of $H(t_0) = H_0$ is the current value of the Hubble parameter, which is measured to be around $67.66 \pm 0.42\,\mathrm{km/s/Mpc}$ by the Planck Mission in 2018.

Extra

The Hubble's law get renamed to Hubble-Lemaître law in 2018 by the International Astronomical Union (IAU) to recognize the contribution of Georges Lemaître, who first derived the linear relationship between the recession velocity and distance of galaxies in 1927, two years before Hubble's observational discovery in 1929. However, the name change is still controversial, and many people still refer to it as Hubble's law.

Extra

The Hubble's constant can not be measured directly, but it can be inferred indirectly through observations of the universe, and there are two main methods: the distance ladder method and the cosmic microwave background (CMB) method.

The distance ladder method uses standard candles like Cepheid variables and Type Ia supernovae to measure the distance to galaxies, it focus on "late universe", which means the universe at low redshiftz < 0.1, and get the value of the Hubble constant to be around $73.04 \pm 1.04\,\mathrm{km/s/Mpc}$ by the SH0ES team in 2021.

The CMB method uses the anisotropies in the cosmic microwave background radiation to infer the value of the Hubble constant, it focus on "early universe", which means the universe at high redshift $z \sim 1100$, and get the value of the Hubble constant to be around $67.66 \pm 0.42\,\mathrm{km/s/Mpc}$ by the Planck Mission in 2018.

The difference results of H_0 obtained by two methods is known as the Hubble tension, which is one of the biggest unsolved problems in cosmology.

2.2.4 Luminousity Distance

PHYSICS INTUITION

In real observations, we cannot directly measure the geometric distance of a distant galaxy, what we directly measure is its brightness (flux).

So we define **luminousity distance**: the distance an object *appears* to have if we infer distance only from how bright it looks.

In a static Euclidean world, brightness only drops because light is spread over larger area ($\propto 1/4\pi r^2$), so we can directly infer distance r.

But in an expanding universe, light is dimmed not only by geometric spreading, but also by redshift effects during travel. Therefore this brightness-based distance is not exactly the same as simple geometric distance, and that is why we introduce d_L as a separate and useful distance definition in cosmology.

FOUNDATION

Start from the observational definition of luminousity distance:

$$F \equiv \frac{L}{4\pi d_L^2},$$

where L is the source intrinsic luminousity (energy emitted per unit source proper time), and F is the observed flux (energy received per unit detector area per unit observer time).

In FLRW, a source at comoving coordinate χ has present-day proper radius $a_0\chi$ from us, so photons are spread over area

$$A = 4\pi(a_0\chi)^2.$$

With $a_0 = 1$, this becomes $A = 4\pi\chi^2$.

The key point is that cosmological expansion changes both photon energy and arrival rate:

(i) **Photon energy redshift:** from $1 + z = \lambda_{obs}/\lambda_{emit} = \nu_{emit}/\nu_{obs}$, each photon has

$$E_{obs} = h\nu_{obs} = \frac{h\nu_{emit}}{1+z} = \frac{E_{emit}}{1+z}.$$

So detected energy gets one suppression factor $1/(1+z)$.

(ii) **Arrival-time dilation:** from previous section,

$$\frac{\Delta t_{obs}}{\Delta t_{emit}} = \frac{a_0}{a_{emit}} = 1+z.$$

So the same emitted wave train is received over a longer observer interval by factor $(1+z)$, giving another suppression factor $1/(1+z)$ in energy-per-time.

Therefore, compared with naive inverse-square spreading, observed flux is reduced by

$$\frac{1}{(1+z)^2} = \frac{1}{1+z} \times \frac{1}{1+z}.$$

Hence

$$\begin{aligned} F &= \frac{L}{4\pi(a_0\chi)^2} \cdot \frac{1}{(1+z)^2} \\ &= \frac{L}{4\pi\left[(1+z)a_0\chi\right]^2}. \end{aligned}$$

Compare with $F = L/(4\pi d_L^2)$, we identify

$$d_L = (1+z)a_0\chi.$$

Using the chapter convention $a_0 = 1$:

$$\boxed{d_L = (1+z)\chi.}$$

THEORY

The luminousity distance d_L is related to the comoving distance χ and the redshift z by the formula:

$$d_L = (1+z)\chi$$

IMPORTANT NOTE

The equation $d_L = (1+z)\chi$ is assuming a flat universe ($k = 0$). For non-flat universe, we need a generalized formula:

$$d_L = (1+z) \cdot S_k(\chi)$$

where $S_k(\chi)$ is the comoving angular diameter distance, which depends on the curvature of the universe:

$$S_k(\chi) = \begin{cases} \chi & ,k = 0 \\ \frac{1}{\sqrt{k}}\sin(\sqrt{k}\chi) & ,k > 0 \\ \frac{1}{\sqrt{-k}}\sinh(\sqrt{-k}\chi) & ,k < 0 \end{cases}$$

2.3 Cosmological Horizons

2.3.1 Hubble Horizon

PHYSICS INTUITION

The Hubble horizon is the distance scale where the recession speed from cosmic expansion equals the speed of light.

If a galaxy is at proper distance D, its recession velocity is $v_{rec} = H(t)D$. So there is a special distance where $v_{rec} = c$.

This gives us a useful characteristic scale of the universe at each epoch, but it is not exactly the same as a strict causal horizon.

FOUNDATION

From Hubble's law in proper distance form:

$$v_{rec}(t) = H(t)D_{proper}(t)$$

define the Hubble horizon by $v_{rec} = c$:

$$c = H(t)D_H(t) \quad \Rightarrow \quad D_H(t) = \frac{c}{H(t)}$$

If we use chapter natural unit $c = 1$:

$$D_H(t) = \frac{1}{H(t)}$$

and corresponding comoving Hubble radius is

$$\chi_H(t) = \frac{D_H(t)}{a(t)} = \frac{c}{a(t)H(t)}.$$

THEORY

Calculation on **Hubble horizon**:

$$D_H(t) = \frac{c}{H(t)} \qquad (c = 1 \Rightarrow D_H = 1/H)$$

and in comoving coordinate:

$$\chi_H(t) = \frac{c}{a(t)H(t)}.$$

Remark

Hubble horizon is a **kinematic scale**, not always a strict causal boundary. In many cosmological models, photons can cross the Hubble horizon when $H(t)$ changes with time.

2.3.2 Event Horizon

Physics Intuition

The cosmological event horizon answers a future question: **which regions can ever send a signal to us, even if we wait forever?**

If a source is currently too far away, light emitted *now* may never reach us because expansion can stretch space too fast at late time.

Foundation

For radial light in FLRW, $ds^2 = 0$ gives:

$$d\chi = \frac{dt}{a(t)}.$$

At cosmic time t, the maximum comoving distance that can still communicate with us in the infinite future is

$$\chi_{eh}(t) = \int_t^\infty \frac{dt'}{a(t')}.$$

Therefore the proper event-horizon distance at time t is

$$D_{eh}(t) = a(t)\chi_{eh}(t) = a(t)\int_t^\infty \frac{dt'}{a(t')}.$$

Theory

Calculation on **event horizon**:

$$\chi_{eh}(t) = \int_t^\infty \frac{dt'}{a(t')}, \qquad D_{eh}(t) = a(t)\int_t^\infty \frac{dt'}{a(t')}.$$

Event horizon exists only if the integral converges.

Important Note

The event horizon here is not the same things as the event horizon of a black hole we usually talk about in general relativity, but they are similar concepts. The event horizon of a black hole is the boundary beyond which nothing can escape from the black hole, while the cosmological event horizon is the boundary beyond which events cannot affect an observer at a given time.

Extra

In a universe with late-time accelerated expansion (e.g. with dark energy domination), an event horizon is expected. In contrast, in a purely decelerating universe, this horizon may not exist.

2.3.3 Particle Horizon

Physics Intuition

The particle horizon answers a past question: **how far can we see by now?**

Because the universe has finite age, light has only had finite time to travel since the Big Bang. So there is a largest observable distance at time t.

Foundation

Again use radial null geodesic in FLRW:

$$d\chi = \frac{dt}{a(t)}.$$

The largest comoving distance from which light could have reached us by time t is

$$\chi_{ph}(t) = \int_0^t \frac{dt'}{a(t')}.$$

Thus the proper particle-horizon distance is

$$D_{ph}(t) = a(t)\chi_{ph}(t) = a(t)\int_0^t \frac{dt'}{a(t')}.$$

Theory

Calculation on **particle horizon**:

$$\chi_{ph}(t) = \int_0^t \frac{dt'}{a(t')}, \qquad D_{ph}(t) = a(t) \int_0^t \frac{dt'}{a(t')}.$$

At present time t_0:

$$\chi_{ph,0} = \int_0^{t_0} \frac{dt'}{a(t')}, \qquad D_{ph,0} = a_0 \chi_{ph,0} = \chi_{ph,0} \quad (a_0 = 1).$$

Remark

Particle horizon is about **past light cone** (what has been observable), while event horizon is about **future light cone** (what can ever become observable).

Question Example 2.1: Numerical Comparison of Three Horizons in Our Universe

Use a flat ΛCDM background with present parameters close to Planck values:

$$H_0 \approx 67.66\,\mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \Omega_m^0 \approx 0.315, \qquad \Omega_\Lambda^0 \approx 0.685.$$

(a). Estimate present-day values of the three horizons.
(b). Compare their order and explain the physical meaning.
(c). In the limit $t \to \infty$, discuss what happens to the three horizons and whether the order swaps.

Solution.
(a). **Present-day values** ($t = t_0$):
Hubble horizon

$$D_H(t_0) = \frac{c}{H_0} \approx \frac{299792\,\mathrm{km/s}}{67.66\,\mathrm{km\,s^{-1}\,Mpc^{-1}}} \approx 4.43\,\mathrm{Gpc} \approx 14.4\,\mathrm{Gly}.$$

Particle horizon

$$D_{ph}(t_0) = a_0 \int_0^{t_0} \frac{dt'}{a(t')} \quad (a_0 = 1).$$

Numerically (for the above ΛCDM parameters):

$$D_{ph}(t_0) \approx 14.2\,\mathrm{Gpc} \approx 46.3\,\mathrm{Gly}.$$

Event horizon

$$D_{eh}(t_0) = a_0 \int_{t_0}^{\infty} \frac{dt'}{a(t')} \quad (a_0 = 1).$$

Numerically:

$$D_{eh}(t_0) \approx 5.0\,\mathrm{Gpc} \approx 16.3\,\mathrm{Gly}.$$

Therefore, at present we have approximately

$$D_{ph}(t_0) > D_{eh}(t_0) > D_H(t_0).$$

(b). **Physical intuition for this order**:
D_{ph} is largest because it accumulates all past light-travel from the Big Bang until now.
D_{eh} is smaller because it is limited by future accelerated expansion, so only a finite region can ever communicate with us.
D_H is a local expansion scale c/H_0, so it is typically smaller than the full causal-history scale D_{ph}.

(c). **Future limit** $t \to \infty$ in our Λ-dominated universe:
At late time, $a(t) \sim e^{H_\Lambda t}$ with

$$H_\Lambda = H_0\sqrt{\Omega_\Lambda^0}.$$

So

$$D_H(t) \to \frac{c}{H_\Lambda}, \qquad D_{eh}(t) \to \frac{c}{H_\Lambda},$$

i.e. Hubble horizon and event horizon approach the same constant (about 17 Gly for the above parameters).
Meanwhile

$$D_{ph}(t) = a(t)\int_0^t \frac{dt'}{a(t')} \to \infty,$$

because $a(t)$ keeps growing exponentially while the comoving particle-horizon integral approaches a finite constant.

Hence the asymptotic order is

$$D_{ph}(\infty) \gg D_{eh}(\infty) = D_H(\infty).$$

So in our universe there is **no full order swap**; instead, the present inequality $D_{eh} > D_H$ gradually shrinks to equality.

CHAPTER 3

Cosmic Evolution (Lecture 5-7)

3.1 Dynamical Equations

3.1.1 1st Friedmann Equation

FOUNDATION

Recall the FLRW metric:

$$g_{\mu\nu} = \begin{pmatrix} -1 & 0 & 0 & 0 \\ 0 & \frac{a^2}{1-kr^2} & 0 & 0 \\ 0 & 0 & a^2r^2 & 0 \\ 0 & 0 & 0 & a^2r^2\sin^2\theta \end{pmatrix}$$

where $a(t)$ is the scale factor, and k is the curvature constant.

By brunches of tedious calculations from Christoffel symbols, Riemann curvature tensor(If you really want to do the calculation, you can used the equations in the appendix B.2, trust me, it was not a good time), we can find out the Ricci tensor and Ricci scalar as below:

$$\begin{cases} R_{00} = -3\frac{\ddot{a}}{a} \\ R = \frac{6}{a^2}\left(a\ddot{a} + \dot{a}^2 + k\right) \end{cases}$$

Now plugging above results into the Einstein's field equation, we can get the 1st Friedmann equation as below:

$$
\begin{aligned}
R_{00} - \frac{1}{2} g_{00} R &= 8\pi G T_{00} \\
-3\frac{\ddot{a}}{a} + \frac{1}{2}\frac{6}{a^2}\left(a\ddot{a} + \dot{a}^2 + k\right) &= 8\pi G\rho \\
-3\frac{\ddot{a}}{a}\frac{a}{a} + 3\frac{1}{a^2}\left(a\ddot{a} + \dot{a}^2 + k\right) &= 8\pi G\rho \\
-3\frac{\ddot{a}a}{a^2} + 3\frac{1}{a^2}\left(a\ddot{a}\right) + 3\frac{1}{a^2}\left(\dot{a}^2 + k\right) &= 8\pi G\rho \\
3\frac{\dot{a}^2 + k}{a^2} &= 8\pi G\rho \\
3H^2 + 3\frac{k}{a^2} &= 8\pi G\rho \\
H^2 &= \frac{8\pi G}{3}\rho - \frac{k}{a^2}
\end{aligned}
$$

THEORY

The 1st Friedmann equation is written as below:

$$
H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{k}{a^2}
$$

where H is the Hubble parameter, ρ is the energy density, k is the curvature constant

This equation tells us how the expansion rate H changes with the energy(mass) density ρ, curvature k

3.1.2 2nd Friedmann Equation

Foundation

Similarly, we can get the 2nd Friedmann equation by using the results of Ricci tensor and Ricci scalar, and finding the trace of the Einstein's field equation:

$$\begin{aligned}
R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R &= 8\pi G T_{\mu\nu} \\
R_{\mu\nu}g^{\mu\nu} - \frac{1}{2}g_{\mu\nu}g^{\mu\nu}R &= 8\pi G T_{\mu\nu}g^{\mu\nu} \\
R - \frac{1}{2}R\delta^{\mu}_{\mu} &= 8\pi G T \\
-R &= 8\pi G T \\
-(\frac{6}{a^2}\left(a\ddot{a} + \dot{a}^2 + k\right)) &= 8\pi G(-\rho + 3p) \\
\frac{a\ddot{a}}{a^2} + \frac{\dot{a}^2 + k}{a^2} &= -\frac{8\pi G}{6}(-\rho + 3p) \\
\frac{\ddot{a}}{a} + \frac{8\pi G}{3}\rho &= \frac{4\pi G}{3}(\rho) - \frac{4\pi G}{3}(3p) \\
\frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3p)
\end{aligned}$$

Theory

The 2nd Friedmann equation is written as below:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)$$

where p is the pressure.

This equation tells us how the expansion acceleration $\ddot{a}$ changes with the energy(mass) density ρ and pressure p.

Physics Intuition

From the 2nd Friedmann equation, we can see that the expansion of universe is decelerating by matter and pressure, as the term $\rho + 3p$ has -ve contribution to the expansion acceleration $\ddot{a}$.

Remark

The Friedmann equations assumed that the universe is homogeneous and isotropic, which means the energy density ρ and pressure p are the same everywhere in the large scale of universe.

3.1.3 Continuity Equation

Physics Intuition

Now we have two equations with three unknowns: a, ρ, and p. Thus, we need the "3rd Friedmann equation" to solve for these variables, and we usually call it the continuity equation.

Foundation

We can first take the time derivative of the 1st Friedmann equation:

$$\begin{aligned}
\frac{\dot{a}^2}{a^2} &= \frac{8\pi G}{3}\rho - \frac{k}{a^2} \\
\dot{a}^2 &= \frac{8\pi G}{3}\rho a^2 - k \\
2\dot{a}\ddot{a} &= \frac{8\pi G}{3}(\dot{\rho}a^2 + \rho 2a\dot{a}) - 0 \\
\dot{a}\ddot{a}\frac{1}{a\dot{a}} &= \frac{4\pi G}{3}(\dot{\rho}a^2\frac{1}{a\dot{a}} + \rho 2a\dot{a}\frac{1}{a\dot{a}}) \\
\frac{\ddot{a}}{a} &= \frac{4\pi G}{3}(\dot{\rho}\frac{a}{\dot{a}} + 2\rho) \\
-\frac{4\pi G}{3}(\rho + 3p) &= \frac{4\pi G}{3}(\dot{\rho}\frac{a}{\dot{a}} + 2\rho) \\
\dot{\rho}\frac{a}{\dot{a}} + 2\rho &= -\rho - 3p \\
\dot{\rho}\frac{a}{\dot{a}} &= -3\rho - 3p \\
\dot{\rho} &= -3\frac{\dot{a}}{a}(\rho + p)
\end{aligned}$$

Theory

The continuity equation is written as below:

$$\dot{\rho} + 3H(\rho + p) = 0$$

It tell us how the energy(mass) density ρ changes with the size of the universe a.

Question Example 3.1: Another proof of the continuity equation

Assuming that the universe is Adiabatic Expansion (i.e. no heat exchange), can you derive the continuity equation by using the first law of thermodynamics?

$$dE + pdV = dQ = 0$$

Solution.

$$\begin{aligned} dE + pdV &= 0 \\ d(\rho V) + pdV &= 0 \\ d(\rho a^3) + pd(a^3) &= 0 \\ a^3 d\rho + 3a^2 \rho da + 3a^2 pda &= 0 \\ a^3 d\rho + 3a^2(\rho + p)da &= 0 \\ d\rho + 3\frac{da}{a}(\rho + p) &= 0 \\ \dot{\rho} + 3H(\rho + p) &= 0 \end{aligned}$$

Extra

There is actually a third way to derive the continuity equation, by using the Local conservation of energy-momentum tensor:

$$\nabla_\mu T^{\mu\nu} = 0$$

In fact, the derivation from the local conservation of energy-momentum tensor suggest that the continuity equation is just a consequence of local energy-momentum conservation, which gives the physical meaning of the continuity equation.

The proof is left as an exercise for the reader.
(Hints: watch the Relativity 110f by Eigenchris, [1])

3.2 Cosmological constant

3.2.1 Origin of Cosmological Constant

Physics Intuition

Recall the 2nd Friedmann equation:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p)$$

Assume that the universe contains normal matter which $p \geq 0$ and $\rho > 0$, then we have $\ddot{a} < 0$.

When Einstein get this result in 1917, he found the result is unacceptable for his belief of a static universe, which requires $\ddot{a} = \dot{a} = 0$. Thus, he introduced the cosmological constant Λ, in order to balance the gravity of matter, but he cant give out a physical meaning of Λ at that time.

Foundation

After introducing the cosmological constant Λ, the Einstein's field equation becomes:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

If we treat the extra term $\Lambda g_{\mu\nu}$ as an energy-momentum tensor, we can rewrite the equation as:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G(T_{\mu\nu} + T^{(\Lambda)}_{\mu\nu})$$

where $T^{(\Lambda)}_{\mu\nu} = -\frac{\Lambda}{8\pi G}g_{\mu\nu}$ is the energy-momentum tensor of the cosmological constant.

Physics Intuition

In a physic intuition, the extra term come from cosmological constant can be treated as a perfect fluid with energy density $\rho_\Lambda = \frac{\Lambda}{8\pi G}$ and pressure $p_\Lambda = -\rho_\Lambda$. Thus, the equation of LHS and RHS of the Einstein's field equation are still balanced, we just add an extra contribution of energy apart from the normal matter.

THEORY

The cosmological constant in energy-momentum tensor form can be written as:

$$T^{(\Lambda)}_{\mu\nu} = \begin{pmatrix} \rho_\Lambda & 0 & 0 & 0 \\ 0 & -\rho_\Lambda & 0 & 0 \\ 0 & 0 & -\rho_\Lambda & 0 \\ 0 & 0 & 0 & -\rho_\Lambda \end{pmatrix}$$

whether:

$$\begin{cases} \rho_\Lambda = \frac{\Lambda}{8\pi G} = \text{constant} \\ p_\Lambda = -\rho_\Lambda \end{cases}$$

And we can see that the energy is constant throughout spacetime with negative pressure, and we said that the cosmological constant is conceptually equivalent to vacuum energy.

EXTRA

When Hubble shows that universe is expanding in 1929, the cosmological constant is no longer needed to balance the gravity of matter, thus it is largely abandoned, and Einstein even call it the "biggest blunder" in his life.

Dramatically, the cosmological constant is revived in 1998, when the observation of Type Ia supernovae shows that the universe is accelerating expanding, Astronomers need to introduce the extra term of energy back in EFE to explain the accelerated expansion, and the cosmological constant are revived with a different physical meaning, which is the dark energy.

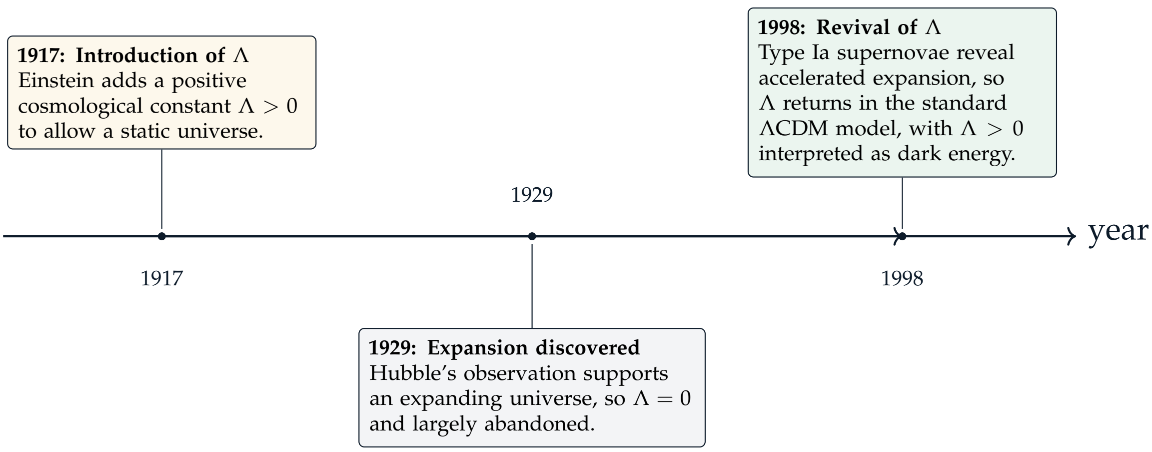

Figure 3.1: Three major changes in the historical role of the cosmological constant.

IMPORTANT NOTE

Although we said they are equivalent, but the cosmological constant and vaccum energy are not exactly the same thing.

They act like the same way that fills the spacetime with constant energy density and negative pressure, but the cosmological constant based on the observation of expansion of universe, we measure the value of Λ as around $10^{-52}\,\mathrm{m}^{-2}$ by using CMB and supernovae.

However, the vacuum energy is based on the quantum field theory, which predicts a value of vacuum energy density as around 10^{120} times larger than the observed value of Λ. This huge discrepancy is called the **cosmological constant problem**, which is one of the biggest unsolved problems in physics.

IMPORTANT NOTE

The existence of cosmological constant yields a curvature 4D-spacetime called de Sitter space. However, it does not implies that the 3d curvature k is non-zero, and vice versa, $k = 0$ does not implies $\Lambda = 0$. Thus, the cosmological constant and curvature are two independent parameters in the Friedmann equations.

3.2.2 Friedmann Equations with Cosmological Constant

THEORY

By considering the cosmological constant in EFE, and using the similar method shown in the previous sections, we can get the modified Friedmann equations as below,

the 1st Friedmann equations becomes:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3}$$

and the 2nd Friedmann equations becomes:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}$$

and the continuity equation is unchanged:

$$\dot{\rho} + 3H(\rho + p) = 0$$

3.2.3 Newtonian Gravity with Cosmological Constant

FOUNDATION

Let $p = 0$ for non-relativistic matter. Recall 00-component of Trace-formed EFE:

$$
\begin{aligned}
R_{00} &= 8\pi G(T_{00} - \frac{1}{2}g_{00}T) + \Lambda g_{00} \\
R_{00} &= 8\pi G(\rho - \frac{1}{2}(-1)(-\rho + 3p)) + \Lambda(-1) \\
R_{00} &= 4\pi G(\rho + 3p) - \Lambda \\
R_{00} &= 4\pi G\rho - \Lambda
\end{aligned}
$$

In the Weak-Field limit (Newtonian limit), we have $R_{00} \approx -\frac{1}{2}\nabla^2 h_{00}$, and $h_{00} = -2\Phi$, where Φ is the Newtonian potential, thus we can rewrite above equation as:

$$\nabla^2\Phi = 4\pi G\rho - \Lambda$$

By using superposition principle, we can solve the above equation in mass term and cosmological constant term separately. Mass term:

$$
\begin{aligned}
\nabla^2\Phi_M &= 4\pi G\rho \\
\Phi_M(r) &= -\frac{GM}{r}
\end{aligned}
$$

Then assuming spherical coordinate, the cosmological constant term can be solved as:

$$
\begin{aligned}
\nabla^2\Phi_\Lambda &= -\Lambda \\
\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d\Phi_\Lambda}{dr}\right) &= -\Lambda \\
r^2\frac{d\Phi_\Lambda}{dr} &= -\frac{1}{3}\Lambda r^3 + C_1 \\
\frac{d\Phi_\Lambda}{dr} &= -\frac{1}{3}\Lambda r + \frac{C_1}{r^2} \\
\Phi_\Lambda(r) &= -\frac{1}{6}\Lambda r^2 - \frac{C_1}{r} + C_2
\end{aligned}
$$

By letting $C_1 = C_2 = 0$, we can get the solution of cosmological constant term as:

$$\Phi_\Lambda(r) = -\frac{1}{6}\Lambda r^2$$

Theory

The introduction of cosmological constant also yields a modification to the Newtonian gravity, which can be written as:

$$\Phi(r) = -\frac{GM}{r} - \frac{1}{6}\Lambda r^2$$

where the first term is the usual Newtonian potential, and the second term is the contribution of cosmological constant, which is a repulsive potential that grows with distance.

We can also hence get the modified Newtonian force as below:

$$F(r) = -m\nabla^2\Phi(r) = -\frac{GMm}{r^2} + \frac{1}{3}\Lambda rm$$

where the first term is the usual Newtonian force, and the second term is the contribution of cosmological constant, which is a repulsive force that grows with distance.

Question Example 3.2: Effect of Cosmological Constant

Consider a Sun-Earth like system,
(a). estimate the threshold of R above which the cosmological replusion force will dominate over the gravitational attraction.
(b). calculate the ratio of the cosmological force to the gravitational force at the distance of Pluto, which is around 40 AU.

Solution.
(a). The threshold of R can be found by setting the two forces equal:

$$\frac{GMm}{R^2} = \frac{1}{3}\Lambda Rm$$

Solving for R, we get:

$$R = \left(\frac{3GM}{\Lambda}\right)^{1/3}$$

Plugging in the values of G, M, and Λ:

$$\begin{aligned} R &= \left(\frac{3 \cdot 6.67430 \times 10^{-11} \cdot 1.989 \times 10^{30}}{2.036 \times 10^{-35}}\right)^{1/3} \\ &= 2.694 \times 10^{18}\text{m} \end{aligned}$$

which is around 90 parsecs, and is much larger than the size of solar system, thus cosmological constant has negligible effect on dynamics of solar system.

(b). The ratio of the cosmological force to the gravitational force at the distance of Pluto is:

$$\begin{aligned}\frac{F_\Lambda}{F_G} &= \frac{\frac{1}{3}\Lambda rm}{\frac{GMm}{r^2}} \\ &= \frac{1}{3}\frac{\Lambda r^3}{GM} \\ &= \frac{1}{3}\frac{2.036\times 10^{-35}\cdot(40\times 1.496\times 10^{11})^3}{6.67430\times 10^{-11}\cdot 1.989\times 10^{30}} \\ &= 1.1\times 10^{-17}\end{aligned}$$

which is an extremely small number, thus the cosmological constant has negligible effect within the solar system, even at the distance of Pluto.

3.3 Evolution of Scale Factor

3.3.1 Equation of State(EoS)

Physics Intuition

The equation of state (EoS) is used to describe the relationship between pressure and energy density of a fluid, in cosmology, we usually use a simple linear EoS as below:

$$p = w\rho$$

where w is the EoS parameter, which can be used to characterize different types of matter and energy in the universe.

Foundation

From the continuity equation:

$$\dot{\rho} = -3H(\rho + p)$$

sub the $p = w\rho$ into the above equation, we can get:

$$\begin{aligned}
\dot{\rho} &= -3H\rho(1+w) \\
\frac{d\rho}{dt} &= -3\frac{\dot{a}}{a}\rho(1+w) \\
\frac{d\rho}{\rho} &= -3(1+w)\frac{da}{a} \\
\int \frac{d\rho}{\rho} &= -3(1+w)\int \frac{da}{a} \\
\ln\rho &= -3(1+w)\ln a + C \\
\rho &= \rho_0 \cdot a^{-3(1+w)} \\
\rho &\propto a^{-3(1+w)}
\end{aligned}$$

THEORY

The energy density evolution with scale factor can be written as:

$$\rho(a) \propto a(t)^{-3(1+w)}$$

where $w = \frac{p}{\rho}$ is the EoS parameter.

For different types of matter and energy, we have different values of w as below:

$$\begin{cases}
w = 0, & \text{for non-relativistic matter (dust)} \\
w = \frac{1}{3}, & \text{for relativistic matter (radiation)} \\
w = -\frac{1}{3}, & \text{for curvature} \\
w = -1, & \text{for cosmological constant (dark energy)}
\end{cases}$$

Therefore, we can get their relationship between energy density and scale factor:

$$\begin{cases}
\rho_d & \propto a^{-3} \\
\rho_r & \propto a^{-4} \\
\rho_k & \propto a^{-2} \\
\rho_\Lambda & = \text{constant}
\end{cases}$$

REMARK

For these who are interested in the calculation of EoS parameter of dust and radiation, you can check chp 5.2.2 about the kinetic theory of non-relativistic and relativistic particles correspondingly in this book. We will just give the result here without the detailed calculation.

Question Example 3.3: Effective Energy Density and Pressure of Curvature

(a). By treating the curvature term in the Friedmann equations as energy density and pressure, can you derive the EoS parameter of curvature?
(b). Rewrite the Friedmann equations and continuity equation by effective energy density and pressure of curvature, which are the combination of the original energy density and pressure and the effective energy density and pressure of curvature. (i.e. $\rho_{\text{eff}} = \rho + \rho_K$, $p_{\text{eff}} = p + p_K$)

Solution.
(a). If we treat the curvature term in the Friedmann equations as some terms in energy density and pressure, we can rewrite the Friedmann equations as below:
The 1st Friedmann equation can be rewritten as:

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2} = \frac{8\pi G}{3}(\rho + \rho_K)$$

And hence we get:

$$\rho_K = -\frac{3}{8\pi G}\frac{k}{a(t)^2}$$

The continuity equation can be rewritten as:

$$\dot{\rho_K} + 3H(\rho_K + p_K) = 0$$

where p_K is the effective pressure of curvature.

By substituting the expression of ρ_K into the above equation, we can get:

$$\begin{aligned}
-\frac{3}{8\pi G}\left(-2\frac{k}{a^3}\dot{a}\right) &= -3H\left(-\frac{3}{8\pi G}\frac{k}{a^2} + p_K\right) \\
\frac{6}{8\pi G}\frac{k}{a^2}H &= \frac{9}{8\pi G}\frac{k}{a^2}H - 3Hp_K \\
3Hp_K &= \frac{3}{8\pi G}\frac{k}{a^2}H \\
p_K &= \frac{1}{8\pi G}\frac{k}{a^2}
\end{aligned}$$

Thus, we can get the EoS parameter of curvature as:

$$w_K = \frac{p_K}{\rho_K} = -\frac{1}{3}$$

(b). using part a result and the equation provided in question, we get:

$$\begin{cases}
H^2 & = \frac{8\pi G}{3}(\rho_{\text{eff}}) \\
\frac{\ddot{a}}{a} & = -\frac{4\pi G}{3}(\rho_{\text{eff}} + 3p_{\text{eff}}) \\
\dot{\rho}_{\text{eff}} & = -3H(\rho_{\text{eff}} + p_{\text{eff}})
\end{cases}$$

PHYSICS INTUITION

- **Dust ($w = 0$):**
  In astrophysics, we often call non-relativistic matter as "dust", for these particles, their energy density is dominated by their rest mass energy, and their pressure is negligible compared to their energy density, on other hand, $v \ll c$, so $p \approx 0$, thus we have $w = 0$.

  In a physical intuition, the energy density of dust will dilute as the universe expands, as the number of particles in a comoving volume is constant, thus $\rho \propto a^{-3}$.

- **Radiation ($w = \frac{1}{3}$):**
  For relativistic matter, their energy density is dominated by their kinetic energy, and their pressure is not negligible compared to their energy density, on other hand, $v \approx c$, so $p \approx \frac{1}{3}\rho$, thus we have $w = \frac{1}{3}$.

  In a physical intuition, the cosmological redshift will cause the wavelength of photons to stretch as the universe expands, thus the energy of radiaition have an extra factor of a^{-1} compared to dust, thus $\rho \propto a^{-4}$.

- **Curvature ($w = -\frac{1}{3}$):**
  The curvature term in the Friedmann equations can be treated as an effective energy density and pressure, and we can get the EoS parameter of curvature as $w = -\frac{1}{3}$.

  In a physical intuition, the curvature term can be expressed as $\frac{1}{R^2}$ in differential geometry, where R is the radius of curvature, and the radius of curvature will grow with the expansion of universe, thus the energy density of curvature will dilute as $\rho \propto a^{-2}$.

- **Cosmological Constant ($w = -1$):**
  The cosmological constant behave like property of empty space itself, when the universe expands, the scale factor a will grow, but the total amount of vacuum energy also grows, thus the energy density of it is scale factor independent, thus we have $\rho =$ constant.

3.3.2 EoS of Cosmologic Constant

THEORY

The equation of state of cosmological constant is:

$$w_\Lambda = \frac{p_\Lambda}{\rho_\Lambda} = -1 < -\frac{1}{3}$$

Assume flat universe with $k = 0$, and the universe is dominated by cosmological constant, then the 1st Friedmann equation becomes:

$$\begin{aligned} H^2 &= \frac{8\pi G}{3}\rho_\Lambda \\ H^2 &= \frac{\Lambda}{3} \\ \frac{\dot{a}}{a} &= \sqrt{\frac{\Lambda}{3}} \\ \dot{a}(t) &= a(t)\sqrt{\frac{\Lambda}{3}} \end{aligned}$$

By solving this ODE, we can get the solution of scale factor as below:

$$\begin{aligned} a(t) &= a_0 e^{H(t-t_0)} = a_0 e^{\sqrt{\frac{\Lambda}{3}}(t-t_0)} \\ \dot{a}(t) &= H \cdot a_0 e^{H(t-t_0)} = \sqrt{\frac{\Lambda}{3}} \cdot a_0 e^{\sqrt{\frac{\Lambda}{3}}(t-t_0)} \end{aligned}$$

PHYSICS INTUITION

From the above solution of the cosmological constant dominated universe, we can see that the scale factor grows exponentially with time, which means the universe will become larger and larger with time (i.e. $a(t) \to \infty$ as $t \to \infty$), and the expansion of universe will become faster and faster with time (i.e. $\dot{a}(t) \to \infty$ as $t \to \infty$), thus we say that the universe is accelerating expanding in this case.

3.3.3 Scale Factor Equation

FOUNDATION

For the case of $w \neq -1$, we can rewrite the 1st Friedmann equation as:

$$H^2 \equiv \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho$$

while we have the matter-energy density as:

$$\rho(a) = \rho_0 a^{-3(1+w)}$$

thus we can get the equation of scale factor as:

$$\begin{aligned}
\left(\frac{\dot{a}}{a}\right)^2 &= \frac{8\pi G}{3}\rho_0 a^{-3(1+w)} \\
\frac{\dot{a}}{a} &= \sqrt{\frac{8\pi G}{3}\rho_0 a^{-\frac{3(1+w)}{2}}} \\
\dot{a} &= \sqrt{\frac{8\pi G}{3}\rho_0 a^{-\frac{3(1+w)}{2}+1}} \\
&= \sqrt{\frac{8\pi G}{3}\rho_0 a^{-\frac{1+3w}{2}}}
\end{aligned}$$

Let $\sqrt{\frac{8\pi G}{3}\rho_0} = C$, we can rewrite the above equation as:

$$\begin{aligned}
\dot{a} &= Ca^{-\frac{1+3w}{2}} \\
\int a^{\frac{1+3w}{2}} da &= \int C dt \\
\frac{2}{3(1+w)} a^{\frac{3(1+w)}{2}} &= Ct + D \\
a(t) &= \left(\frac{3(1+w)}{2}(Ct+D)\right)^{\frac{2}{3(1+w)}}
\end{aligned}$$

If we consider the boundary condition that at time $t = t_0$, the scale factor is $a(t_0) = a_0$, we can get the constant D as:

$$\begin{aligned}
0 &= \left(\frac{3(1+w)}{2}(0+D)\right)^{\frac{2}{3(1+w)}} \\
D &= 0
\end{aligned}$$

Now we set the reference scale factor a_0 at time t_0 as:

$$a_0 = \left(\frac{3(1+w)}{2} (Ct_0) \right)^{\frac{2}{3(1+w)}}$$

And we will conclude the solution of scale factor as:

$$\begin{aligned} a(t) &= \left(\frac{3(1+w)}{2} (Ct) \right)^{\frac{2}{3(1+w)}} \\ &= \left(\frac{3(1+w)}{2} (Ct_0) \right)^{\frac{2}{3(1+w)}} \left(\frac{t}{t_0} \right)^{\frac{2}{3(1+w)}} \\ &= a_0 \left(\frac{t}{t_0} \right)^{\frac{2}{3(1+w)}} \end{aligned}$$

THEORY

In fact, in general case that $w \neq -1$, we can get the scale factor equation as below:

$$a(t) = a_0 \left(\frac{t}{t_0} \right)^{\frac{2}{3(1+w)}}$$

where t_0 is the reference time, and a_0 is the scale factor at time t_0

REMARK

You may notice that at $t = 0$, the scale factor $a(t) = 0$, and hence $\rho(t) = \infty$, which means the energy density diverges at the beginning of universe, and the spacetime singularity is called the Big Bang singularity.

In this singularity, the spacetime did not continue before $t = 0$, and the general relativity breaks down due to infinite energy density and infinite small scale factor, thus we need a theory of quantum gravity to describe the physics at very near the beginning of universe, which is still an open problem in physics.

THEORY

In summary, we can get the energy density sources yield table:

Energy Density Source	w	$\rho(a)$	$a(t)$
Dust	0	$\propto a^{-3}$	$\propto t^{2/3}$
Radiation	$\frac{1}{3}$	$\propto a^{-4}$	$\propto t^{1/2}$
Curvature	$-\frac{1}{3}$	$\propto a^{-2}$	$\propto t$
Cosmological Constant	-1	Constant	$\propto e^{Ht}$

3.4 SCDM and ΛCDM Models

3.4.1 Critical Energy Density and Density Parameters

Physics Intuition

The Friedmann equations tell us that for the evolution of universe depends on energy density of different components, but they contribute different value at the same time, thus we introduce the concept of critical energy density and density parameters to characterize the ratio of contribution of different components to the total energy density of universe.

Theory

By defining the critical energy density as:

$$\rho_c = \frac{3H^2}{8\pi G}$$

and the density parameter of each component as:

$$\Omega_i = \frac{\rho_i}{\rho_c}$$

where i can be matter, radiation, cosmological constant, curvature, etc.

We can rewrite the 1st Friedmann equation as:

$$\begin{aligned} H^2 &= \frac{8\pi G}{3}\rho - \frac{k}{a^2} + \frac{\Lambda}{3} \\ \frac{3H^2}{8\pi G} &= \rho - \frac{3k}{8\pi G a^2} + \frac{\Lambda}{8\pi G} \\ \rho_c &= \rho + \rho_K + \rho_\Lambda \\ 1 &= \Omega_{mr} + \Omega_\Lambda + \Omega_K \\ &= \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_K \end{aligned}$$

with:

$$\begin{aligned} \Omega_{mr} &= \frac{\rho}{\rho_c} = \Omega_m + \Omega_r \\ \Omega_\Lambda &= \frac{\rho_\Lambda}{\rho_c} = \frac{\Lambda}{3H^2} \\ \Omega_K &= \frac{\rho_K}{\rho_c} = -\frac{k}{a^2 \cdot H^2} = -\frac{k}{\dot{a}^2} \end{aligned}$$

where Ω_{mr} is the contribution of matter and radiation, Ω_Λ is the contribution of cosmological constant, and Ω_K is the contribution of curvature.

And we can get the total energy density of universe as:

$$\Omega = \sum_i \Omega_i = \Omega_m + \Omega_r + \Omega_\Lambda + \Omega_K$$

with $\Omega = 1$ obviously.

Remark

In some textbooks, they define the Ω as the sum of all density parameters except curvature, i.e. $\Omega(t) = \Omega_m(t) + \Omega_r(t) + \Omega_\Lambda(t)$, and then the curvature density parameter can be written as $\Omega_K(t) = 1 - \Omega(t)$.

Further, we can also get the relationship between curvature density parameter and total energy density as:

$$\Omega_K(t) = 1 - \Omega(t)$$

Thus:

- If $\Omega(t) < 1$, then $\Omega_K(t) > 0$, and $k < 0$, which means the universe is open and negatively curved.
- If $\Omega(t) = 1$, then $\Omega_K(t) = 0$, and $k = 0$, which means the universe is flat.
- If $\Omega(t) > 1$, then $\Omega_K(t) < 0$, and $k > 0$, which means the universe is closed and positively curved.

3.4.2 Total Energy Density of Universe

Theory

The total energy density of the universe can be expressed as:

$$1 = \Omega = \Omega_m(t) + \Omega_r(t) + \Omega_\Lambda(t) + \Omega_K(t)$$

where Ω_m is the contribution of matter, which is the sum of dark matter (d) and ordinary/baryonic matter (b): $\Omega_m = \Omega_d + \Omega_b$, Ω_r is the contribution of radiation, Ω_Λ is the contribution of cosmological constant, and Ω_K is the contribution of curvature.

THEORY

In a Einstein-de Sitter universe, which is a flat universe with only matter and no cosmological constant, i.e.:

$$\Omega_m = 1, \quad \Omega_r = 0, \quad \Omega_\Lambda = 0, \quad \Omega_K = 0$$

, the total energy density is dominated by matter:

$$\Omega = \Omega_m(t) = 1$$

The model are also known as the **Standard Cold Dark Matter (SCDM)** model, which is the simplest cosmological model, and it predicts a decelerating universe with $\ddot{a} < 0$.

At the late 1990s, the SCDM model is challenged by the acceleration of universe, thus the Λ**CDM** model is proposed, which is a flat universe with matter and cosmological constant $\Omega_m + \Omega_\Lambda = 1$, with $\Omega_\Lambda > 0$.

THEORY

In a ΛCDM model, the friedmann equations can be written as:

$$H^2 = \frac{8\pi G}{3}\left(\frac{\rho_m^0}{f_m(a)} + \frac{\rho_r^0}{f_r(a)} + \frac{\rho_k^0}{a^2} + \rho_\Lambda^0\right)$$

where $f_m(a)$ and $f_r(a)$ are the functions of scale factor that describe the evolution of matter and radiation energy density with scale factor. Note we can not treat $f_m(a) = a^3$ and $f_r(a) = a^4$ without the assumption of independent evolution of matter and radiation, which is not the case in the early universe when matter and radiation are coupled together.

IMPORTANT NOTE

Actually, we can still have $\Lambda = 0$ and exist some other kind of special matter with EoS parameter $w < -\frac{1}{3}$ to drive the acceleration of universe, so that we don not need to introduce the cosmological constant to explain the acceleration of universe.

Therefore, the ΛCDM model is not the unique model to explain dark energy, but just the most pervasive one among all dark energy models.

3.4.3 Related Equations in Energy Density Parameters

Foundation

We now derive the Hubble parameter as a function of redshift by using the 1st Friedmann equation and the density parameters defined in previous section. Starting from:

$$H^2 = \frac{8\pi G}{3}(\rho_m + \rho_r + \rho_K + \rho_\Lambda)$$

and using:

$$\rho_m = \rho_m^0 a^{-3}, \qquad \rho_r = \rho_r^0 a^{-4}, \qquad \rho_K = \rho_K^0 a^{-2}, \qquad \rho_\Lambda = \rho_\Lambda^0,$$

with

$$\rho_c^0 = \frac{3H_0^2}{8\pi G}, \quad \Omega_m^0 = \frac{\rho_m^0}{\rho_c^0}, \quad \Omega_r^0 = \frac{\rho_r^0}{\rho_c^0}, \quad \Omega_K^0 = \frac{\rho_K^0}{\rho_c^0}, \quad \Omega_\Lambda^0 = \frac{\rho_\Lambda^0}{\rho_c^0},$$

we get:

$$\frac{H^2}{\rho_c^0} = \frac{8\pi G}{3}\left(\frac{\rho_m^0}{a^3} + \frac{\rho_r^0}{a^4} + \frac{\rho_K^0}{a^2} + \rho_\Lambda^0\right) \cdot \frac{1}{\rho_c^0}$$

$$\frac{H^2}{H_0^2} = \Omega_m^0 a^{-3} + \Omega_r^0 a^{-4} + \Omega_K^0 a^{-2} + \Omega_\Lambda^0$$

Using $a_0 = 1$, we have $a = \frac{1}{1+z}$. Therefore:

$$H(z) = H_0 E(z),$$

where:

$$E(z) = \sqrt{\Omega_m^0(1+z)^3 + \Omega_r^0(1+z)^4 + \Omega_K^0(1+z)^2 + \Omega_\Lambda^0}$$

Foundation

Next, we derive the emitting time in both scale-factor form $t(a)$ and redshift form $t(z)$. From the definition of Hubble parameter:

$$H \equiv \frac{\dot{a}}{a},$$

we get:

$$\dot{a} = aH(a)$$

$$dt = \frac{da}{aH(a)}$$

Therefore, for a source emitted at a_{emit} and observed today at $a_0 = 1$:

$$t_0 - t(a_{emit}) = \int_{a_{emit}}^{1} \frac{da'}{a'H(a')}$$

Equivalently, by choosing Big Bang as $t = 0$ at $a = 0$, we have:

$$t(a) = \int_0^a \frac{da'}{a' H(a')}$$

Now use the redshift relation:

$$z = \frac{1}{a} - 1$$

Differentiate with respect to time:

$$\frac{dz}{dt} = -\frac{1}{a^2}\dot{a} = -\left(\frac{1}{a}\right)\left(\frac{\dot{a}}{a}\right) = -(1+z)H(z)$$

Therefore:

$$dt = -\frac{dz}{(1+z)H(z)}.$$

For a source emitted at redshift z and observed today ($z = 0$):

$$t_0 - t(z) = \int_{t(z)}^{t_0} dt = \int_0^z \frac{dz'}{(1+z')H(z')}$$

So equivalently:

$$t(z) = t_0 - \int_0^z \frac{dz'}{(1+z')H(z')}$$

Foundation

Finally, we convert the comoving distance from the time form in chapter 2.1.3 to the redshift form. Recall:

$$\chi = \int_{t_{emit}}^{t_0} \frac{dt}{a(t)}$$

Using $a = \frac{1}{1+z}$ and $dt = -\frac{dz}{(1+z)H(z)}$, we have:

$$\frac{dt}{a(t)} = (1+z)dt = -\frac{dz}{H(z)}$$

Therefore:

$$\chi(z) = \int_{t(z)}^{t_0} \frac{dt}{a(t)} = \int_z^0 \left(-\frac{dz'}{H(z')}\right) = \int_0^z \frac{dz'}{H(z')}$$

Writing $H(z) = H_0 E(z)$, we get the common form:

$$\chi(z) = \frac{1}{H_0} \int_0^z \frac{dz'}{E(z')}$$

THEORY

In real-world cosmological research, the redshift is one of the easiest observable quantity, thus we can express the equations we have learned in previous sections in terms of redshift, which is more convenient for data analysis and comparison with observations.

The Hubble parameter as a function of redshift can be expressed as:

$$H(z) = H_0\sqrt{\Omega_m^0(1+z)^3 + \Omega_r^0(1+z)^4 + \Omega_K^0(1+z)^2 + \Omega_\Lambda^0}$$
$$H(a) = H_0\sqrt{\Omega_m^0 a^{-3} + \Omega_r^0 a^{-4} + \Omega_K^0 a^{-2} + \Omega_\Lambda^0}$$

The emitting time can be expressed as:

$$t(a_{emit}) = t_0 - \int_{a_{emit}}^{1} \frac{da'}{a'H(a')}$$
$$t(z) = t_0 - \int_0^z \frac{dz'}{(1+z')H(z')}$$

The comoving distance can be expressed as:

$$\chi(z) = \int_0^z \frac{dz'}{H(z')} = \frac{1}{H_0}\int_0^z \frac{dz'}{E(z')}$$

And hence the luminosity distance can be expressed as:

$$d_L(z) = (1+z)\cdot\chi(z) = (1+z)\cdot\frac{1}{H_0}\int_0^z \frac{dz'}{E(z')}$$

Question Example 3.4: Age of Universe from

Consider a flat ΛCDM-like universe with parameters:

$$\Omega_d^0 = 0.3, \qquad \Omega_r^0 = 0, \qquad \Omega_K^0 = 0, \qquad \Omega_\Lambda^0 = 0.7,$$

and let:

$$H_0 = 70\,\mathrm{km\,s^{-1}\,Mpc^{-1}}.$$

Use the $t(a)$ equation to estimate the present age t_0 of the universe.

Hints.
In this problem, using the redshift form directly is less convenient because the Big Bang limit corresponds to $z \to \infty$. Using a-integration gives cleaner boundary conditions: $a = 0 \to t = 0$ and $a = 1 \to t = t_0$.

Solution.
Here we treat the matter term as $\Omega_m^0 = 0.3$ (using the given Ω_d^0 value), so:

$$H(a) = H_0\sqrt{0.3a^{-3} + 0.7}.$$

Hence:

$$\begin{aligned} t_0 = t(a=1) &= \int_0^1 \frac{da}{aH(a)} \\ &= \frac{1}{H_0}\int_0^1 \frac{da}{a\sqrt{0.3a^{-3}+0.7}} \\ &= \frac{2}{3H_0\sqrt{0.7}}\sinh^{-1}\left(\sqrt{\frac{0.7}{0.3}}\right) \\ &\approx 0.964\,H_0^{-1}. \end{aligned}$$

Convert H_0 to SI unit:

$$H_0 = 70\frac{\mathrm{km}}{\mathrm{s\,Mpc}} = 70\frac{10^3\,\mathrm{m/s}}{3.086\times10^{22}\,\mathrm{m}} \approx 2.268\times10^{-18}\,\mathrm{s}^{-1},$$

so:

$$H_0^{-1} \approx 4.41\times10^{17}\,\mathrm{s} \approx 13.97\,\mathrm{Gyr}.$$

Therefore:

$$t_0 \approx 0.964\times13.97\,\mathrm{Gyr} \approx 13.5\,\mathrm{Gyr} \approx 1.35\times10^{10}\,\mathrm{yr}.$$

Question Example 3.5: LCDM Energy-Budget Milestones

Assume a near-flat universe with:

$$\Omega_d^0 = 0.315, \qquad \Omega_r^0 = 9.4 \times 10^{-5}, \qquad \Omega_\Lambda^0 = 0.685, \qquad \Omega_K^0 = -9.4 \times 10^{-5} \approx 0$$

(a). Find z_1 at radiation–dust equality.
(b). Find z_2 at dust–Λ equality.
(c). Write down the equations of $\Omega_i(z)$ in terms of $E(z)$ and Ω_i^0.

Solution.
First write:

$$E^2(z) = \Omega_r^0(1+z)^4 + \Omega_d^0(1+z)^3 + \Omega_K^0(1+z)^2 + \Omega_\Lambda^0$$

(a). Radiation–dust equality means:

$$\Omega_r^0(1+z_1)^4 = \Omega_d^0(1+z_1)^3 \Rightarrow 1+z_1 = \frac{\Omega_d^0}{\Omega_r^0}$$

So:

$$1+z_1 = \frac{0.315}{9.4 \times 10^{-5}} \approx 3351 \Rightarrow z_1 \approx 3.35 \times 10^3$$

(b). Dust–Λ equality means:

$$\Omega_d^0(1+z_2)^3 = \Omega_\Lambda^0 \Rightarrow 1+z_2 = \left(\frac{\Omega_\Lambda^0}{\Omega_d^0}\right)^{1/3}$$

Therefore:

$$1+z_2 = \left(\frac{0.685}{0.315}\right)^{1/3} \approx 1.295 \Rightarrow z_2 \approx 0.295$$

(c). In general, each density parameter at redshift z is:

$$\Omega_i(z) = \frac{\rho_i(z)}{\rho_c(z)} = \frac{\Omega_i^0 f_i(z)}{E^2(z)},$$

with

$$f_r(z) = (1+z)^4, \quad f_d(z) = (1+z)^3, \quad f_K(z) = (1+z)^2, \quad f_\Lambda(z) = 1$$

Hence:

$$\Omega_r(z) = \frac{\Omega_r^0(1+z)^4}{E^2(z)},$$

$$\Omega_d(z) = \frac{\Omega_d^0(1+z)^3}{E^2(z)},$$

$$\Omega_K(z) = \frac{\Omega_K^0(1+z)^2}{E^2(z)},$$

$$\Omega_\Lambda(z) = \frac{\Omega_\Lambda^0}{E^2(z)}$$

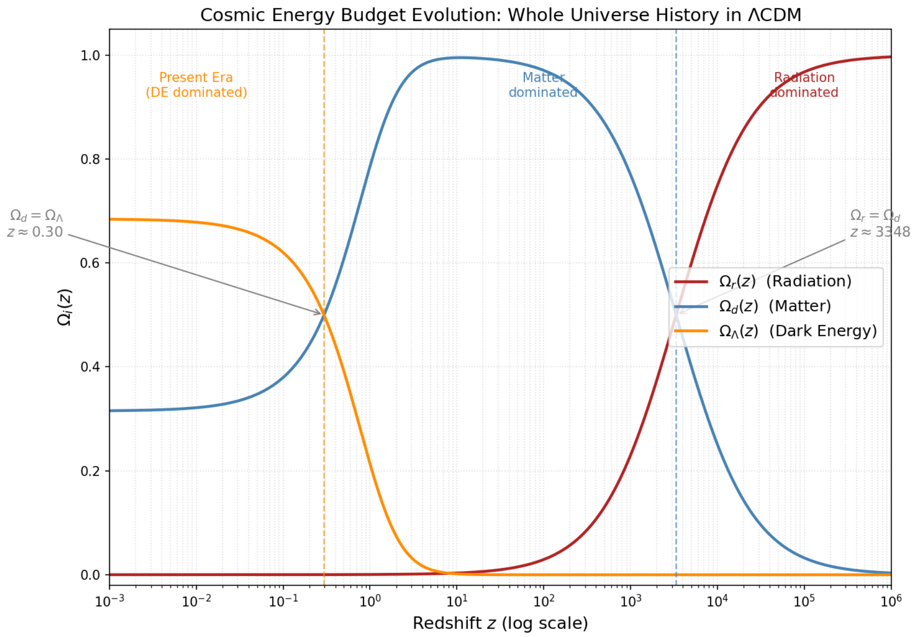

Figure 3.2: Visual interpretation of the LCDM energy-budget transitions: radiation–dust equality, dust–Λ equality, and redshift evolution of $\Omega_i(z)$.

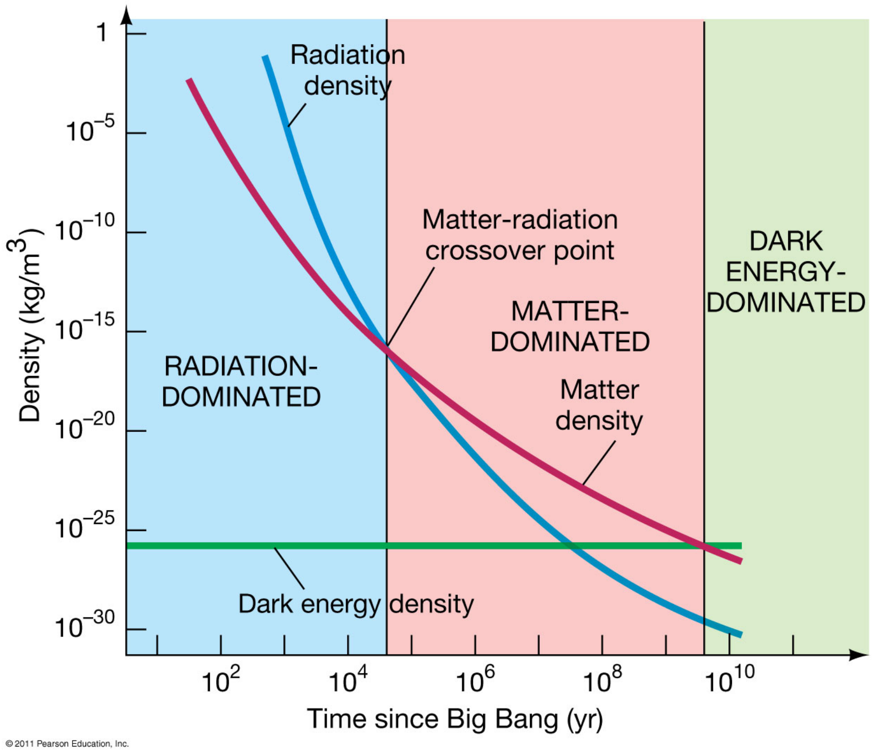

Figure 3.3: Same as previous figure but it showing the exact values of energy density $\rho_i(z)$ instead of density parameters $\Omega_i(z)$.

3.5 Dark Energy and the Accelerating Universe

3.5.1 Observational Evidence for Dark Energy

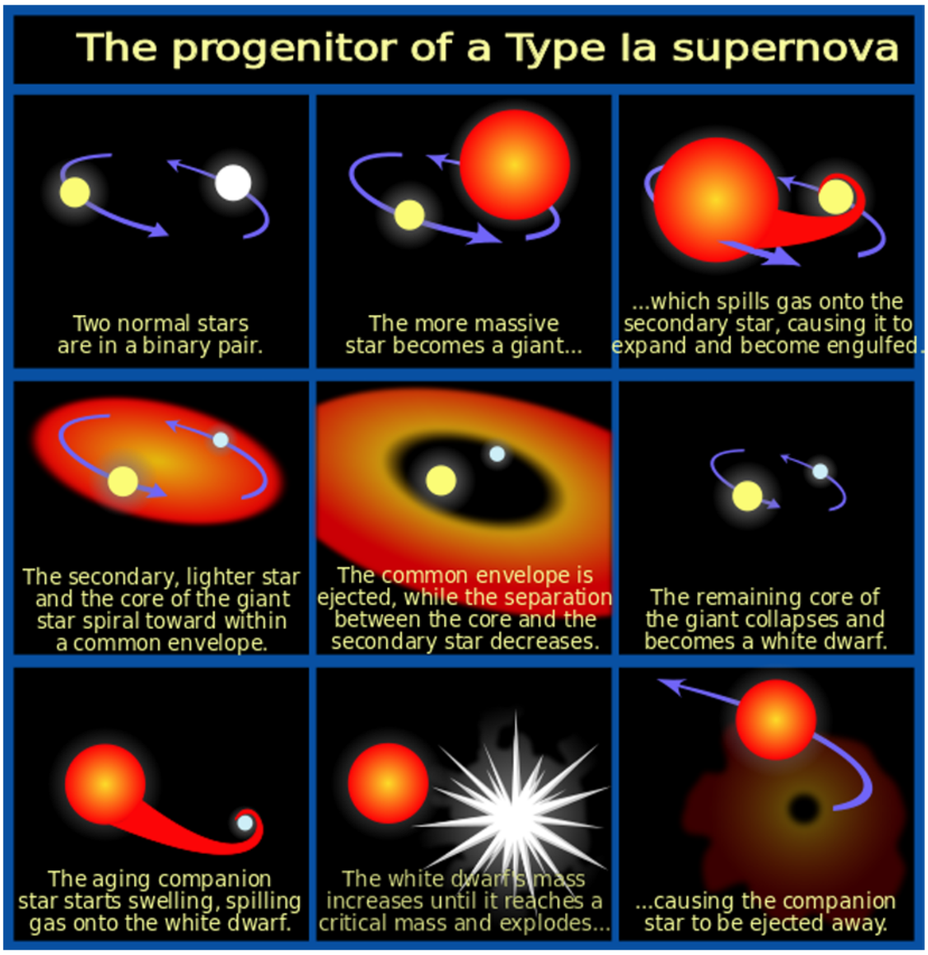

Figure 3.4: The progenitor of type Ia supernova.

PHYSICS INTUITION

A Type Ia supernova is a stellar explosion that happens when a white dwarf in a binary system gains too much mass (close to the Chandrasekhar limit) and undergoes runaway thermonuclear burning. Because this explosion is triggered at nearly the same critical mass(the Chandrasekhar limit), Type Ia supernovae have very similar intrinsic peak luminosities after calibration.

This makes them "standard candles": if we know how bright they truly are, and we measure how bright they look from Earth, then the inverse-square law of light gives their distance. By comparing this distance with the observed redshift, astronomers can map the expansion history of the universe.

Foundation

In astronomy, we express absolute magnitude M and apparent magnitude m of an object as:

$$M = -2.5 \log_{10}\left(\frac{L}{L_0}\right)$$
$$m = -2.5 \log_{10}\left(\frac{F}{F_0}\right)$$

where L is the intrinsic luminousity of the object, F is the observed flux:

$$F = \frac{L}{4\pi d_L^2}$$

Let L_0 and F_0 are reference luminousity and flux, which are $L_0 = 3.0128 \times 10^{28}\text{W}$ and $F_0 = \frac{L_0}{4\pi d_0^2} = 2.518 \times 10^{-8}\text{W}/\text{m}^2$ with $d_0 = 10\text{pc}$.

By using the above definition, we can get the distance modulus μ as:

$$\begin{aligned}
\mu &= m - M \\
&= -2.5 \log_{10}\left(\frac{F}{F_0}\right) + 2.5 \log_{10}\left(\frac{L}{L_0}\right) \\
&= -2.5 \log_{10}\left(\frac{L}{4\pi d_L^2}\frac{4\pi d_0^2}{L_0}\right) + 2.5 \log_{10}\left(\frac{L}{L_0}\right) \\
&= -2.5\left(\log_{10}\left(\frac{L d_0^2}{L_0 d_L^2}\right) - \log_{10}\left(\frac{L}{L_0}\right)\right) \\
&= -2.5 \log_{10}\left(\frac{d_0^2}{d_L^2}\right) \\
&= -2.5 \log_{10}\left(\frac{(10\text{pc})^2}{d_L^2}\right) \\
&= 5 \log_{10}\left(\frac{d_L}{10\text{pc}}\right) \\
&= 5 \log_{10}\left(\frac{d_L}{1\text{pc}}\right) - 5
\end{aligned}$$

Recall that the comoving distance χ is defined as:

$$\chi = \int_{t_e}^{t_0} \frac{dt}{a(t)} = \int_0^z \frac{dz'}{H_0 E(z')}$$

where $E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_r(1+z)^4 + \Omega_\Lambda + \Omega_K(1+z)^2 + \Omega_\Lambda}$.

At the sametime, we also know that:

$$d_L = (1+z)a_0\chi = (1+z)\chi$$

Therefore, we know that:

$$d_L = (1+z)\int_0^z \frac{dz'}{H_0 E(z')}$$

Theory

By using equation of distance modulus μ :

$$\mu = m - M = 5\log_{10}\left(\frac{d_L}{1\text{pc}}\right) - 5$$

and the equation of luminousity distance d_L:

$$d_L(z) = \frac{(1+z)}{H_0}\int_0^z \frac{dz'}{E(z')}$$

with E(z) as the energy density evolution function in different cosmological models, we can get the theoretical prediction of distance modulus μ for different cosmological models, and then we can compare the theoretical prediction with the observational data of type Ia supernovae to test the cosmological models.

Physics Intuition

From the above figure, we can see that the distance modulus μ of type Ia supernovae v.s. logarithm of redshift z, the upper sub-figure shows the distance modulus of supernovae, and the lower sub-figure shows distance modulus difference $\Delta\mu$ between the observed data and the theoretical prediction of model with $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.0$.

For each sub-figure, we have 3 curves/lines, the upper curve **(curve 1)** is $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$ which including the dark energy, the middle curve **(curve 2)** is $\Omega_m = 0.3$ and $\Omega_\Lambda = 0$ which is the SCDM model without dark energy, and the lower curve **(curve 3)** is $\Omega_m = 1$ and $\Omega_\Lambda = 0$ which is the Einstein-de Sitter universe.

We can see that the data points of type Ia supernovae are more consistent with the upper curve, which means the universe is accelerating expanding, and the SCDM model without dark energy is disfavored by the data.

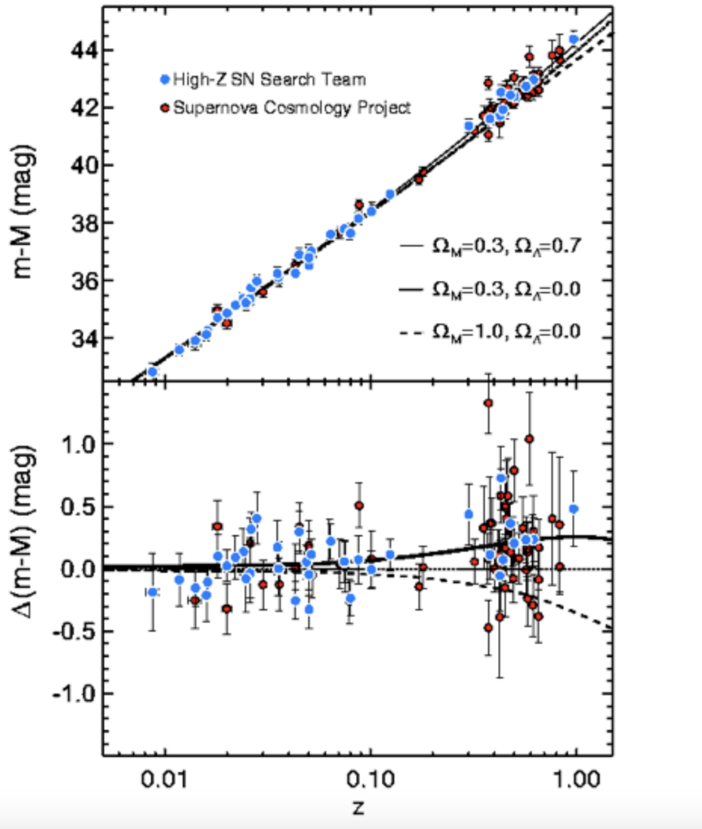

Figure 3.5: $m - M$ and $\Delta(m - M)$ v.s. z of the type Ia supernovae.

Physics Intuition

For the curve 1 with $\Omega_m = 0.3$ and $\Omega_\Lambda = 0.7$, we get E(z) as:

$$E_1(z) = \sqrt{0.3(1+z)^3 + 0.7}$$

and for the curve 2 with $\Omega_m = 0.3$ and $\Omega_\Lambda = 0$, we get:

$$E_2(z) = \sqrt{0.3(1+z)^3 + 0.7(1+z)^2}$$

where the extra term $0.7(1+z)^2$ is the contribution of curvature.
and for the curve 3 with $\Omega_m = 1$ and $\Omega_\Lambda = 0$, we get:

$$E_3(z) = \sqrt{1(1+z)^3} = (1+z)^{3/2}$$

It is easily to see that for the same redshift z, we have:

$$E_1(z) < E_2(z) < E_3(z)$$

Hence by $d_L(z) = \frac{(1+z)}{H_0} \int_0^z \frac{dz'}{E(z')}$, we can get:

$$d_{L,1}(z) > d_{L,2}(z) > d_{L,3}(z)$$

And thus:

$$\mu_1(z) > \mu_2(z) > \mu_3(z)$$

which is consistent with the above figure.

Physics Intuition

In the other words, since we know that $\ddot{a}_1(t) > \ddot{a}_2(t) > \ddot{a}_3(t)$ for $t < t_0$, and $\dot{a}_0 = H_0$ for all models, by:

$$\dot{a}(t) = \dot{a}_0 - \int_t^{t_0} \ddot{a}(t')dt'$$

we can get:

$$\dot{a}_1(t) < \dot{a}_2(t) < \dot{a}_3(t)$$

and using $a_0 = a(t_0) = 1$, by:

$$a(t) = a_0 - \int_t^{t_0} \dot{a}(t')dt'$$

we can get, for same time $t < t_0$:

$$a_1(t) > a_2(t) > a_3(t)$$

Hence, for the same redshift $z = \frac{1}{a} - 1$, we have:

$$a_1(t_1) = a_2(t_2) = a_3(t_3) = a$$

and thus:

$$t_1 < t_2 < t_3$$

Note if we received a light that with a known redshift z, we can know that in three models, the light is emitted at time t_1, t_2 and t_3 respectively;

Let light travel time be $t_L^i = t_0 - t_s^i$ with t_s^i is the time when the light is emitted, we can get:

$$t_L^1 > t_L^2 > t_L^3$$

Therefore, the light travel time is longer in the model with dark energy, which means the luminousity distance is larger in the model with dark energy, and thus the distance modulus is larger(dimmer for the same redshift) in the model with dark energy, which is consistent with the above figure.

Remark

Note although the density of dark energy is very small $\approx 1.67 \times 10^{-27}\text{kg/m}^3$, but it is still the dominant component of energy density in the universe, which is around 68% of the total energy density of universe, that mainly because the energy density of matter and radiation dilute with the expansion of universe, while the energy density of dark energy is constant, thus it will become dominant at late time.

Extra

The discovery of the acceleration mainly attributed to two independent teams, the High-z Supernova Search Team(1998) and the Supernova Cosmology Project(1999), for this work, the leaders of these two teams, Saul Perlmutter, Brian P. Schmidt and Adam G. Riess are awarded the Nobel Prize in Physics in 2011.

3.5.2 Dark Energy Models

We have mentioned that the ΛCDM model is one of the model that using cosmological constant to explain the acceleration of universe, but there are also some other models that can explain the acceleration of universe without introducing the cosmological constant, such as quintessence model, phantom energy model, etc.

THEORY

The **Quintessence model** is a scalar field model that can explain the acceleration of universe, the scalar field ϕ has a potential $V(\phi)$, and the energy density and pressure of the scalar field can be written as:

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$$
$$p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi)$$

Thus, the EoS parameter of quintessence can be written as:

$$w_\phi = \frac{p_\phi}{\rho_\phi} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)}$$

If the potential energy dominates over the kinetic energy, i.e. $V(\phi) \gg \frac{1}{2}\dot{\phi}^2$, then we have $w_\phi \approx -1$, which can drive the acceleration of universe.

Unlike the cosmological constant, the quintessence model can have a time-varying EoS parameter, which suggested the quintessence can be either attracting or repulsive depending on the value of $\dot{\phi}^2$ and $V(\phi)$, and thus produce a dynamic dark energy that can evolve with time.

THEORY

The **phantom energy model** is a scalar field model that can explain the acceleration of universe, but it has a EoS parameter $w < -1$, which means the energy density of phantom energy will increase with the expansion of universe, and the acceleration of universe will become faster and faster with time, which can lead to a future singularity called "Big Rip", where the scale factor and energy density will diverge in a finite time.

3.6 Fate of Universe

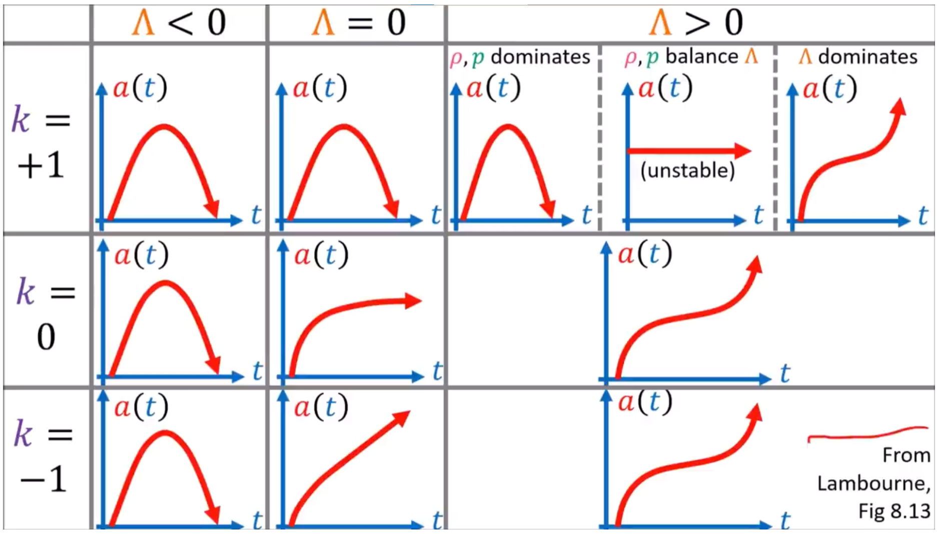

Figure 3.6: The evolution of scale factor with different combination of energy density.

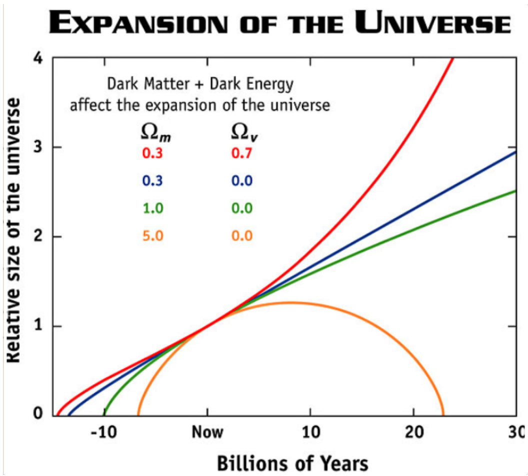

Figure 3.7: The evolution of scale factor with different combination of energy density.

Question Example 3.6: Matter-Only ($\Lambda = 0$) Universe: $k > 0,\ k = 0,\ k < 0$ Cases

Assume a matter-only universe ($p = 0$) and $\Lambda = 0$. Recall the 1st Friedmann equation:

$$H^2 = \frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{k}{a^2}$$

and use $\rho(a) = \rho_* a^{-3}$, where ρ_* is a constant.
(a). For each curvature case $k > 0$, $k = 0$, $k < 0$, calculate the late-time behavior of $\dot{a}$ (as $t \to \infty$, or explain if that limit does not exist) and describe the physical meaning.
(b). For each case, solve the scale factor function $a(t)$.
(c). In the $k > 0$ case, find $a_{\max}$ and t_f where a reaches maximum and then returns to zero; explain the physical meaning of t_f.

Solution.
Define:

$$A \equiv \frac{8\pi G}{3}\rho_* > 0$$

Then:

$$\dot{a}^2 = \frac{A}{a} - k$$

(a). Late-time behavior of $\dot{a}$:

- $k = 0$:

  $$\dot{a}^2 = \frac{A}{a} \Rightarrow \dot{a} \to 0 \quad (t \to \infty)$$

  Physical meaning: expansion never stops, but keeps decelerating and becomes slower and slower.

- $k < 0$ (write $k = -|k|$):

  $$\dot{a}^2 = \frac{A}{a} + |k| \Rightarrow \dot{a} \to \sqrt{|k|} \quad (t \to \infty)$$

  Physical meaning: expansion continues forever and asymptotically approaches a constant expansion speed (curvature-dominated coasting).

- $k > 0$:

  $$\dot{a}^2 = \frac{A}{a} - k$$

  There is a turning point $\dot{a} = 0$; the universe reaches a maximum size and then recollapses. So a physical $t \to \infty$ branch does not exist.

(b). Scale factor solutions (step-by-step):

- $k = 0$: Start from:

  $$\dot{a}^2 = \frac{A}{a} \Rightarrow \dot{a} = \frac{\sqrt{A}}{\sqrt{a}} \Rightarrow \sqrt{a}\,da = \sqrt{A}\,dt$$

Integrate both sides:

$$\int \sqrt{a}\, da = \int \sqrt{A}\, dt \Rightarrow \frac{2}{3}a^{3/2} = \sqrt{A}\, t + C$$

Use Big Bang initial condition $a(0) = 0$ so $C = 0$:

$$a^{3/2} = \frac{3}{2}\sqrt{A}\, t \Rightarrow a(t) = \left(\frac{3}{2}\sqrt{A}\, t\right)^{2/3} \propto t^{2/3}$$

- $k > 0$ **(closed, parametric form):** Start from

$$\dot{a}^2 = \frac{A}{a} - k \Rightarrow dt = \frac{da}{\sqrt{A/a - k}} = \frac{\sqrt{a}\, da}{\sqrt{A - ka}}.$$

Use substitution

$$a = \frac{A}{2k}(1 - \cos\eta) = \frac{A}{k}\sin^2\frac{\eta}{2}, \qquad da = \frac{A}{2k}\sin\eta\, d\eta.$$

Then

$$\sqrt{a} = \sqrt{\frac{A}{k}}\sin\frac{\eta}{2}, \qquad \sqrt{A - ka} = \sqrt{A}\cos\frac{\eta}{2},$$

so

$$dt = \frac{\sqrt{a}\, da}{\sqrt{A - ka}} = \frac{A}{2k^{3/2}}(1 - \cos\eta)\, d\eta.$$

Integrate and choose $t = 0$ at $\eta = 0$:

$$t(\eta) = \frac{A}{2k^{3/2}}\int_0^{\eta}(1 - \cos u)\, du = \frac{A}{2k^{3/2}}(\eta - \sin\eta).$$

Therefore

$$a(\eta) = \frac{A}{2k}(1 - \cos\eta), \qquad t(\eta) = \frac{A}{2k^{3/2}}(\eta - \sin\eta), \quad 0 \le \eta \le 2\pi$$

- $k < 0$ **(open, parametric form):** Write $k = -|k|$, then

$$\dot{a}^2 = \frac{A}{a} + |k| \Rightarrow dt = \frac{da}{\sqrt{A/a + |k|}} = \frac{\sqrt{a}\, da}{\sqrt{A + |k|a}}.$$

Use substitution

$$a = \frac{A}{2|k|}(\cosh\eta - 1), \qquad da = \frac{A}{2|k|}\sinh\eta\, d\eta.$$

From this substitution, one gets

$$dt = \frac{A}{2|k|^{3/2}}(\cosh\eta - 1)\, d\eta.$$

Integrate and choose $t = 0$ at $\eta = 0$:

$$t(\eta) = \frac{A}{2|k|^{3/2}} \int_0^{\eta} (\cosh u - 1)\, du = \frac{A}{2|k|^{3/2}} (\sinh \eta - \eta).$$

Therefore

$$a(\eta) = \frac{A}{2|k|}(\cosh \eta - 1), \qquad t(\eta) = \frac{A}{2|k|^{3/2}}(\sinh \eta - \eta), \quad \eta \geq 0$$

(c). For $k > 0$:

$$\dot{a} = 0 \Rightarrow \frac{A}{a_{\text{max}}} - k = 0 \Rightarrow a_{\text{max}} = \frac{A}{k}$$

In the closed-universe parametric solution, recollapse to $a = 0$ occurs at $\eta = 2\pi$, hence:

$$t_f = t(2\pi) = \frac{A\pi}{k^{3/2}}$$

Physical meaning of t_f: it is the finite lifetime from Big Bang to Big Crunch for this matter-only, closed, $\Lambda = 0$ universe.

EXTRA

So, what is the fate of our universe?

Based in the current CMB measurements, we know that our universe is basically flat $k \approx 0$, and the energy density proportion is showed as below:

$$\begin{cases} \Omega_\Lambda \approx 0.68 \\ \Omega_d \approx 0.27 \\ \Omega_b \approx 0.05 \\ \Omega_r \approx 10^{-4} \\ \Omega_K \approx 0 \end{cases}$$

Thus, our universe is dominated by the cosmological constant, and the scale factor will grow exponentially in the future, which means the universe will expand forever and become colder and darker, which is called the "Big Freeze" scenario.

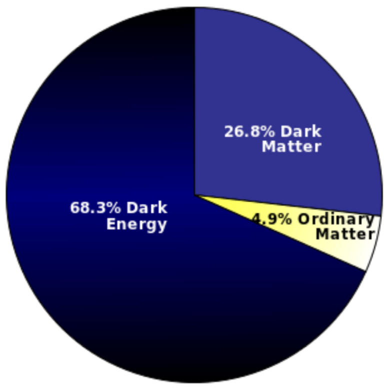

Figure 3.8: The energy density budget of our universe.

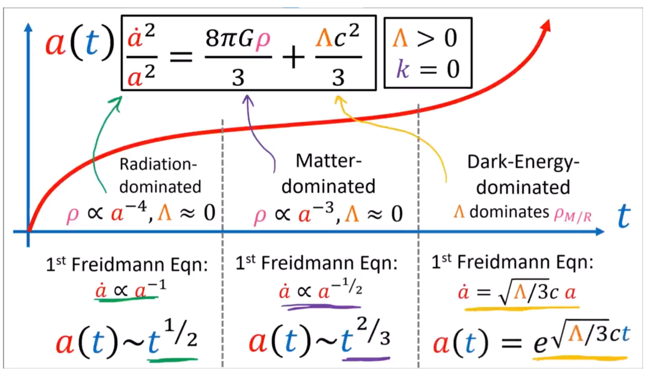

Figure 3.9: The evolution of scale factor of our universe.

CHAPTER 4

The Standard Model (Lecture 8)

Particle physics is not the main focus of this course, but it is important to have a basic understanding of the fundamental particles and interactions that govern the universe, as they play a crucial role in cosmology, especially in the early universe.

4.1 The Standard Model of Particle Physics

PHYSICS INTUITION

The Standard Model is a theoretical framework that describes the fundamental particles and their interactions, except for gravity. The fundamental particles are the smallest building blocks we have know so far, that build up matter(and energy) of the universe, and the interactions between them govern the behaviour of all physical phenomena.

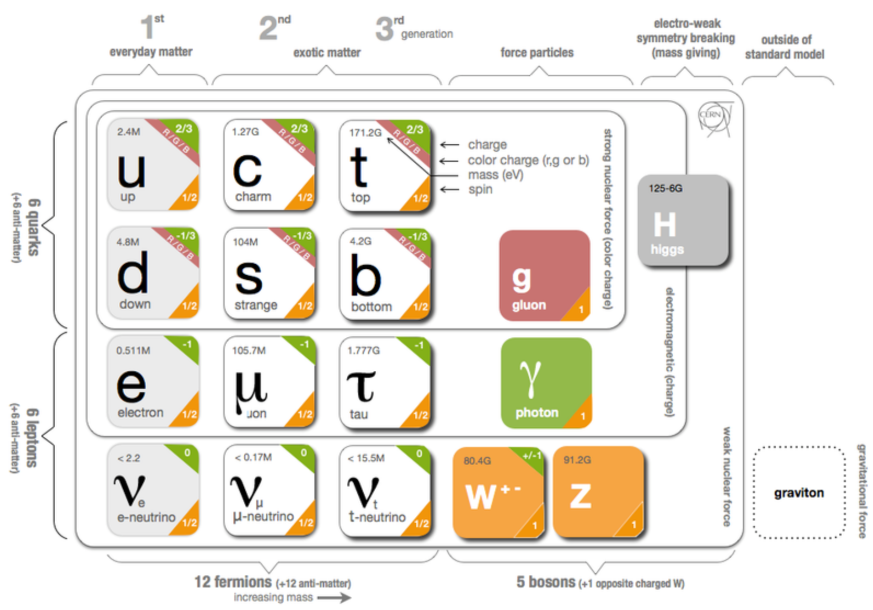

Figure 4.1: The Standard Model of particle physics, showing the fundamental particles and their interactions.

Foundation

From the above figure, we can see that for each particles, they have below properties:

- **Spin**: This is an intrinsic form of angular momentum that particles possess. It determines the statistics that particles obey (Fermi-Dirac for fermions and Bose-Einstein for bosons) and their behaviour under rotations.
  - **Spin 0 (scalar)**: e.g. Higgs boson.
  - **Spin 1/2 (fermions)**: e.g. electrons, quarks.
  - **Spin 1 (vector)**: e.g. photon, W and Z bosons.
- **Charge**: This is a fundamental property that determines how particles interact with electromagnetic fields. Particles can have positive, negative, or zero charge. The same type of particles with different charges are said as particles-antiparticles pair, such as electron and positron, proton and antiproton, etc.
- **Mass**: This is a measure of the amount of matter in a particle. It determines how particles respond to gravitational fields and how they interact with the Higgs field.
- **Color Charge**: This is a property of quarks that determines how they interact with the strong force. Each quark can have 1 of 3 color charges (red, green, blue), and the strong force is mediated by gluons that carry 1 of the 8 color charges.
- **Chiralility**: This describes the handedness of particles, which is important for weak interactions. Left-handed particles and right-handed particles can have different interactions with the weak force.

  For a group of particles with similar properties, we can classify them into a family, such as the three **generations** of fermions (electron, muon, tau and their corresponding neutrinos; up, down, charm, strange, top, bottom quarks), and the gauge bosons (photon, W and Z bosons, gluons).
  Sometimes, we will also use the term **flavour** to refer to the different types of particles within a family, such as the different flavours of quarks (up, down, charm, strange, top, bottom) and leptons (electron, muon, tau).
- **Interactions**: The fundamental particles interact with each other through four fundamental forces: gravity, electromagnetism, weak nuclear force, and strong nuclear force. The Standard Model describes the electromagnetic, weak, and strong interactions, but it does not include gravity.

Foundation

The Standard Model consists of three main components:

- **Fermions**: These are the matter particles, which include quarks and leptons. Quarks combine to form protons and neutrons, while leptons include electrons and neutrinos.
- **Gauge Bosons**: These are the force carriers that mediate the fundamental interactions. They include photons (electromagnetic force), W and Z bosons (weak force), and gluons (strong force).
- **Higgs Boson**: This particle is responsible for giving mass to other particles through the Higgs mechanism.

Foundation

Due to the strong interaction is asymptotically free, the quarks usually exist in forms of hadrons with 2 types:

- **Baryons**: These are made up of three quarks, and are subatomic particles such as protons $(p = uud)$ and neutrons $(n = udd)$.
- **Mesons**: These are made up of a quark and an antiquark, such as pions (π^+, π^0, π^-) and kaons (K^+, K^0, K^-).

Note not all quarks are confined in hadrons, such as the top quark, which is too heavy to form hadrons before it decays.

Remark

The Standard Model(SM) just a zoo of fundamental particles discovered so far, but it is not a complete theory of everything. It does not include gravity, and it cannot explain dark matter, dark energy, or the matter-antimatter asymmetry in the universe. Therefore, physicists are still searching for a more fundamental theory that can unify all forces and particles in the universe, which is often referred to as the "Theory of Everything(TOE)".

4.2 Higgs Mechanism and Mass Generation

Physics Intuition

For the boson particles, such as W and Z bosons, they are massive, but if we want to write down a gauge invariant Lagrangian for them, we cannot include a mass term for them, as it will break the gauge symmetry.

In a nutshell, allow these force carriers to have mass would means they are moving slower than the speed of light, which provide a extra degree of freedom for them; The extra degree of freedom provides extra force(e.g. for W and Z bosons, it is the weak force), which causing infinite force results at high energy in calculations, which make the theory broken. Thus, we need to fix our theory to make it consistent with the existence of massive gauge bosons, and this is where the Higgs mechanism comes in.

Important Note

One of the common mistakes is that people think the Higgs mechanism gives mass to all particles, but in fact, it only play a dominant role in giving mass to the W and Z bosons;

while for fermions, such as quarks, only $\sim 1\%$ of their mass come from the Higgs mechanism, the rest $\sim 99\%$ of their mass come from the strong interaction that bind them together in hadrons, such as protons and neutrons.

Physics Intuition

The Higgs mechanism is a process by which particles acquire mass through their interactions with the Higgs field. The Higgs field is a scalar field that permeates all of space, and it has a non-zero vacuum expectation value (VEV). When particles interact with the Higgs field, they acquire mass proportional to the strength of their interaction with the field.

On a easier-to-understand level, we can think of the Higgs field as a kind of "cosmic molasses" that fills the universe. When particles move through this molasses, they experience resistance, which manifests as mass. The stronger the interaction with the Higgs field, the more resistance they experience, and thus the greater their mass.

For example, the W and Z bosons acquire mass through their interactions with the Higgs field, while photons remain massless because they do not interact with the Higgs field.

In Higgs mechanism, the mass property of gauge bosons no longer come from them selves(like the fermions), but come from the interaction with the Higgs field.

By this tricks, the gauge bosons extra degree of freedom is provided by the Higgs field, thus allow the gauge bosons seems to be "massive" without breaking the gauge symmetry.

Theory

the total number of fundamental degrees of freedom in phase transition with the Higgs mechanism:

$$N_{\text{dof}} = 2 \times N_G + N_{\text{Higgs}} = 3 \times N_G + (N_{\text{Higgs}} - N_G)$$

where N_G is the number of gauge bosons that acquire mass through the Higgs mechanism, and N_{Higgs} is the number of Higgs fields. The first term $2 \times N_G$ accounts for the two transverse polarizations of the gauge bosons, while the second term $(N_{\text{Higgs}} - N_G)$ accounts for the remaining degrees of freedom from the Higgs fields after giving mass to the gauge bosons.

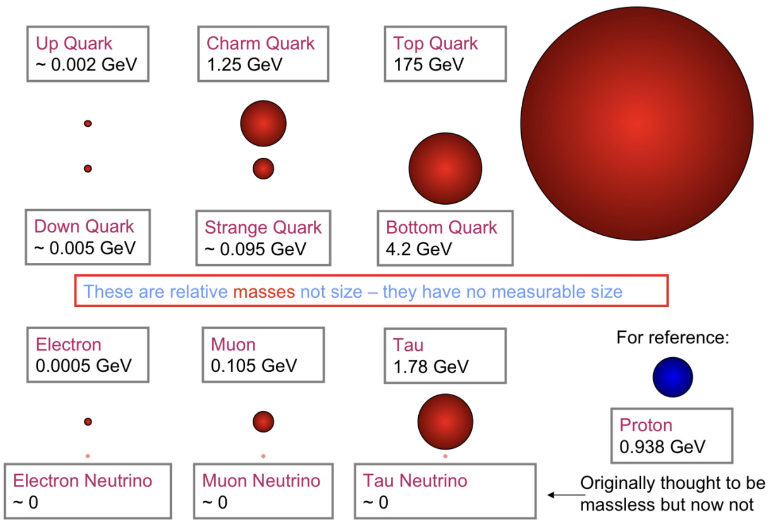

Figure 4.2: Some examples of masses of fundamental particles in the Standard Model.

4.3 SM, GUT, and TOE

EXTRA

The currently standard model(SM) only describes three of the four fundamental forces (electromagnetic, weak, and strong interactions), and have 2 theories for the strong interaction (QCD) and the electroweak interaction (EW).

However, physicists are still searching for a more fundamental theory that can unify the forces; The first steps is trying to unify the strong and electroweak interactions into a single theory, as they are described by similar gauge symmetries. Howard Georgi and Sheldon Glashow proposed the first Grand Unified Theory (GUT) in 1974, which is based on the gauge group $SU(5)$. In this theory, the strong and electroweak interactions are unified at high energies, and they predict the existence of new particles and interactions that have not yet been observed.

In 1981, MSSM (Minimal Supersymmetric Standard Model) was proposed, which is a supersymmetric extension of the Standard Model. It act as a model to led towards the GUT. MSSM introduces a new symmetry called supersymmetry, which relates bosons and fermions, and it predicts the existence of superpartners for all the particles in the Standard Model. One of the main differences between MSSM and GUT is that MSSM still keep the same gauge group as the SM $SU(3)_C \times SU(2)_L \times U(1)_Y$. But both of them still in the realm of theory instead of established physics, as they have not yet been experimentally confirmed.

Upon the above theories, physicists hypothesize that there is a more fundamental theory that can unify all forces, including gravity, which is often referred to as the "Theory of Everything (TOE)". One of the leading candidates for a TOE is string theory, which posits that the fundamental building blocks of the universe are not point particles, but rather one-dimensional "strings" that vibrate at different frequencies to give rise to the various particles and forces we observe.

In the below contents, we will mainly focus on the SM, and we will briefly touch the MSSM if need.

CHAPTER 5

Thermodynamics and Primordial Soup (Lecture 9-12)

To get the factors of energy density of radiation and matter, so that we can know the Evolution of the universe in a more general way, we will need to find out how radiation and matter mixture evolves.

Thermodynamics allow us to know the contribution of different particle species to the total energy density of the universe, and thus determine the expansion rate and thermal history of the universe.

5.1 Statistical Distributions

5.1.1 Energy-Momentum Relation

To calculate the value of various thermodynamic quantities, we need first clarify some basic concepts and notations first for convenience.

In below contents, we will use the $E(p)$ notation to represent the energy of a particle with momentum p and mass m.

Foundation

Recall special relativity, the energy of a particle with mass m and three-momentum p is given by

$$E(\vec{p}) = \sqrt{|\vec{p}|^2 c^2 + m^2 c^4}.$$

Taken $c = 1$ and $p = |\vec{p}|$, we have $E(p) = \sqrt{p^2 + m^2}$.

In a Non-relativistic limit ($p \ll m$):

$$E(p) = m\sqrt{1 + \frac{p^2}{m^2}} \approx m + \frac{p^2}{2m}.$$

In an Ultra-relativistic limit ($p \gg m$):

$$E(p) \approx p.$$

5.1.2 Chemical Potential

Physics Intuition

Imagine we have a box of gas, and we want to add a particle into the box. The chemical potential μ is the energy cost of adding that particle to the system, which can be thought of as the "price" we need to pay to add one more particle to the system.

However, when we add a particle to the system, we also increase the entropy of the system, which can be thought of as the "benefit" we get from adding that particle, since it allows for more possible microstates and thus increases the disorder of the system.

Furthermore, the temperature T of the system determines how much we value that increase in entropy, since at higher temperatures, we need to pay more energy to achieve the same increase in entropy.

Therefore, the chemical potential can be thought of as a balance between the energy cost of adding a particle and the entropy gain from adding that particle, which is given by the formula:

$$\mu = \text{energy cost} - T \cdot \text{entropy gain}$$

Foundation

μ is the chemical potential, which measures the change in internal energy when a particle is added to the system at constant entropy and volume. It is defined as:

$$\mu = \left(\frac{\partial U}{\partial N}\right)_{S,V} = \left(\frac{\partial F}{\partial N}\right)_{T,V},$$

Or in other way:

$$\delta\mu = \delta U - T\delta S,$$

where U is the internal energy, F is the Helmholtz free energy, N is the number of particles, S is the entropy, and T is the temperature.

For ideal gases, $\mu \to 0$, and we will assume $\mu = 0$ in below contents unless otherwise specified.

Remark

The chemical potential μ is typically zero for photons and other massless bosons, while it can be non-zero for massive particles depending on the context (e.g., in a gas of electrons).

while for particles and their own anti-particles, $\mu_+ = -\mu_-$.

5.1.3 Thermal Equilibrium Distributions

Foundation

For a state with energy E, and particle number N, the probability of the state is given by the Boltzmann factor:

$$P \propto e^{-(E-\mu N)/k_B T}$$

While for a single-particle state with energy $E(p)$, each momentum state can have occupation number n:

- Fermions: $n = 0$ or 1 ,
- Bosons: $n = 0, 1, 2, \ldots$

Then the average occupation number for a single-particle state with energy $E(p)$ is given by:

$$\langle n \rangle = \frac{\sum_n n e^{-(E(p)-\mu n)/k_B T}}{\sum_n e^{-(E(p)-\mu n)/k_B T}}$$

For fermions, we have:

$$\langle n \rangle = \frac{0 \cdot e^{-E(p)/k_B T} + 1 \cdot e^{-(E(p)-\mu)/k_B T}}{e^{-E(p)/k_B T} + e^{-(E(p)-\mu)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

For bosons, we have:

$$\langle n \rangle = \frac{\sum_{n=0}^{\infty} n e^{-(E(p)-\mu n)/k_B T}}{\sum_{n=0}^{\infty} e^{-(E(p)-\mu n)/k_B T}} = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

By letting $f(p) = \langle n \rangle$, we can calculate the number density by density of states $\times$ occupation number, which is given by:

$$\frac{dN}{d^3 p} = \overbrace{\frac{g}{(2\pi)^3}}^{\text{density of states}} \overbrace{f(p)}^{\text{occupation number}} = \frac{g}{(2\pi)^3} \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1}$$

The term $\frac{g}{2\pi^2}$ come from the Heisenberg uncertainty principle, which states that density states, i.e. the number of quantum states in a given volume of phase space as $(2\pi\hbar)^3$ under natural unit, and the factor g accounts for the spin degeneracy of the particle species.

THEORY

In thermal equilibrium, the distribution of particle energies follows specific statistical distributions:

- **Fermi-Dirac distribution** for fermions:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} + 1}$$

- **Bose-Einstein distribution** for bosons:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} - 1}$$

- **Maxwell-Boltzmann distribution** for classical particles:

$$f(p) = e^{-(E(p)-\mu)/k_B T}$$

In general, the distribution can be written as:

$$f(p) = \frac{1}{e^{(E(p)-\mu)/k_B T} \pm 1},$$

where the $+$ sign is for fermions and the $-$ sign is for bosons.

THEORY

In non-relativistic limit($E \gg T$), the Fermi-Dirac and Bose-Einstein distributions reduce to the Maxwell-Boltzmann distribution:

$$e^{(E(p)-\mu)/k_B T} \gg 1 \implies f(p) \approx e^{-(E(p)-\mu)/k_B T}$$

5.2 Number Density, Energy Density, and Pressure

5.2.1 Spin Degeneracy Factor

Physics Intuition

Imagine in a hotel, there are multiple rooms with different price(Energy level), the spin is like the no. of beds in the room, which can accommodate multiple guests(particles) with the same energy level. When 2 particles enter the same room(share the same energy level), we call it a degenerate state.

The degeneracy factor g states how many particles can stay in the same energy level, which is determined by the intrinsic spin quantum number s of the particle. On the other hand, it can be treat as the degree of freedom of the particle, which is the number of independent ways the particle can exist in a given energy state.

Foundation

In quantum mechanics, the magnetic quantum number m_s can take on values from $-s$ to $+s$ in integer steps($-s, -s+1, \ldots, s-1, s$). And we can get g by counting the number of possible m_s values, which is given by:

$$g = 2s + 1$$

However, for massless particles, such as photons, the spin is always aligned with the direction of motion, which means they only have 2 possible spin states (helicity states), thus $g = 2$ for photons.

Remark

The g here only counts the intrinsic spin degeneracy, but in general, we also need to consider other types of degeneracy, such as color charge for quarks, which can further increase the effective degeneracy factor for those particles: For example, if we treat the 8 type of gulons as the same one and calculate the g for gluons, we will get $g_{\text{total}} = 16$.

$$g_{\text{effective}} = g_{\text{spin}} \times g_{\text{color}}$$

In the below contents, when we see g_i, it only represents the intrinsic spin degeneracy, and we will need to multiply it with other degeneracy factors if there are any.

EXTRA

The g values for some common particles are:

Particle	symbol	g
Photon	g_γ	2
Electron/Positron	g_e^-, g_e^+	2
Z boson	g_Z	3
Higgs boson	g_H	1

5.2.2 Derivation of the 3 equations

THEORY

The number density n, energy density ρ, and pressure P of some particle i are given by:

$$n_i = \frac{g_i}{(2\pi)^3} \int f_i(p)\, d^3p,$$

$$\rho_i = \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p,$$

$$P_i = \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p$$

By substituting the Fermi-Dirac, Bose-Einstein, or Maxwell-Boltzmann distribution functions into the above integrals, we can derive explicit expressions for n_i, ρ_i, and P_i for different particle species under various conditions (e.g., relativistic vs. non-relativistic limits).

FOUNDATION

(1) Relativistic Fermions ($E \approx p$):

- Number density:

$$
\begin{aligned}
n_i &= \frac{g_i}{(2\pi)^3} \int \frac{d^3p}{e^{(E(p)-\mu)/k_BT}+1} \\
&= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} && \text{, using spherical coordinates} \\
&= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \int_0^\infty \frac{u^2 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\
&= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \cdot \frac{3}{2}\zeta(3) \\
&= \frac{3}{4}\frac{\zeta(3)}{\pi^2} g_i T^3
\end{aligned}
$$

- Energy density:

$$
\begin{aligned}
\rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p \\
&= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} && \text{, E = p for relativistic particles} \\
&= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\
&= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot \frac{7}{8} \cdot 6\zeta(4) \\
&= \frac{7}{8}\frac{\pi^2}{30} g_i T^4
\end{aligned}
$$

- Pressure:
  using:

$$
\frac{p^2}{3E(p)} = \frac{1}{3}\frac{p^2}{p} = \frac{1}{3}p,
$$

  we get:

$$
\begin{aligned}
P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p \\
&= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{\frac{1}{3}p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT}+1} \\
&= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u+1} && , u = \frac{p-\mu}{k_BT} \\
&= \frac{7}{8}\frac{\pi^2}{30} g_i T^4 \cdot \frac{1}{3} \\
&= \frac{1}{3}\rho_i
\end{aligned}
$$

(2) Relativistic Bosons ($E \approx p$):

- Number density:

$$\begin{aligned} n_i &= \frac{g_i}{(2\pi)^3} \int \frac{d^3p}{e^{(E(p)-\mu)/k_BT} - 1} \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{4\pi p^2 dp}{e^{(p-\mu)/k_BT} - 1} \\ &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \int_0^\infty \frac{u^2 du}{e^u - 1} \qquad , u = \frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2} k_B^3 T^3 \cdot 2\zeta(3) \\ &= \frac{\zeta(3)}{\pi^2} g_i T^3 \end{aligned}$$

- Energy density:
  using $E(p) \approx p$, we have:

$$\begin{aligned} \rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3p \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT} - 1} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u - 1} \qquad , u = \frac{p-\mu}{k_BT} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot 6\zeta(4) \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \cdot \frac{\pi^4}{15} \\ &= \frac{\pi^2}{30} g_i T^4 \end{aligned}$$

- Pressure:

$$\begin{aligned} P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3p \\ &= \frac{g_i}{(2\pi)^3} \int_0^\infty \frac{\frac{1}{3} p \cdot 4\pi p^2 dp}{e^{(p-\mu)/k_BT} - 1} \\ &= \frac{g_i}{2(\pi)^2} k_B^4 T^4 \int_0^\infty \frac{u^3 du}{e^u - 1} \qquad , u = \frac{p-\mu}{k_BT} \\ &= \frac{\pi^2}{30} g_i T^4 \cdot \frac{1}{3} \\ &= \frac{1}{3} \rho_i \end{aligned}$$

(3) Non-relativistic Particles ($E \approx m + p^2/2m$):

- Number density:

$$
\begin{aligned}
n_i &= \frac{g_i}{(2\pi)^3} \int e^{-(E(p)-\mu)/k_B T} d^3 p \\
&= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int e^{-p^2/2mk_B T} d^3 p \\
&= \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int 4\pi p^2 e^{-p^2/2mk_B T} dp \\
&= \frac{g_i}{2(\pi)^2} e^{-(m-\mu)/k_B T} \cdot \frac{\sqrt{\pi}}{4} (2mk_B T)^{3/2} \qquad \text{, using Gaussian integral} \\
&= g_i \left(\frac{mk_B T}{2\pi} \right)^{3/2} e^{-(m-\mu)/k_B T}
\end{aligned}
$$

- Energy density: using $E(p) \approx m + \frac{p^2}{2m}$, we have:

$$
\begin{aligned}
\rho_i &= \frac{g_i}{(2\pi)^3} \int E(p) f_i(p)\, d^3 p \\
&\approx \frac{g_i}{(2\pi)^3} \int \left(m + \frac{p^2}{2m} \right) e^{-(E(p)-\mu)/k_B T} d^3 p \\
&= \frac{g_i m}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int e^{-p^2/2mk_B T} d^3 p \\
&\quad + \frac{g_i}{(2\pi)^3} e^{-(m-\mu)/k_B T} \int \frac{p^2}{2m} e^{-p^2/2mk_B T} d^3 p \\
&= n_i m + \frac{3}{2} n_i k_B T \qquad \text{, using Gaussian integral} \\
&= n_i m \qquad \text{, since } m \gg T \text{ for non-relativistic particles}
\end{aligned}
$$

- Pressure: using $\frac{p^2}{3E(p)} \approx \frac{p^2}{3m}$, we have:

$$
\begin{aligned}
P_i &= \frac{g_i}{(2\pi)^3} \int \frac{p^2}{3E(p)} f_i(p)\, d^3 p \\
&\approx \frac{g_i}{(2\pi)^3} \cdot \frac{2}{3} \int \frac{p^2}{2m} e^{-(E(p)-\mu)/k_B T} d^3 p \\
&= n_i T \qquad \text{, by using the result from energy density calculation}
\end{aligned}
$$

THEORY

Number density, energy density, and pressure for particles with different thermal equilibrium distributions:

	Relativistic Fermions	Relativistic Bosons	Non-relativistic Particles (Fermions/ Bosons)
Number Density n_i	$\frac{3}{4}\frac{\zeta(3)}{\pi^2}g_i T^3$	$\frac{\zeta(3)}{\pi^2}g_i T^3$	$g_i\left(\frac{m_i T}{2\pi}\right)^{3/2}e^{-m_i/T}$
Energy Density ρ_i	$\frac{7}{8}\frac{\pi^2}{30}g_i T^4$	$\frac{\pi^2}{30}g_i T^4$	$n_i m_i$
Pressure P_i	$\frac{1}{3}\rho_i$	$\frac{1}{3}\rho_i$	$n_i T$

where $\zeta(3) \approx 1.202$ is the Riemann zeta function evaluated at 3.

We can see once the particles no longer relativistic, the number density and energy density will be exponentially suppressed.
It is because in the thermal plasma, the energy of particle are $E \sim T$, and when $T \ll 2m$, the probability of pair production which require energy $E \geq 2m$ will be very low, thus the number density and energy density will be suppressed.

PHYSICS INTUITION

It also explains why very few positrons exist in the current universe!
In the early universe, $T \gg m_e \approx 0.511\,\text{MeV}$, e^+e^- pairs were abundant.
As the universe cooled below $T \sim m_e$, the number density of positrons (and electrons) dropped exponentially, leaving only a tiny fraction that survived annihilation, which is why we observe so few positrons today.

PHYSICS INTUITION

We can also observed the ferminos have lower number density and energy density than bosons.

Actually it also match our expectation in previous section, we know that bosons can occupied the same quantum state but fermions cannot (Pauli exclusion principle), thus we can expect that fermions have lower number density and energy density than bosons as lower occupation number.

PHYSICS INTUITION

For non-relativistic particles, we can see the energy density is dominated by the rest mass energy $n_i m_i$, while the pressure is given by $n_i T$, which is just the ideal gas law $P = nk_B T$ with $k_B = 1$ under natural unit.

Theory

We can see the relationship between p and ρ for different relativistic limits:

$$P = \begin{cases} \frac{1}{3}\rho & , T \gg m \\ nT \ll \rho & , T \ll m \end{cases}$$

In fact, it also consistent with the assumption of state parameter of radiation and matter, which are $w = 1/3$ and $w = 0$ when in relativistic and non-relativistic limits respectively.

Theory

The average particle energy is given by:

$$\langle E \rangle = \frac{\rho_i}{n_i} = \begin{cases} \frac{7\pi^4}{180\zeta(3)} T \approx 3.151 T & , T \gg m \text{ for relativistic fermions} \\ \frac{\pi^4}{30\zeta(3)} T \approx 2.701 T & , T \gg m \text{ for relativistic bosons} \\ m + \frac{3}{2} T & , T \ll m \text{ for non-relativistic particles} \end{cases}$$

Thats mean we can have a simplification of $\langle E \rangle \sim T$ for relativistic particles, as they have same order of magnitude, which is low enough error in astronomy :)

5.3 Effective Number of Relativistic DOF

Theory

The Friedmann equation shows that the expansion rate of the universe is determined by the total energy density ρ:

$$\rho(T) = \sum_i \rho_i(T)$$

where i runs over all particle species in the thermal plasma.

Considering in the the early universe, energy density is dominated by relativistic particles, we can directly sum over the relativistic species, and define the effective number of relativistic degrees of freedom g_* as:

$$\rho(T) = \frac{\pi^2}{30} g_*(T) T^4,$$

where:

$$g_*(T) = \sum_{\text{bosons}} g_i + \frac{7}{8} \sum_{\text{fermions}} g_i$$

Important Note

Note the above equation is only valid in early universe which radiation dominates. When universe cools down and matter dominates, the energy density will be dominated by non-relativistic particles, and above equation will no longer valid. In that case, we need to consider the contribution of non-relativistic particles to the energy density, which is given by $\rho_i = n_i m_i$.

Remark

g_* is depends on T(i.e. $g_*(T)$), as the sum runs over only relativistic species, with $m_i \ll T$. As the universe cools down, more and more species become non-relativistic and drop out of the sum, thus g_* decreases with time.

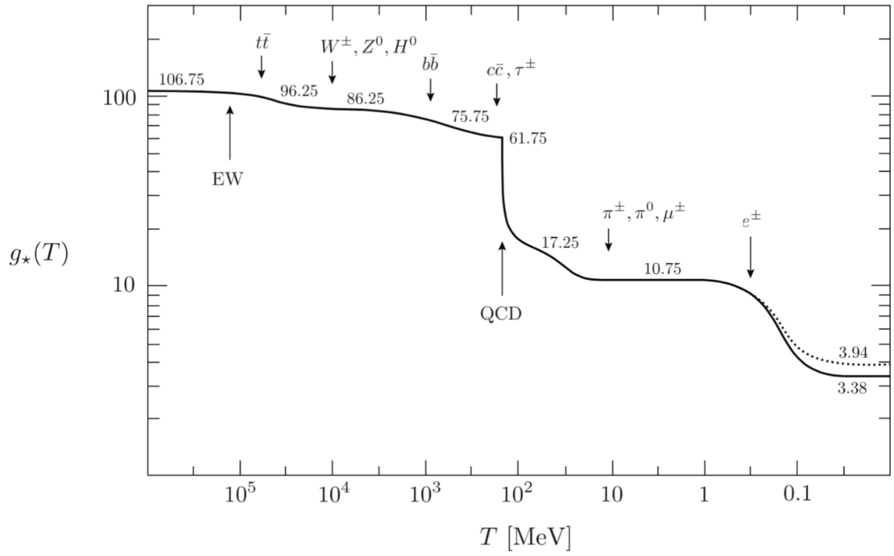

Figure 5.1: Evolution of g_* with temperature.

EXTRA

In fact, the above equation of g_* assume that all particles are not decoupled and in same temperature, to be more general, we need to consider the contribution of decoupled particles with different temperature, and the equation will be:

$$g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4$$

5.4 Entropy

5.4.1 Definition of Entropy

PHYSICS INTUITION

Far from above topic, we only discuss the macrostate of a system(e.g., temperature, pressure, volume), but we haven't discussed the microstate of a system, which is the specific configuration of particles(i.e. the positions and momenta of every particles) that make up the system.

In a given macrostate, there can be many different microstates that correspond to the same macrostate. For example, a macrostate of 2 coins combined with 1 H and 1 T can be achieved by 2 different microstates: HT and TH. While a macrostate of 2 coins combined with 2 H can only be achieved by 1 microstate: HH.

The entropy of a system is a measure of the number of microstates that correspond to a given macrostate. The more microstates there are, the higher the entropy.

In the context of thermodynamics, the concentrated energy or particles(mass) in a small region of space has low entropy, while the dispersed energy or particles in a large region of space has high entropy.

FOUNDATION

State Function is function of state variables, which only depends on the current state of the system, and is independent of the path taken to reach that state. Below are some examples of state functions:

- Internal Energy (U)
- Volume (V)
- Particle Number (N)
- Temperature (T)

And the Entropy (S) is also a state function.

REMARK

The entropy value is not absolute, but relative. We can only measure the change of entropy ΔS between two states, but not the absolute value of entropy S itself.

This is because the absolute value of entropy depends on the choice of reference state, which is arbitrary. For example, we can choose the reference state to be the state of zero entropy, which corresponds to a perfectly ordered system with only one microstate.

In that case, the entropy of any other state will be positive, and the change in entropy ΔS will be equal to the absolute value of entropy S for that state. However, if we choose a different reference state, the absolute value of entropy will be different, but the change in entropy ΔS between two states will remain the same, as it only depends on the difference in the number of microstates between the two states, which is independent of the choice of reference state.

FOUNDATION

There is 2 different definitions of entropy depending on the context:

- **classical thermodynamics definition:**

$$dS = \frac{dQ}{T}$$

where dQ is the heat added to the system, and T is the temperature of the system.

It is based on the idea below:
For a system, if we add heat(dQ), the system gain more energy, and particles in the system will have more energy to move around, which allow them to access more velocities microstates. Thus, the number of microstates accessible to the system will increase, which means the entropy of the system will increase.

$$dS \propto dQ$$

On the other hand, if the system has higher temperature, the average energy of particles in the system already high, and their already larger number of microstates accessible to them. Thus, adding the same amount of heat dQ to a system with higher temperature will result in a smaller increase in the number of microstates, and thus a smaller increase in entropy.

$$dS \propto \frac{1}{T}$$

- **statistical mechanics definition:**

$$S = k_B \ln \Omega$$

where k_B is the Boltzmann constant, and Ω is the number of microstates corresponding to the macrostate of the system.

It is based on the idea below:
Entropy(S) has extensive property, which means that if we have two independent systems A and B, the total entropy of the combined system is the sum of the entropies of the individual systems:

$$S_{\text{total}} = S_A + S_B$$

On the other hand, the number of microstates(Ω) has multiplicative property, which means that if we have two independent systems A and B, the total number of microstates of the combined system is the product of the number of microstates of the individual systems:

$$\Omega_{\text{total}} = \Omega_A \cdot \Omega_B$$

Therefore, to fulfill both properties, we can define the entropy as the logarithm of the number of microstates:

$$S \propto \ln \Omega$$

Now we can use the above definition of entropy to prove the 2nd law of thermodynamics.

Foundation

Start from Carnot cycle, assume we have a reversible carnot engine operating between two heat reservoirs at temperatures T_H and T_C ($T_H > T_C$).

$$\text{Efficiency}\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{Q_C}{Q_H}\frac{Q_C}{T_C} = \frac{Q_H}{T_H}$$

In a complete cycle:

$$\oint_{\text{rev}} \frac{dQ}{T} = 0$$

Note carnot cycle is theoretical most efficient engine, which means for any cycle:

$$\oint \frac{dQ}{T} \leq 0$$

And for a heat loss irreversible process, we have:

$$\oint_{\text{irr}} \frac{dQ}{T} < 0$$

So only if the process is reversible, the change in entropy will be equal to the integral of dQ/T;

And if the process is irreversible, the change in entropy will be greater than the integral of dQ/T, or equivalently, the integral of dQ/T will be less than the change in entropy due to the heat loss in the irreversible process.

Theory

The 2nd law of thermodynamics can been expressed by the general entropy inequality:

$$\Delta S = S_{\text{final}} - S_{\text{initial}} \geq \int_{\text{initial}}^{\text{final}} \frac{dQ}{T}|_{\text{path}}$$

where ΔS is the change in entropy of the system,
If the process is reversible, then lhs is $=$ to rhs;
and if the process is irreversible, then lhs is $>$ than rhs.

In an isolated system, there is no heat exchange with the surroundings, so $dQ = 0$, thus the inequality reduces to:

$$\Delta S_{\text{isolated}} \geq 0$$

and we get:

$$\begin{cases} \Delta S = 0 & \text{, for reversible processes} \\ \Delta S > 0 & \text{, for irreversible processes} \end{cases}$$

Physics Intuition

According to the second law of thermodynamics, the total entropy of an isolated system can never decrease over time, which means that the universe tends to evolve towards states with higher entropy.

In the macro level, it shows the universe tends to spread out energy and matter, and become more homogeneous and isotropic over time.

5.4.2 Conservation of Entropy

Physics Intuition

Assuming the universe is an isolated system (adiabatic), by 2nd law of thermodynamics, the total entropy of the universe is either constant or increasing over time.

Further, if we assume the evolution of the universe is quasi-static and reversible, thats mean the universe evolves through a series of equilibrium states, and there is no heat exchange with the surroundings, so the ΔS of the universe is zero, and entropy is conserved.

Theory

The fundamental thermodynamic relation is given by:

$$dU = TdS - PdV + \mu dN$$

by rearranging the above equation, we can get:

$$dS = \frac{dU + PdV - \mu dN}{T}$$

or

$$s = \frac{\rho + p}{T},$$

where $s = \frac{dS}{dV}$, $\rho = \frac{dU}{dV}$, and p are the equilibrium entropy density, energy density, and pressure of the system respectively.

In a more general case, considering $U = \rho V$ and $V \propto a^3$, and assuming $\mu = 0$, we can get:

$$\begin{aligned} dU &= Tds - pdV \\ d\rho V + \rho dV &= Tds - pdV \\ T\frac{ds}{dt} &= V\frac{d\rho}{dt} + (\rho + p)\frac{dV}{dt} \end{aligned}$$

By substituting the continuity equation $\frac{d\rho}{dt} = -3H(\rho + p)$ into the above equation, we can get:

$$\begin{aligned} T\frac{ds}{dt} &= V(-3H(\rho + p)) + 3HV(\rho + p) \\ &= 0 \end{aligned}$$

Thus:

$$\frac{ds}{dt} = 0$$

which match our intuition that the entropy is conserved in the universe.

And by knowing the relationship between energy density and entropy density, we can derive the effective number of relativistic degrees of freedom for entropy density $g_{*s}(T)$ in the early universe.

Theory

For a relativistic particle species in thermal equilibrium, we have:

$$\rho = \frac{\pi^2}{30} g_*(T) T^4,$$

and using:

$$s = \frac{\rho + p}{T},$$

we get:

$$s = \frac{2\pi^2}{45} g_{*s}(T) T^3,$$

where $g_{*s}(T)$ is the effective number of relativistic degrees of freedom for entropy density, which is defined as:

$$g_{*s}(T) = \sum_{\text{bosons}} g_i + \frac{7}{8} \sum_{\text{fermions}} g_i$$

From the above equation, we can get the entropy conservation in another way:

FOUNDATION

Assume all particles remains relativistic and not decoupled when universe evolves, we have $g_{*s}(T) = \text{constant}$, thus we can get:

$$\rho \propto T^4,$$

Recall:

$$\rho \propto a^{-4},$$

we can get:

$$T \propto a^{-1},$$

At the same time, we also know that:

$$s \propto T^3,$$

Therefore:

$$sV \propto sa^3 = \frac{2\pi^2}{45} g_{*s}(T) T^3 a^3 = \text{constant}$$

REMARK

The 2nd proof of entropy conservation is based on the assumption that all particles remains relativistic and not decoupled when universe evolves, which is only valid in the radiation dominated era of early universe.

However, in the 1st proof of entropy conservation, we only assume the universe is an isolated system and evolves through a series of equilibrium states, which is more general(both radiation and matter) and can be applied to the whole history of the universe, including the matter dominated era and dark energy dominated era.

5.4.3 g_* and g_{*s} in the early universe

Foundation

In the early universe, if we assume all standard model particles are relativistic and in thermal equilibrium, we can calculate the effective number of relativistic degrees of freedom for energy density $g_*(T)$ and for entropy density $g_{*s}(T)$:

(1) For bosons:

- Photon: $g_\gamma = 2$
  (2 polarization states)
- Gluons: $g_g = 2 \times 8 = 16$
  (2 polarization states $\times$ 8 color states)
- W$\pm$ bosons: $g_W = 3 \times 2 = 6$
  (3 spin states $\times$ 2 charge states)
- Z boson: $g_Z = 3$
  (3 spin states)
- Higgs boson: $g_H = 1$
  (1 spin state)

(2) For fermions:

- leptons ((e, μ, τ), (e^+, μ^+, τ^+)): $g_l = 2 \times 3 \times 2 = 12$
  (2 spin states $\times$ 3 generations $\times$ 2 charge states)
- Neutrinos (ν_e, ν_μ, ν_τ): $g_v = 2 \times 3 = 6$
  (2 spin states $\times$ 3 generations, but neutrinos anti-particles are itself, so no charge states)
- Quarks ((u, d, c, s, t, b), ($\bar{u}, \bar{d}, \bar{c}, \bar{s}, \bar{t}, \bar{b}$)): $g_q = 2 \times 3 \times 6 \times 2 = 72$
  (2 spin states $\times$ 3 color states $\times$ 6 flavors $\times$ 2 charge states)

By summing over all relativistic species, we can get:

$$
\begin{aligned}
g_b &= 2 + 16 + 6 + 3 + 1 = 28, \\
g_f &= 12 + 6 + 72 = 90
\end{aligned}
$$

And thus we can get:

$$
g_*(T) = g_{*s}(T) = g_b + \frac{7}{8} g_f = 28 + \frac{7}{8} \times 90 = 106.75
$$

Theory

In early universe, by only considering particles in standard model, and assuming all particles are relativistic and not decoupled to other particles, we can get:

$$g_*(T) = g_{*s}(T) = 106.75$$

Important Note

The equivalence of $g_*(T) = g_{*s}(T)$ is only valid when all the relativistic particles are in thermal equilibrium and have the same temperature.

Again, if we consider the contribution of decoupled particles with different temperature, the equation will be:

$$\begin{cases} g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4 \\ g_{*s}(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8} \sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3 \end{cases}$$

and thus $g_*(T)$ and $g_{*s}(T)$ will no longer be always equal, as they have different power of temperature ratio.

Question Example 5.1: Higgs Mechanism and g_*

We know that in Higgs mechanism, DoF of Higgs boson and W and Z bosons will changed, but our calculation of g_* in early universe is still valid, why?

Solution.
The answer is that both of them are still boson before and after Higgs mechanism, thus they will contribute to g_* with the same weight, and their total DoF is still remain unchanged, thus the value of g_* will not change.

5.5 Decoupling of Particles

5.5.1 Asymptotic Freedom

Physics Intuition

Back to standard model, we have seen a lone electron,neutrino, photon, but we haven't seen a lone quark, why?

Unlike gravitational or electromagnetic interaction, the strong interaction interact in reverse way, when we pull quarks apart, the strong interaction become stronger, while when we push quarks together, the strong interaction become weaker.

The "Asymptotic" refers to limit of high energy or short distance, and "Free" refers to the fact that quarks behave like free particles in this limit. Thus, asymptotic freedom means that at very high energies or short distances, the strong interaction becomes weak, and quarks can move freely without being confined within hadrons.

In early universe with high energy scale, the strong interaction is weak due to asymptotic freedom, thus quarks can move freely and exist as free particles, but as the universe cools down and energy scale decreases, the strong interaction becomes stronger, thus quarks will be confined within hadrons and we will not see free quarks anymore(or in other words, **Decoupling of quarks**).

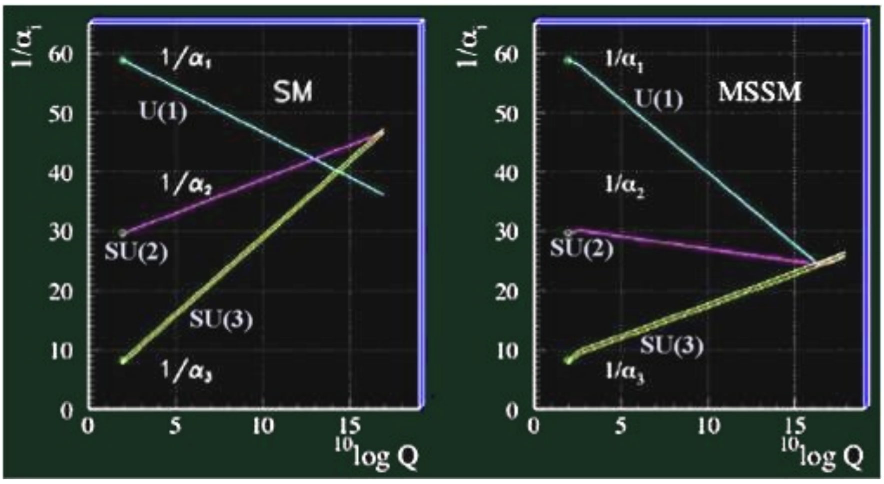

Figure 5.2: The decoupling of particles in the early universe.

Extra

From the above figure, we can see there is 3 lines:

- The cyan line $U(1)$: Hypercharge force interaction
- The magenta line $SU(2)$: Weak force interaction
- The yellow line $SU(3)$: Strong force interaction

The hypercharge force here is the precursor of the electromagnetic force, after electroweak symmetry breaking(at low energy scale), the hypercharge field mixes with the $SU(2)$ field to give us electromagnetic force and weak force.

The α here means the intereaction strength of the corresponding force, for example, $\alpha_{EM,0} \approx 1/137$ is the interaction strength of electromagnetic force at low energy scale(nowadays).

In right diagram, We can see that unlike the hypercharge force and weak force, the strong force interaction strength decreases with increasing energy scale, which is the asymptotic freedom of strong interaction.

Extra

The above figure also points out the limitation of standard model, which is the non-unification of coupling constants at high energy scale.

On the left panel, we can see that in SM, the 3 lines do not meet at a single point, which means the coupling constants of the 3 interactions do not unify at high energy scale, thus we need to introduce new physics beyond SM if we wants achieve the unification of coupling constants.

On the right panel, we can see that in MSSM, the 3 lines meet at a single point, which means the coupling constants of the 3 interactions unify at high energy scale, thus MSSM is a good candidate for new physics beyond SM to achieve the unification of coupling constants.

Extra

The asymptotic freedom discovered by Gross, Wilczek, and Politzer in 1973, which is a fundamental property of quantum chromodynamics (QCD), the theory that describes the strong interaction between quarks and gluons.

5.5.2 Decoupling

Physics Intuition

In physics, decoupling refers to the process by which a particle species stops interacting with other particles in the thermal plasma and becomes effectively non-interacting.

If we want to talk about the evolution of thermal history of the universe, we need to consider the decoupling of particles, when the particles decouple, they will no longer be in thermal equilibrium with other particles, and their number density and energy density will evolve differently from the particles that are still in thermal equilibrium.

When universe cools down and energy scale decreases, more and more particles will decouple from the thermal plasma, thus we will see a series of decoupling events in the thermal history of the universe, such as electroweak phase transition, QCD phase transition, neutrino decoupling, electron-positron annihilation, etc. And these decoupling provides information for us how the universe evolves for a "primordial soup" to the current universe we see today.

Physics Intuition

The decoupling of particles can be understood by comparing the interaction rate Γ of the particle species with the expansion rate of the universe H.

If $\Gamma \gg H$, the particle species is in thermal equilibrium with other particles, and it will continue to interact and exchange energy with other particles.

If $\Gamma \ll H$, the particle species is decoupled from other particles, and it will stop interacting and exchange energy with other particles, thus it will evolve independently from other particles.

Therefore, the condition for decoupling can be expressed as:

$$\Gamma < H$$

5.6 Thermal History of the Universe

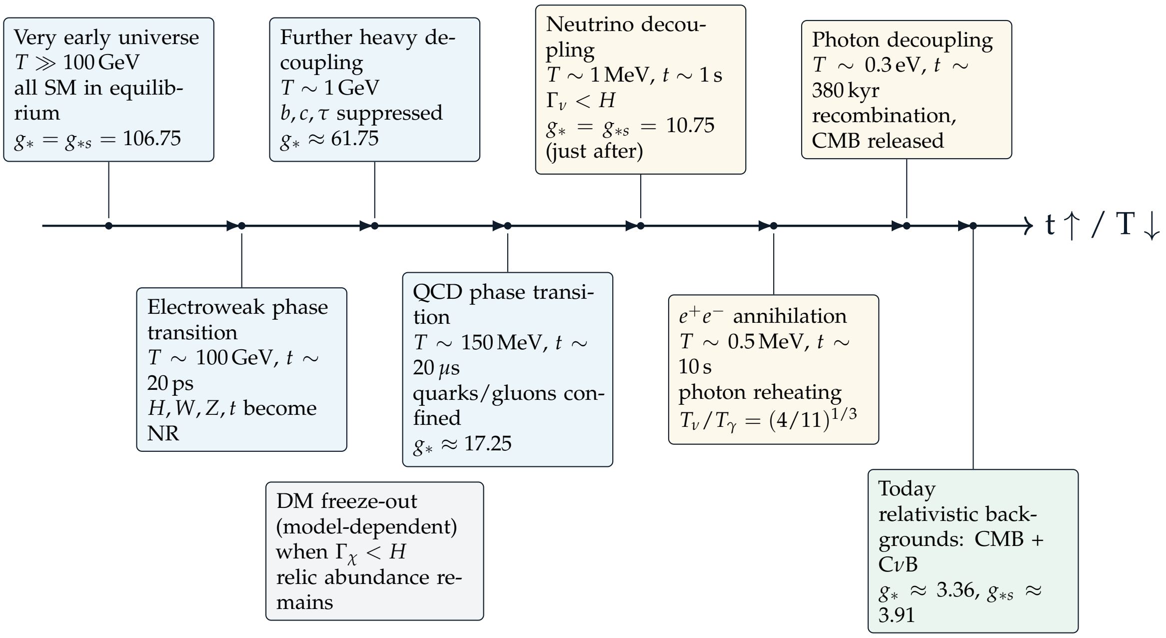

Figure 5.3: Thermal history timeline of major decoupling events.

5.6.1 Electroweak Phase Transition

At around $T \sim 100\,\mathrm{GeV}$, which $t \sim 20\,\mathrm{ps}$, the temperature of the universe drops below the energy scale of electroweak interaction, thus the electroweak symmetry is spontaneously broken and the Higgs field condenses, which gives mass to the W and Z bosons, and also gives mass to fermions.

For heavier particles like **Higgs (H)**, **W**, **Z** bosons, **top (t)** quark, they will decouple from the thermal plasma after electroweak phase transition, as their number density and energy density will be exponentially suppressed, and so does their abundance and interaction rate (Γ). In this stage, the g_* become 86.25.

At around $T \sim 1\,\mathrm{GeV}$, the temperature of universe further drops, and some heavier fermions like **bottom (b)** quark, **charm (c)** quark, **tau (τ)** lepton will also decouple from the thermal plasma, as their number density and energy density will be exponentially suppressed, and so does their abundance and interaction rate (Γ).

while for lighter particles like electron, muon, tau, they will remain in thermal equilibrium for a while and decouple at later time when the temperature drops below their mass scale. In this stage, the g_* become 61.75.

5.6.2 QCD Phase Transition

The asymptotic freedom of quarks and gluons prevents their decoupling in the early universe, as the strong interaction not strong enough to confine them within hadrons, thus they can exist as free particles and interact with other particles in the thermal plasma.

At around $T \sim 150\,\text{MeV}$, which $t \sim 20\,\mu\text{s}$, the temperature of the universe drops below the energy scale of QCD, the strong coupling constant α_s becomes large, thus the strong interaction becomes strong enough to confine **quarks** and **gluons** within hadrons, they are decoupled from the thermal plasma as their abundance and interaction rate (Γ) will be exponentially suppressed, and we will not see free quarks and gluons anymore, but only hadrons such as protons and neutrons.

After the QCD phase transition, the quark-gluon plasma condenses into hadrons plasma, we get many different species of baryons and mesons, but all except **pions** as one of the mesons are non-relativistic in this stage.

And given $g_\pi = 3$ for pions, we can said in this stage, the g_* become 17.25.

Remark

Before the QCD phase transition, quark and gluon are massless, then gain mass through the confinement after the QCD phase transition.

5.6.3 Neutrino Decoupling

After the forming of pions, they soon decouple again $m_\pi \sim 139.6\,\text{MeV}$. And the next decoupling event is the neutrino decoupling, which happens at around $T \sim 1\,\text{MeV}$, which $t \sim 1\,\text{s}$.

Theory

The decoupling of neutrinos can be expressed by the following reaction:

$$\nu + \bar{\nu} \leftrightarrow e^+ + e^-$$

with the creation rate:

$$\Gamma_\nu = n_e \sigma_\nu c$$

where n_e is the number density of electrons, σ_ν is the cross section of neutrino interaction, and c is the speed of light.

FOUNDATION

The number density can be expressed as:

$$n_e \sim \frac{\text{Energy Density}}{\text{Energy per particle}} \sim \frac{T^4}{T} = T^3$$

At the same time, we know neutrino interaction is weak interaction, at energy scale much lower than the mass of W and Z bosons, we used Fermi Theory, which is governed by the Fermi constant G_F with dimension of $[\text{Energy}]^{-2}$, By the born rules, we know that the cross section of neutrino interaction can be expressed as:

$$\sigma_\nu \propto G_F^2$$

while we know the cross section has dimension of $[\text{Energy}]^{-2}$, by dimensional analysis, we can get:

$$\sigma_\nu \sim G_F^2 T^2$$

And combining the above equations, we can get:

$$\Gamma_\nu = n_e \sigma_\nu c \sim T^3 \times G_F^2 T^2 = G_F^2 T^5$$

THEORY

Note the interaction rate has below relationship with temperature:

$$\Gamma_\nu \sim G_F^2 T^5$$

where G_F is the Fermi constant.

Recall the Friedmann equation:

$$\begin{aligned} H^2 &= \frac{8\pi G}{3}\rho \\ H &\sim \sqrt{G\rho} \\ H &\sim \sqrt{\frac{\rho}{M_{\text{Pl}}^2}} \\ H &\sim \frac{T^2}{M_{\text{Pl}}} \end{aligned}$$

where M_{Pl} is the Planck mass, and we have used the fact that in the early universe, the energy density ρ is dominated by radiation, which scales as $\rho \propto T^4$.

Therefore, we have:

$$\begin{aligned}\frac{\Gamma_\nu}{H} &\sim \frac{G_F^2 T^5}{T^2/M_{\text{Pl}}} \\ &\sim G_F^2 M_{\text{Pl}} T^3\end{aligned}$$

At around $T \sim 1\,\text{MeV}$, which $t \sim 1\,\text{s}$, the temperature of the universe drops below the energy scale of weak interaction, thus neutrinos decouple from the thermal plasma as their interaction rate (Γ_ν) becomes smaller than the expansion rate of the universe (H), and they will free-stream through the universe without interacting with other particles.

After decoupling, neutrinos no longer interact with other particles, their temperature only influenced by the expansion of universe. And these decoupled neutrinos are what we call the Cosmic Neutrino Background (CνB), which has a temperature of $T_\nu \approx 1.95\,\text{K}$.

Just after neutrino decoupling, the g_* and g_{*s} become:

$$\begin{aligned}g_*(T) &= g_\gamma + \frac{7}{8}(g_\nu + g_e) = 2 + \frac{7}{8}(6+4) = 10.75, \\ g_{*s}(T) &= g_*(T) = 10.75\end{aligned}$$

Important Note

Unlike other particles, the neutrinos still remain **relativistic** after decoupling, so they will still contribute to the effective number of relativistic degrees of freedom g_* and g_{*s}.

Start from here, the g_* and g_{*s} will become no longer equal, as temperature of neutrinos is not same as the photons, while both of them are still relativistic and contribute to g_* and g_{*s} with different weight due to their different temperature.

5.6.4 Electron-Positron Annihilation

THEORY

The interaction of electrons, positrons, and photons can be expressed by the following reaction:

$$e^+ + e^- \leftrightarrow \gamma + \gamma$$

At around $T \sim 0.5\,\text{MeV}$, which $t \sim 10\,\text{s}$, the temperature of the universe drops below the mass of electron, the thermal photons no longer create e^+e^- pairs, while existing e^+e^- pairs will continue to annihilate into photons, thus the number density of e^+e^- pairs will drop significantly, and so does their interaction rate (Γ), thus they decouple from the thermal plasma and free-stream through the universe.

The decoupling of e^+e^- pairs lead to the increase of temperature of photons because the annihilation of e^+e^- pairs into photons will release energy into the photon bath, which will increase the energy density of photons, and thus increase the temperature of photons.

THEORY

Notes the total entropy of the universe is conserved, thus we have:

$$s_1 V_1 = s_2 V_2$$
$$g_{*s}(T_1)T_1^3 a_1^3 = g_{*s}(T_2)T_2^3 a_2^3$$

And in our case:

$$2T_{\gamma,2}^3 a_2^3 + \frac{7}{8} \times 6T_{\nu,2}^3 a_2^3 = 10.75 T_1^3 a_1^3$$

Before e^+e^- annihilation, photons, neutrinos, and electrons are all in thermal equilibrium with the same temperature, thus we have:

$$T_1^3 a_1^3 = T_{\nu,2}^3 a_2^3$$

and after e^+e^- annihilation, only photons are heated up, thus we have:

$$2T_{\gamma,2}^3 a_2^3 + 5.25 T_{\nu,2}^3 a_2^3 = 10.75 T_{\nu,2}^3 a_2^3$$
$$2T_{\gamma,2}^3 a_2^3 = 5.5 T_{\nu,2}^3 a_2^3$$
$$\frac{T_{\gamma,2}}{T_{\nu,2}} = \left(\frac{5.5}{2}\right)^{1/3}$$
$$= \left(\frac{4}{11}\right)^{-1/3}$$

Therefore, after e^+e^- annihilation, the temperature of neutrinos is that of the photons multiplied by a factor of $\left(\frac{4}{11}\right)^{1/3}$.

REMARK
why nowadays we can observed only electrons but not positrons?

The answer is the small asymmetry between matter and antimatter in the early universe(baryon asymmetry), which means there are slightly more electrons than positrons in the early universe, thus after the annihilation of e^+e^- pairs, there are still some residual electrons left in the universe, which is what we observed today.

we will explain the baryon asymmetry in more detail in later chapters.

5.6.5 Photon Decoupling

At around $T \sim 0.3\,\text{eV}$, which $t \sim 380\,\text{kyr}$, the temperature of the universe drops below the hydrogen binding energy$(\sim 13.6\,\text{eV})$. In this epoch, the universe undergoes **recombination**, which means free electrons and protons combine to form neutral hydrogen atoms.

THEORY
The interaction of photons via Thomson scattering off free electrons is given by:

$$\gamma + e^- \to \gamma' + e^-,$$

with reaction rate:

$$\Gamma_\gamma = n_e \sigma_T c$$

where n_e is the number density of free electrons, σ_T is the Thomson scattering cross section, and c is the speed of light.

And we can see that before recombination, the number density of free electrons is high, thus $\Gamma_\gamma \gg H$, photons are in thermal equilibrium with matter; After recombination, the number density of free electrons drops significantly, thus $\Gamma_\gamma \ll H$, photons decouple from matter and free-stream through the universe.

After decoupling, the photons are no longer interacting with matter, their temperature only influenced by the expansion of universe. And these decoupled photons are what we observe today as the Cosmic Microwave Background (CMB) radiation $T_{\text{CMB}} \approx 2.725\,\text{K}$.

Important Note

The $C\nu B$ is the cosmic neutrino background, while the CMB is the cosmic microwave background, they have different value and do not mix together, as they decouple at different time and have different temperature.

Question Example 5.2: Current Entropy Density of the Universe

In the universe nowadays, the entropy density is dominated by photons bath due to its huge amount compared to other particles.

Given the temperature of CMB is $T_{\text{CMB}} \approx 2.725\,\text{K}$,
(a). What is the effective number of relativistic degrees of freedom ($g_*\&g_{*s}$) in the current universe?
(b). What is the energy density and entropy density of the universe nowadays?
(c). The temperature of $C\nu B$ is $T_\nu \approx 1.95\,\text{K}$, divide it by the temperature of CMB, what you get? What is the physical meaning of this ratio?

Solution.
(a). Given the temperature of CMB is $T_{\text{CMB}} \approx 2.725\,\text{K} \approx 2.348\,\text{eV}$, only photons and neutrinos are still relativistic, thus we only need to consider the contribution of photons and neutrinos to the entropy density of the universe.

Note the DoF of energy density g_* and entropy density g_{*s} are defined as:

$$\begin{cases} \rho = \frac{\pi^2}{30} g_*(T) T^4 \\ s = \frac{2\pi^2}{45} g_{*s}(T) T^3 \end{cases}$$

with:

$$\begin{cases} g_*(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^4 + \frac{7}{8}\sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^4 \\ g_{*s}(T) = \sum_{\text{bosons}} g_i \left(\frac{T_i}{T}\right)^3 + \frac{7}{8}\sum_{\text{fermions}} g_i \left(\frac{T_i}{T}\right)^3 \end{cases}$$

Note $T = T_{\text{CMB}} = T_\gamma$, and $T_\nu = \left(\frac{4}{11}\right)^{1/3} T_{\text{CMB}}$.

By above equations, we can get:

$$\begin{aligned} g_*(T) &= 2 + \frac{7}{8} \times 6 \left(\frac{4}{11}\right)^{4/3} \\ &\approx 3.363, \\ g_{*s}(T) &= 2 + \frac{7}{8} \times 6 \left(\frac{4}{11}\right)^{3/3} \\ &\approx 3.909 \end{aligned}$$

(b). Given $g_*(T)$ and $g_{*s}(T)$, we can get the energy density and entropy density of the universe nowadays:

$$\begin{aligned}\rho &= \frac{\pi^2}{30} g_*(T) T^4 \\ &\approx 61.0\mathrm{K}^4 \\ &\approx 7.012\mathrm{J/m}^3 \\ s &= \frac{2\pi^2}{45} g_{*s}(T) T^3 \\ &\approx 34.70\mathrm{K}^3 \\ &\approx 2.888 \times 10^{-9} \mathrm{m}^{-3}\end{aligned}$$

(c). The ratio of temperature of $C\nu B$ to the temperature of CMB is:

$$\frac{T_\nu}{T_{\mathrm{CMB}}} = \frac{1.95\,\mathrm{K}}{2.725\,\mathrm{K}} \approx 0.716$$

The value is close to $\frac{4}{11}^{1/3} \approx 0.714$, which is the theoretical prediction of the ratio of temperature of $C\nu B$ to the temperature of CMB after e^+e^- annihilation, thus this ratio provides a strong evidence for the standard cosmological model and our understanding of the thermal history of the universe.

5.6.6 Dark Matter Freeze-out

The dark matter is still a hypothetical matter, but if it exists, it should also have a role in thermal history in the early universe.

one of the candidates for dark matter is Weakly Interacting Massive Particles (WIMPs χ), which are particles arised beyond the standard model, and is massive($\sim 10 \sim 1000\,\mathrm{GeV}$) and weakly interacting with other particles.

Theory

In some time of early universe, the WIMPS interact as:

$$\chi + \bar{\chi} \leftrightarrow \text{SM particles}$$

The above reaction is in two-side, the left-to-right reaction is the annihilation of WIMPs into SM particles, while the right-to-left reaction is the production of WIMPs from SM particles.

When universe cools down and energy scale drop below the scales of mass of WIMPs, the creation of WIMPs from SM particles will be exponentially suppressed, thus the right-to-left reaction will be suppressed, while the left-to-right reaction will still continue to annihilate WIMPs into SM particles, thus the number density of WIMPs will drop significantly, and so does the reaction rate Γ_χ.

Extra

why the WIMPS didn't annihilate completely through the time?

The reaction rate of annihilation of WIMPs can be expressed as:

$$\Gamma_\chi = n_\chi \langle \sigma v \rangle,$$

where n_χ is the number density of WIMPs, σ is the cross section of WIMPs annihilation into SM particles, and v is the relative velocity of WIMPs.
When universe expands, the annihilation rate Γ_χ will drop due to the drop of number density n_χ, and when Γ_χ drops below the expansion rate of universe H, the WIMPs will freeze-out, and they will nearly stop annihilating into SM particles, thus there will still be some residual WIMPs left in the universe, which can account for the observed dark matter abundance today.

CHAPTER 6

Big Bang Nucleosynthesis(BBN) (Lecture 13-14)

After electron-positron annihilation, light nuclei can form and survive without being immediately photodissociated by the high-energy photons in the early universe.

This process, known as Big Bang Nucleosynthesis (BBN), occurred within the first few minutes after the Big Bang($T \sim 0.1$ MeV) and is responsible for the formation of light elements such as deuterium, helium-3, helium-4, and lithium-7.

The abundances of these elements provide crucial evidence for the Big Bang model and allow us to test our understanding of the early universe's conditions and evolution.

6.1 Equilibrium

6.1.1 Binding Energy

PHYSICS INTUITION

Due to exist of strong interaction fore, the nucleus is more stable than its constituent nucleons in free state, so if we want to disassemble a nucleus into free protons and neutrons, we need to provide energy to overcome the binding energy of the nucleus, and we call it **binding energy**.

On the other hand, when we said a nucleus is more stable than its constituent nucleons, thats means it is in a lower energy state than the free nucleons, so we will expect the mass of the nucleus m_i is less than the sum of the masses of its constituent protons and neutrons, which is $Z_i m_{^1H} + N_i m_n$.

FOUNDATION

For a nuclei of species i with A_i Atomic Mass Number, it has below relationship:

$$A_i = Z_i + N_i$$

, which Z_i and N_i are the number of protons and neutrons in the nucleus respectively.

THEORY

The binding energy of the nucleus of species i is given by:

$$B_i = Z_i m_{1H} + N_i m_n - m_i$$

where Z_i and N_i are the number of protons and neutrons in the nucleus, m_{1H} is the mass of a hydrogen atom, m_n is the mass of a neutron, and m_i is the mass of the nucleus of species i.

We sometime also use the approximation of $m_{1H} \approx m_p$ where m_p is the mass of a proton, since the difference between the two is negligible for our calculations.

6.1.2 Number Density of Nuclei in Chemical Equilibrium

FOUNDATION

Notes the proton, neutron in this stage are all non-relativistic, so we can use the Maxwell-Boltzmann distribution to describe their number densities. Recall:

$$n_i = g_i \left(\frac{m_i k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-(m_i c^2 - \mu_i)/k_B T}$$

with $g_p = g_n = 2$, we can further rearrange the expression to get:

$$n_i = g_i \left(\frac{m_i T}{2\pi}\right)^{3/2} e^{(\mu_i - m_i)/T}$$
$$e^{\mu_i/T} = \frac{n_i}{g_i} \left(\frac{2\pi}{m_i T}\right)^{3/2} e^{m_i/T}$$

At the sametime, if the nuclear reactions are occur fast enough, we can assume the system is in chemical equilibrium:

$$(A_i - Z_i)n + Z_i p \leftrightarrow i$$

And hence the chemical potentials satisfy:

$$(A_i - Z_i)\mu_n + Z_i \mu_p = \mu_i$$

Sub the chemical potential relation into the number density expression, we can get:

$$
\begin{aligned}
n_i &= g_i \left(\frac{m_i T^{3/2}}{2\pi} \right) e^{\frac{Z_i \mu_p + (A_i - Z_i)\mu_n - m_i}{T}} \\
&= g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{\frac{Z_i \mu_p}{T}} e^{\frac{(A_i - Z_i)\mu_n}{T}} e^{-\frac{m_i}{T}} \\
&= g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} \left(\frac{n_p}{g_p} (\frac{2\pi}{m_p T})^{3/2} e^{m_p/T} \right)^{Z_i} \left(\frac{n_n}{g_n} (\frac{2\pi}{m_n T})^{3/2} e^{m_n/T} \right)^{A_i - Z_i} e^{-m_i/T} \\
&= g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} \frac{n_p}{g_p}^{Z_i} \frac{n_n}{g_n}^{A_i - Z_i} \frac{2\pi}{m_p T}^{3Z_i/2} \frac{2\pi}{m_n T}^{3(A_i - Z_i)/2} e^{(Z_i m_p + (A_i - Z_i) m_n - m_i)/T} \\
&= g_i \left(\frac{m_i T}{2\pi} (\frac{2\pi}{m_p T})^{Z_i} (\frac{2\pi}{m_n T})^{A_i - Z_i} \right)^{3/2} \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) \left(g_p^{-Z_i} g_n^{-(A_i - Z_i)} \right) \left(e^{B_i/T} \right) \\
&= g_i \left((\frac{2\pi}{T})^{A_i - 1} \frac{m_i}{m_p^{Z_i} m_n^{A_i - Z_i}} \right)^{3/2} \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) \left(2^{-Z_i} 2^{-(A_i - Z_i)} \right) \left(e^{B_i/T} \right) \\
&= g_i \left((\frac{2\pi}{T})^{A_i - 1} \frac{m_N A_i}{m_N^{Z_i} m_N^{A_i - Z_i}} \right)^{3/2} \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) \left(2^{-A_i} \right) \left(e^{B_i/T} \right) \\
&= g_i \left((\frac{2\pi}{T})^{A_i - 1} \frac{A_i}{m_N^{A_i - 1}} \right)^{3/2} \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) 2^{-A_i} e^{B_i/T} \\
&= g_i A_i^{3/2} \left(\frac{2\pi}{m_N T} \right)^{\frac{3}{2}(A_i - 1)} \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) 2^{-A_i} e^{B_i/T}
\end{aligned}
$$

where we have used the approximation of $m_p \approx m_n \approx \frac{m_i}{A_i} \equiv m_N$ in the last part.

Theory

The number density of species i can be expressed as:

$$
n_i = g_i \cdot \left(A_i^{3/2} 2^{-A_i} \right) \cdot \left(n_p^{Z_i} n_n^{A_i - Z_i} \right) \cdot \left((\frac{2\pi}{m_N T})^{\frac{3}{2}(A_i - 1)} e^{B_i/T} \right)
$$

where m_N is the average nucleon mass.

6.1.3 Abundance of Nuclei in Chemical Equilibrium

THEORY

The abundance of species i is defined as the ratio of its number density to the total baryon number density:

$$X_i = \frac{A_i n_i}{n_B}$$

where n_B is the total baryon number density, which can be expressed as:

$$n_B = \sum_i A_i n_i = n_p^* + n_n^*$$

where n_p^* and n_n^* are the total number densities of protons and neutrons, respectively, including **both free and nuclei states**. To distinguish between the two, we often use the notation n_p and n_n for the free nucleon densities.

FOUNDATION

Start from equations of number density of a nucleus i:

$$n_i = g_i \cdot \left(A_i^{3/2} 2^{-A_i}\right) \cdot \left(n_p^{Z_i} n_n^{A_i - Z_i}\right) \cdot \left(\left(\frac{2\pi}{m_N T}\right)^{\frac{3}{2}(A_i-1)} e^{B_i/T}\right)$$

Let n_p and n_n be the number density of free protons and neutrons, we can get:

$$\begin{cases} X_i = \frac{A_i n_i}{n_B} \\ X_p = \frac{n_p}{n_B} \\ X_n = \frac{n_n}{n_B} \end{cases}$$

with $n_B = n_p + n_n + \sum_i A_i n_i$.

Substituting the expression of n_i into the number density of species i, we can get:

$$\begin{aligned} \frac{X_i n_B}{A_i} &= g_i \cdot \left(A_i^{3/2} 2^{-A_i}\right) \cdot \left((X_p n_B)^{Z_i} (X_n n_B)^{A_i - Z_i}\right) \cdot \left(\left(\frac{2\pi}{m_N T}\right)^{\frac{3}{2}(A_i-1)} e^{B_i/T}\right) \\ \frac{X_i}{A_i} &= g_i \cdot \left(A_i^{3/2} 2^{-A_i}\right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i}\right) \cdot \left(\left(\frac{2\pi}{m_N T}\right)^{\frac{3}{2}(A_i-1)} e^{B_i/T}\right) n_B^{A_i-1} \\ X_i &= g_i \cdot \left(A_i^{5/2} 2^{-A_i}\right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i}\right) n_B^{A_i-1} \cdot \left(\left(\frac{2\pi}{m_N T}\right)^{\frac{3}{2}(A_i-1)} e^{B_i/T}\right) \end{aligned}$$

Now introduce the dimensionless parameter ϵ that characterizes the nucleon density phase space:

$$\epsilon = \frac{1}{2} n_B \left(\frac{2\pi}{m_N T} \right)^{3/2}$$

By using definition of Baryon-photon ratio $n_B = n_\gamma \eta$, and given the blackbody photon number density $n_\gamma = \frac{2\zeta(3)}{\pi^2} T^3$, we can further express ϵ as:

$$\begin{aligned}
\epsilon &= \frac{1}{2} n_B \left(\frac{2\pi}{m_N T} \right)^{3/2} \\
&= \frac{1}{2} n_\gamma \eta \left(\frac{2\pi}{m_N T} \right)^{3/2} \\
&= \frac{1}{2} \cdot \frac{2\zeta(3)}{\pi^2} T^3 \cdot \eta \cdot \left(\frac{2\pi}{m_N T} \right)^{3/2} \\
&= \frac{\zeta(3)\eta}{\pi^2} \left(\frac{2\pi T}{m_N} \right)^{3/2}
\end{aligned}$$

Hence, we can further rearrange the expression to get:

$$\begin{aligned}
X_i &= g_i \cdot \left(A_i^{5/2} 2^{-A_i} \right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i} \right) \cdot \left(2^{A_i - 1} (\frac{n_B}{2})^{A_i - 1} (\frac{2\pi}{m_N T})^{\frac{3}{2}(A_i - 1)} e^{B_i/T} \right) \\
&= g_i \cdot \left(A_i^{5/2} 2^{-A_i + A_i - 1} \right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i} \right) \cdot \left(\epsilon^{A_i - 1} e^{B_i/T} \right) \\
&= \frac{1}{2} g_i \cdot \left(A_i^{5/2} \epsilon^{A_i - 1} \right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i} \right) \cdot e^{B_i/T}
\end{aligned}$$

THEORY

the relic abundance of species i can be expressed as:

$$X_i = \frac{A_i n_i}{n_B} = \frac{1}{2} g_i \cdot \left(A_i^{5/2} \epsilon^{A_i - 1} \right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i} \right) \cdot e^{B_i/T}$$

with:

$$\epsilon = \frac{1}{2} n_B \left(\frac{2\pi}{m_N T} \right)^{3/2} = \frac{\zeta(3)\eta}{\pi^2} \left(\frac{2\pi T}{m_N} \right)^{3/2}$$

, where ϵ is the dimensionless measure of density in phase space, and η is the baryon-to-photon ratio.

And we can get the relationship of ϵ and η:

$$\epsilon \sim \frac{T}{m_N}^{3/2} \eta$$

Remark

The relic abundance X_i depends on two important factors: the Baryon-to-photon ratio η and the Neutron-to-proton ratio X_n/X_p. The former determines the overall density of baryons in the universe, while the latter determines the relative abundance of neutrons and protons, which in turn affects the formation of different nuclei during BBN.

6.2 Baryon-to-photon ratio

Theory

The baryon-to-photon ratio η is a crucial parameter in cosmology that quantifies the relative abundance of baryons (protons and neutrons) to photons in the universe. It is defined as:

$$\eta = \frac{n_B}{n_\gamma}$$

By the CMB measurements, we have $\eta \approx 6 \times 10^{-10}$, which is a extremely small number.

Remark

In fact, the non-zero value of η is a key piece of evidence for the matter-antimatter asymmetry in the universe, which allow matter to exist and form structures like galaxies, stars, and planets.

If η were zero, we would expect equal amounts of matter and antimatter to annihilate each other, leaving behind a universe filled with radiation and no matter. The observed value of η indicates that there is a slight excess of baryons over antibaryons in the early universe, which has profound implications for our understanding of the fundamental physics governing the universe's evolution.

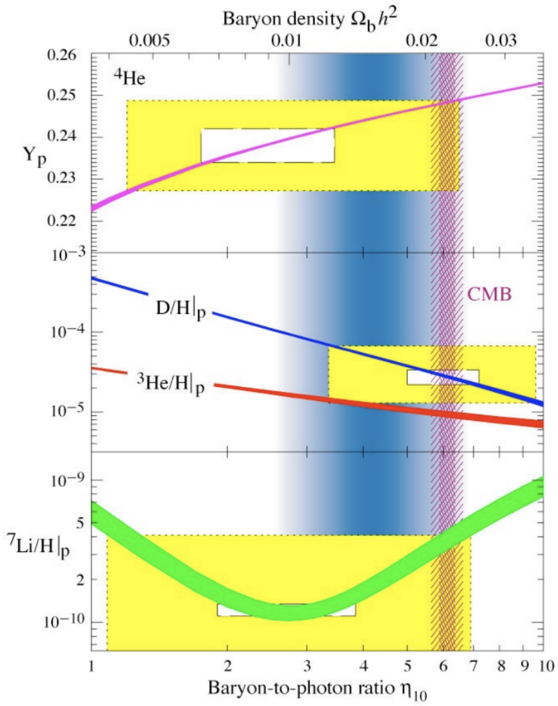

Figure 6.1: Baryon-to-photon ratio η, with x-axis is in unit of 10^{-10}.

EXTRA

From the figure above, the x-axis is different initial condition of universe that with different baryon-to-photon ratio η, and the y-axis is the corresponding relic abundance that we can observe if the universe has that initial condition. The 4 different lines from up to down represent the abundance of helium-4, deuterium, helium-3, and lithium-7 predicted by the BBN theory respectively.

The vertical band in the figure represents the observed value of η from CMB($t \sim 380,000$ years) measurements, which is around 6×10^{-10}. While the yellow/white box represents the observed abundances($t \sim 3 - 20$ minutes) of the 3 light elements (helium-4, deuterium, and lithium-7) from astrophysical observations. The White box represents a precise measurement values, while the yellow box cover a higher confidence level.

We can see that the for the upper two box the theoretical line cross the box at the CMB band, the upper and lower bound of measurement values(i.e. top and bottom edge of the white box) are similar to the value of the line at the CMB band($0.23 - 0.24$ v.s. 0.25 for helium-4, and both around 5×10^{-5} for deuterium); However, for the lithium-7, the value of the line at the CMB band 4×10^{-10} is much higher than the observed value, which is around 1×10^{-10}, and this discrepancy is known as the "lithium problem" in cosmology.

6.3 Neutron-to-Proton Ratio

THEORY

The neutron-proton interaction occurs through weak interaction:

$$n + e^+ \leftrightarrow p + \bar{\nu}_e$$
$$n + \nu_e \leftrightarrow p + e^-$$

Consider the chemical equilibrium of the above reactions, we can get:

$$\mu_n + \mu_{\nu_e} = \mu_p + \mu_{e^-}$$

and by dividing the two number density expressions of neutron and proton, we can get:

$$\frac{n_n}{n_p} = e^{\frac{-(m_n - m_p)}{T} + \frac{\mu_{\nu_e} - \mu_{e^-}}{T}}$$

FOUNDATION

To estimate the chemical potential of neutrino and electron, we assume $T \gg m_e$, and have:

$$\begin{aligned} n_{e^-} - n_{e^+} &= \frac{2}{2\pi^2} \int_0^\infty \left(\frac{1}{e^{\frac{E_{e^-} - \mu_{e^-}}{T}} + 1} - \frac{1}{e^{\frac{E_{e^+} - \mu_{e^+}}{T}} + 1} \right) E^2 dE \\ &= \frac{2T^3}{6\pi^2} \left(\pi^2 \frac{\mu_{e^-}}{T} + \left(\frac{\mu_{e^-}}{T} \right)^3 \right) \end{aligned}$$

and as the universe is electrically neutral, we have $n_{e^-} - n_{e^+} = n_p^*$, and by $n_B = n_p^* + n_n^*$, we can get:

$$\begin{aligned} n_{e^-} - n_{e^+} &= n_p^* < n_p^* + n_n^* \\ n_{e^-} - n_{e^+} &< n_B \\ n_{e^-} - n_{e^+} &< \eta n_\gamma \\ n_{e^-} - n_{e^+} &< \eta \cdot \frac{2\zeta(3)}{\pi^2} T^3 \end{aligned}$$

Now sub the inequality back to the expression of $n_{e^-} - n_{e^+}$, we can get:

$$\begin{aligned} \frac{2T^3}{6\pi^2} \left(\pi^2 \frac{\mu_{e^-}}{T} + \left(\frac{\mu_{e^-}}{T} \right)^3 \right) &< \eta \cdot \frac{2\zeta(3)}{\pi^2} T^3 \\ \frac{1}{6} \left(\pi^2 \frac{\mu_{e^-}}{T} + \left(\frac{\mu_{e^-}}{T} \right)^3 \right) &< \eta \cdot \zeta(3) \\ \frac{\mu_{e^-}}{T} + \frac{1}{\pi^2} \left(\frac{\mu_{e^-}}{T} \right)^3 &< \frac{6\eta}{\pi^2} \zeta(3) \end{aligned}$$

Note that η is around 6×10^{-10}, we can get $\frac{\mu_{e^-}}{T} < 10^{-9}$, which is negligible; At the same time, we know that the neutrino asymmetry is small through observations of CMB and BBN, so through a similar calculation, we can also get $\frac{\mu_{\nu e}}{T} < 10^{-9}$, which is also negligible.

THEORY

If further considering the energy scale of BBN $T \sim 0.1$ MeV, we can just ignore the chemical potential of neutrino and electron, and hence we can get the neutron-to-proton ratio as:

$$\frac{n_n}{n_p} = e^{\frac{-(m_n - m_p)}{T}}$$

where $m_n - m_p \approx 1.293$ MeV.

The reaction continue until expansion of universe outcomes the reaction rate, which happens at around $T \sim 0.7$ MeV, which called "freeze-out" temperature; and at freeze-out, the ratio is around $1/7$ at the start of BBN.

6.4 BBN processes

6.4.1 Deuterium Bottleneck Effect and Start of BBN

PHYSICS INTUITION

All the light nuclei(except hydrogen) are formed through the intermediate state of deuterium(^{2}H), which formed by the reaction:

$$p + n \leftrightarrow {}^2\mathrm{H} + \gamma$$

However, at the early universe, the universe is filled with high-energy photons. Due to the low binding energy of deuterium, these high-energy photons can easily photodissociate the deuterium back into free protons and neutrons, which prevent the formation of heavier nuclei. This phenomenon is known as the **"deuterium bottleneck effect"**.

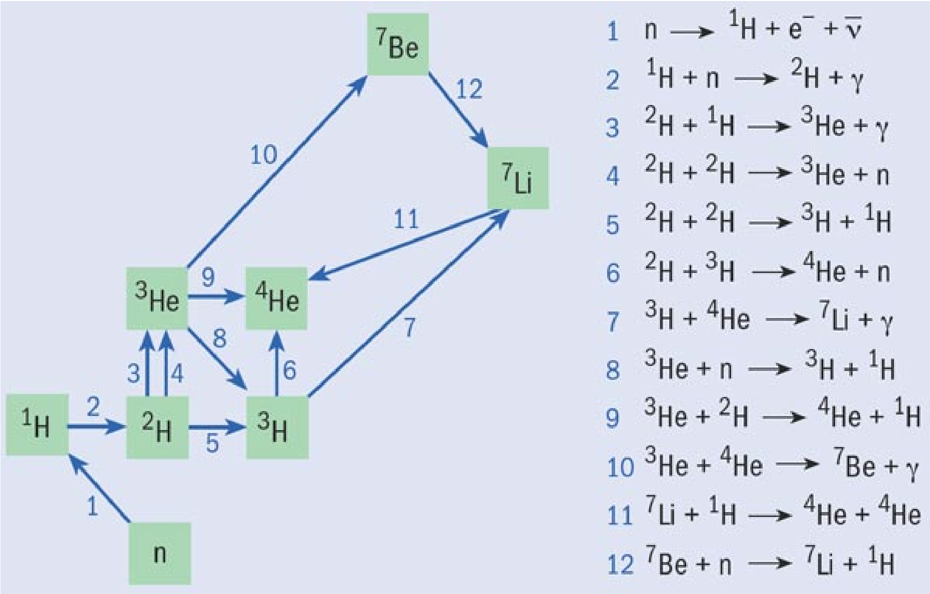

Figure 6.2: BBN interaction chain.

Extra

	B(MeV)	B/A (MeV per u)
^{2}H	2.2245	1.11
^{3}H	8.4820	2.83
^{3}He	7.7186	2.57
^{4}He	28.2970	7.07
^{6}Li	31.9965	5.33
^{7}Li	39.2460	5.61
^{7}Be	37.6026	5.37
^{12}C	92.1631	7.68
^{56}Fe	492.2623	8.79

Theory

As a result, we defined the start of the BBN as the time when large amount of deuterium can survive without being photodissociated, which can be calculated by the relic abundance equation in above section:

$$X_i = \frac{A_i n_i}{n_B} = \frac{1}{2} g_i \cdot \left(A_i^{5/2} \epsilon^{A_i - 1}\right) \cdot \left(X_p^{Z_i} X_n^{A_i - Z_i}\right) \cdot e^{B_i/T}$$

given $B_{^2\mathrm{H}} = 2.2245\,\mathrm{MeV}$, we can get $\epsilon e^{B_{^2\mathrm{H}}/T} \sim 1$ at around $T \sim 0.06 - 0.07\,\mathrm{MeV}$, which is the start of BBN.

FOUNDATION

The detailed steps of getting the above result is as below:

$$X_{^2\mathrm{H}} = \frac{1}{2} g_{^2\mathrm{H}} \cdot \left(A_{^2\mathrm{H}}^{5/2} \epsilon^{A_{^2\mathrm{H}}-1} \right) \cdot \left(X_p^{Z_{^2\mathrm{H}}} X_n^{A_{^2\mathrm{H}}-Z_{^2\mathrm{H}}} \right) \cdot e^{B_{^2\mathrm{H}}/T}$$

By given $g_{^2\mathrm{H}} = 3$, $A_{^2\mathrm{H}} = 2$, $Z_{^2\mathrm{H}} = 1$, and $B_{^2\mathrm{H}} = 2.2245$ MeV, we can further rearrange the expression to get:

$$X_{^2\mathrm{H}} \approx 0.928 \cdot \epsilon \cdot e^{B_{^2\mathrm{H}}/T}$$
$$X_{^2\mathrm{H}} \approx 1 \cdot \frac{\zeta(3)\eta}{\pi^2} \left(\frac{2\pi T}{m_N} \right)^{3/2} e^{B_{^2\mathrm{H}}/T}$$

And using $\eta \approx 6 \times 10^{-10}$, $\zeta(3) \approx 1.202$, and $m_N \approx 939$ MeV:

$$X_{^2\mathrm{H}} \approx 1 \cdot \frac{1.202 \cdot 6 \times 10^{-10}}{\pi^2} \left(\frac{2\pi T}{939} \right)^{3/2} e^{2.2245/T}$$
$$X_{^2\mathrm{H}} \approx 7.3 \times 10^{-11} \cdot 5.474 \times 10^{-4} \cdot T^{3/2} \cdot 9.44^{1/T}$$

Hence, to solve $X_{^2\mathrm{H}} \sim 1$, we get $T \sim 0.0642$ MeV through the above expression.

EXTRA

The history of BBN research began with George Gamow's work in abundances of elements in the universe, he suggested that the early universe was hot and dense enough for nuclear reactions to occur, leading to the formation of light elements. His student Ralph Alpher further developed the theory through calculations. Later, in April 1, 1948, the results was announced in the letter to Physical Review , which signed by Alpher, Bethe, and Gamow, where Hans Bethe's name was added as a pun of "alpha, beta, gamma".

6.4.2 Shutdown of BBN

PHYSICS INTUITION

The shutdown of BBN is mainly due to the expansion and cooling of the universe. As the universe expands, the temperature falls below $T \sim 30$ keV, the energy of light nuclei and protons becomes too low to overcome the Coulomb barrier for nuclear reactions, and hence the nuclear reactions effectively stop, and the heavier nuclei($A > 7$) did not have enough time to form before the universe cools down, which is why we only have light elements from BBN.

6.4.3 Evolution of Elemental Abundances

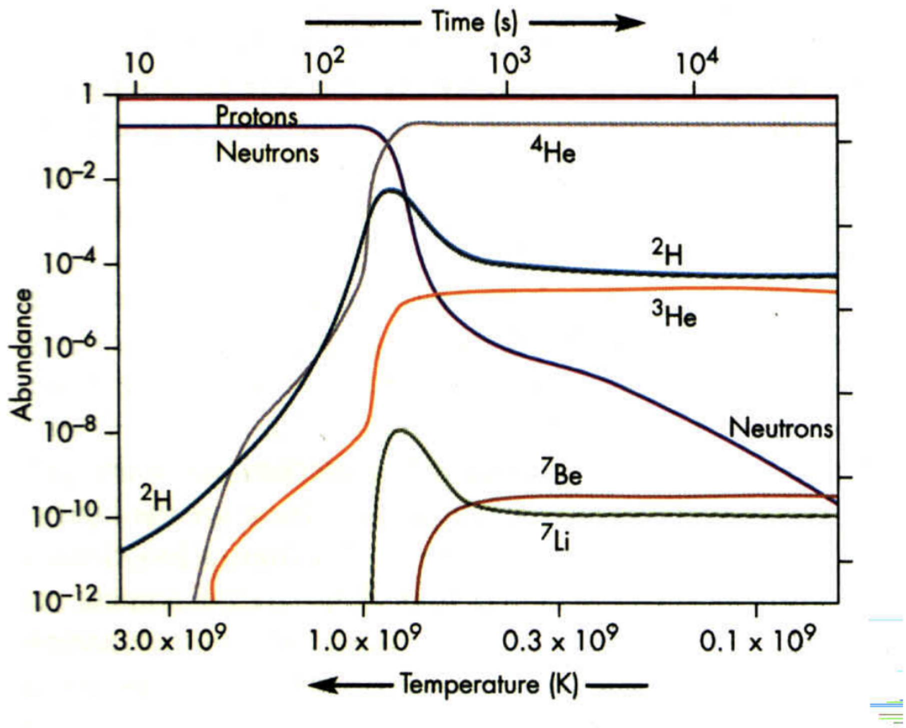

Figure 6.3: Evolution of elemental abundances during BBN.

Physics Intuition

From the figure above, we can have some qualitative understanding of the evolution of elemental abundances during BBN:

- At the beginning of BBN, the neutron-to-proton ratio is around 1/7, but when nuclearsynthesis starts, the neutrons are quickly used up to form light nuclei, and hence the neutron abundance compared to proton abundance drops rapidly; And at the end of BBN, the neutron abundance to proton abundance can be negligible, which is around $10^{-}10$.
- The ^{2}H (deuterium) and ^{3}H (tritium) is intermediate states through reaction towards ^{4}He, so their abundance quickly rise at the beginning of BBN, and reach highest at the time when the reaction rate of forming ^{4}He is the highest, and then quickly drop to a very low level at the end of BBN.
- The ^{3}He is also an intermediate state towards ^{4}He, however, it required ^{2}H or n to do the reactions, which these two reactants are quickly used up at the beginning of BBN, so the abundance of ^{3}He didn't fall as the ^{2}H and ^{3}H, but instead keep a similar level until the end of BBN.

- The ^{4}He is the most stable nucleus among the light nuclei, so almost all neutrons end up in ^{4}He, and hence the abundance of ^{4}He is around 25% at the end of BBN;
- The ^{7}Li is produced through ^{7}Be, and have a reaction $^7\text{Li} + p \leftarrow {}^4\text{He} + {}^4\text{He}$; However, the average binding energy per nucleon of ^{7}Li is around 5.6 MeV, which is much lower than the binding energy per nucleon of ^{4}He, which is around 7.1 MeV, so ^{7}Li tends to return into ^{4}He; As a result, its abundance quickly raised at the beginning of BBN due to high temperature, but then soon drop due to the formation of ^{4}He.
- At the sametime, as the reformation of ^{4}He by ^{7}Be required the middle steps of $^7\text{Be} + n \leftarrow {}^7\text{Li} + p$, but the neutron abundance is quickly used up at the beginning of BBN, so the abundance of ^{7}Be did not drop as the ^{7}Li, but instead keep a similar level until the end of BBN.

Physics Intuition

Now you may have a question that even our observational result match the theoretical relic abundance predicted by BBN, how can we be sure that the observed abundance is indeed the relic abundance from BBN, rather than the abundance produced by some other processes after BBN?(e.g. the abundance of helium-4 can be produced by stars, and the abundance of deuterium can be produced by cosmic ray spallation).

The answer is that we can check the variance of the abundance in different astrophysical environments, for example, all star and interstellar gas contain at least 23% of ^{4}He; To compare, other elements like C,N,O that need to be produced by stars have a significant variance in their abundance in different stars and interstellar gas, with some regions contain almost no C,N,O. The universality minimum abundance of ^{4}He strongly suggest that the observed abundance of ^{4}He is indeed the relic abundance from BBN, rather than the abundance produced by some other processes after BBN.

Another key evidence is the existence of significant amount of deuterium in the universe, as deuterium is fragile and can be easily destroyed due to its low binding energy, the fusion in stars will destroy deuterium rather than produce it, while the cosmic ray spallation can only produce a very small amount of deuterium, therefore the observed abundance of deuterium can only be explained by the relic abundance from BBN, rather than the abundance produced by stars and cosmic ray spallation.

Remark

In fact, the nucleosynthesis process in the early universe suggest that universe is once hot and dense, which is also a key piece of evidence for the Big Bang model.

Foundation

As we know that the neutron is the bottleneck for the formation of light nuclei, and almost all neutrons end up in ^{4}He, so we can estimate the relic abundance of ^{4}He by the neutron-to-proton ratio at the start of BBN:

$$\begin{aligned} X_{^4\mathrm{He}} &= \frac{4n_{^4\mathrm{He}}}{n_B} \\ &\approx \frac{4n_n/2}{n_p + n_n} \\ &= \frac{2n_n}{n_p + n_n} \\ &= \frac{2}{1 + n_p/n_n} \\ &\approx \frac{2}{1 + e^{\frac{m_n - m_p}{T}}} \\ &\approx \frac{2}{1+7} \\ &= \frac{2}{8} \\ &= 0.25 \end{aligned}$$

Note as ^{4}He is formed by 2 protons and 2 neutrons, so the number density of ^{4}He is around half of the neutron number density, and we have $n_{^4\mathrm{He}} \approx \frac{1}{2} n_n$.

Important Note

The chemical equilibrium assumption is only valid at the beginning of BBN; During the stage of BBN, the chemical equilibrium is broken and relic abundance is changed, and we cant simply use the chemical equilibrium expression to calculate the relic abundance;

To calculate the relic abundance during BBN, we need to solving the group of diffusion equations that describe the time evolution of the number density of each species, which is a more complex calculation and require numerical methods to solve. As a matter of fact, the cross sections of these reactions are usually measured in laboratory experiments.

Chapter 7

Cosmic Microwave Background (CMB) (Lecture 15-19)

7.1 Recombination and photon decoupling

7.1.1 Recombination

Physics Intuition

Recombination is the process where free electrons and protons combine to form neutral hydrogen atoms.

$$p + e^- \leftrightarrow H + \gamma$$

This occurs when the universe cools down to about $T \sim 3000$ K, which corresponds to a redshift of $z \sim 1100$.

Before recombination, the universe is a hot plasma that is opaque to photons due to Thomson scattering. After recombination, the universe becomes transparent, allowing photons to travel freely. This is the epoch when the CMB photons were last scattered, and hence we can observe them today as a snapshot of the universe at that time.

Foundation

Recall, the number density of particles i in non-relativistic thermal equilibrium is given by:

$$n_i = g_i \left(\frac{m_i T}{2\pi}\right)^{3/2} e^{-(m_i - \mu_i)/T}$$

Using $m_p \approx m_H$ and $\mu_p + \mu_e = \mu_H$ (as $\mu_\gamma = 0$), we can get the number density of H atoms:

$$\begin{aligned}
n_H &= g_H \left(\frac{m_H T}{2\pi}\right)^{3/2} e^{-(m_H - \mu_H)/T} \\
&= g_H \left(\frac{m_p T}{2\pi}\right)^{3/2} e^{-(m_p + m_e - B - \mu_p - \mu_e)/T} \\
&= \frac{g_H}{g_p g_e} \left(g_p \left(\frac{m_p T}{2\pi}\right)^{3/2} e^{-(m_p - \mu_p)/T} \right) \left(g_e \left(\frac{m_e T}{2\pi}\right)^{3/2} e^{-(m_e - \mu_e)/T} \right) \frac{2\pi}{m_e T}^{3/2} e^{B/T} \\
&= \frac{g_H}{g_p g_e} n_p n_e \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}
\end{aligned}$$

where $B = m_p + m_e - m_H \approx 13.6$ eV is the binding energy of hydrogen.

For a H atom, it is combined by the nucleus (with spin 1/2) and the electron (with spin 1/2), thus it has a total degeneracy spin state as:

$$g_H = g_p + g_e = 2 + 2 = 4$$

Defining the ionization fraction for hydrogen only plasma as:

$$Y_p = \frac{n_p}{n_B}$$

and using the above results of g_H, and approximation $n_H = n_B - n_p = (1 - Y_p)n_B$, we can rewrite it as:

$$(1 - Y_p)n_B = \frac{4}{2 \times 2} n_p n_e \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$
$$(1 - Y_p)n_B = n_p n_e \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$

Note using the charge neutrality condition $n_p = n_e$, and the baryon-to-photon ratio $\eta = n_B/n_\gamma$, we can simplify the equation as:

$$(1 - Y_p)n_B = (Y_p n_B)(Y_p n_B) \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$
$$\frac{1 - Y_p}{Y_p^2} = \eta_\gamma \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$

Further, using the photon number density for a blackbody radiation $n_\gamma = 2\zeta(3)T^3/\pi^2$, we can get the Saha equation for recombination:

$$\frac{1 - Y_p}{Y_p^2} = \eta \frac{2\zeta(3)}{\pi^2} T^3 \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$
$$= \eta \frac{2^{5/2}\zeta(3)}{\pi^{1/2}} \left(\frac{T}{m_e}\right)^{3/2} e^{B/T}$$

THEORY

Under the assumption of chemical equilibrium of the recombination process, we get:

$$n_H = \frac{g_H}{g_p g_e} n_p n_e \left(\frac{2\pi}{m_e T}\right)^{3/2} e^{B/T}$$

And hence the **Saha equation** for ionization fraction of hydrogen is:

$$\frac{1 - Y_p}{Y_p^2} = \eta \frac{2^{5/2}\zeta(3)}{\pi^{1/2}} \left(\frac{T}{m_e}\right)^{3/2} e^{B/T}$$

PHYSICS INTUITION

From the above equations, we know that when $B \ll T$, the RHS $\ll 1$, thus $Y_p \approx 1$, which means almost all the hydrogen is ionized (i.e. all protons and electrons are free).

When temperature drops to $T \ll B$, the ratio of Y_p finally starts to drop. Notice that as $\eta \sim 10^{-9}$ is very small, there are very many photons per baryon, so even when $E > B$, there are still enough photons with energy greater than B to ionize the hydrogen atoms, until $E \ll B$ (around $T \sim 0.3$ eV $\ll 13.6$ eV).

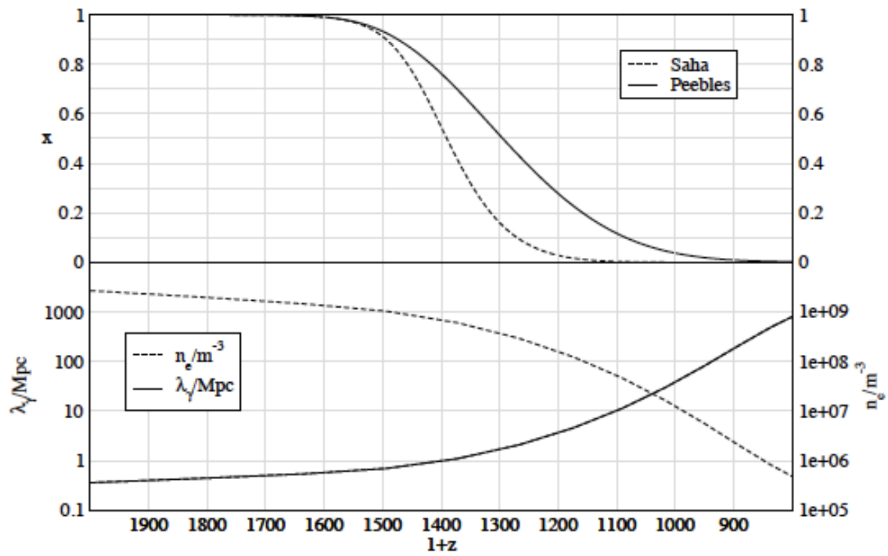

Figure 7.1: The ionization fraction of hydrogen x (upper) and the exact value of $n_e \& \lambda_\gamma$ (lower) as a function of redshift z.

PHYSICS INTUITION

From the above figure, the upper panel show the ionization fraction, the dashed curve represents the Saha equation, which calculate assuming universe is in **perfect chemical equilibrium**, while the solid curve represent the Peebles equation, which more accurate as accounting for the fact that the universe is expanding rapidly and faster than the atoms can recombine, thus the chemical equilibrium is **not perfectly** maintained, and the recombination process is "lagging behind". Hence, the Peebles equation gives a more gentle drop of the ionization fraction, and a slightly later recombination time.

In the lower panel, the dotted curve represents the number density of free electrons n_e, which as the universe expands and recombines, the number density of free electrons drops significantly (from $1e+09$ to $1e+06$); At the same time, the mean free path of photons λ_γ increases dramatically (from 1 Mpc to 1000 Mpc), which means the universe becomes transparent to photons, and the CMB photons can travel freely without being scattered by free electrons.

Theory

Let us define the recombination T_{rec} as the temperature when the ionization fraction drops to $Y_p = 0.5$, which means half of the hydrogen is ionized and half is neutral.

Note that by Wien's law, temperature is inversely proportional to the wavelength, and hence inversely proportional to scale factor, thus we can also define the recombination redshift z_{rec} as:

$$1 + z_{rec} = \frac{a_0}{a_{rec}} = \frac{T_{rec}}{T_0}$$

From the above figure, we can find that:

$$z_{rec} \approx 1300, \quad T_{rec} \approx 0.3 \text{ eV}$$

7.1.2 Photon Decoupling and CMB

Physics Intuition

Although the recombination process starts at $z \sim 1300$, the universe is still not completely transparent to photons, as there are still some free electrons that can scatter photons. The photon decoupling occurs when the mean free path of photons λ_γ becomes larger than the Hubble radius H^{-1}, which in other words, the characteristic interaction become comparable to the expansion rate of the universe $\Gamma_\gamma \sim H$ (**see chapter 5.6.5**).

Thus, the decoupling temperature occurred at $Y_p \sim 0.1$, which is around:

$$T_{dec} \approx 0.26 \text{ eV}, \quad z_{dec} \approx 1100$$

so we can say after the photon decoupling, the "first light" in the universe is emitted at around $z \sim 1100$ with a temperature of $T \sim 3000$ K, and as it no longer interacts with matter, it can travel through the spacetime and reach us today.

However, due to the expansion of the universe, the wavelength of the photons is stretched, and hence the temperature of the light ray is much lower than the original temperature at the time of decoupling, which is now redshifted to the microwave band, which is what we call the Cosmic Microwave Background (CMB) radiation.

THEORY

We can use the relationship between the redshift and temperature to calculate the CMB temperature today. Let $T_{dec} = 0.26$ eV $= 3000$ K and $z_{dec} = 1100$, we can get:

$$T_{CMB} = \frac{T_{dec}}{1 + z_{dec}} \approx 2.727 \text{ K}$$

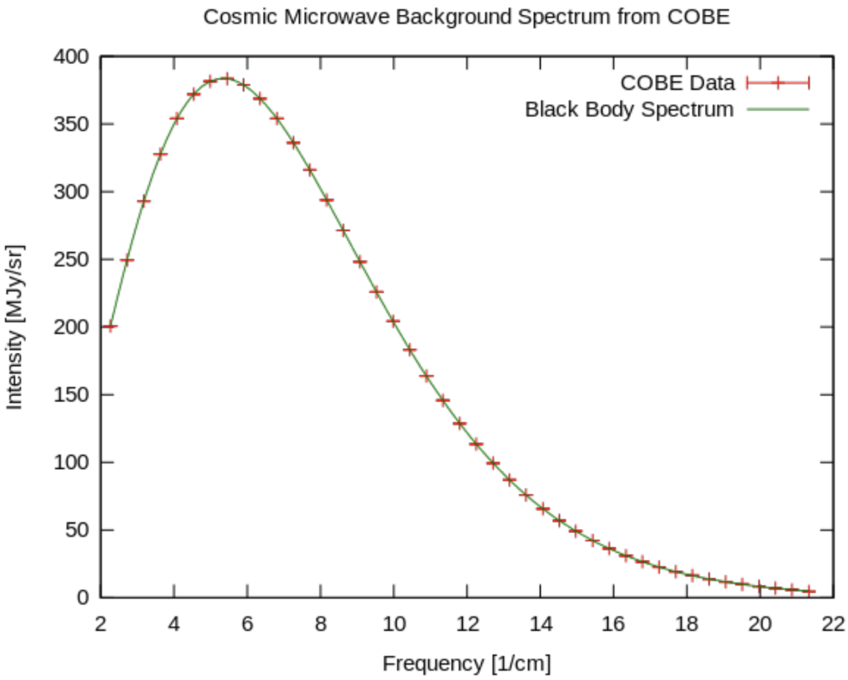

Figure 7.2: The CMB frequency spectrum measured by the COBE satellite with FIRAS instruments.

EXTRA

From the figure, the red crosses (COBE data) shows the actual measurement taken by FIRAS instruments on the COBE satellite, the tiny bars represent the incredibly small error margins; Meanwhile, the green curve represent the theoretical curve calculated by the Planck blackbody radiation formula for perfect blackbody radiation, which is given by:

$$I_\nu = \frac{2h\nu^3}{c^2}\frac{1}{e^{h\nu/k_BT}-1}$$

where I_ν is the intensity of radiation at frequency ν, h is the Planck constant, c is the speed of light, k_B is the Boltzmann constant, and T is the temperature of the blackbody.

We can see that red crosses perfectly match the green curve, which means the CMB radiation is an almost perfect blackbody radiation, and the temperature of the CMB can be determined by fitting the data to the Planck formula, which gives $T \approx 2.725 \pm 0.002$ K with 95% confidence level.

In fact, the result was so significant that it led to John Mather and George Smoot being awarded the Nobel Prize in Physics in 2006 for their work on the COBE satellite and the measurement of the CMB spectrum.

REMARK

The CMB is a snapshot of the universe at the time of photon decoupling, which occurred about 380,000 years after the Big Bang. In fact, it is the oldest light we can observe, and hence is the earliest direct evidence we can "observed".

EXTRA

The CMB was first predicted by Ralph Alpher and Robert Herman in 1948, as a consequence of the Big Bang theory. Later, it was accidentally discovered by Arno Penzias and Robert Wilson in 1965, who were awarded the Nobel Prize in Physics in 1978 for their discovery. The CMB is a crucial piece of evidence for the Big Bang theory, and has been extensively studied by various experiments, such as COBE, WMAP, and Planck satellites.

7.1.3 Dark Ages and Reionization

Physics Intuition

After photon decoupling, the universe becomes transparent to photons, and is still hot ($T \sim 3000$ K), hence showing a uniform red glow colour. After that, the universe become cooler and darker, and the photons are redshifted towards to infrared and microwave bands. As a results, the universe enters a "dark age" as nearly no visible light is emitted.

The dark age last several hundred million years until the first stars and galaxies form ($z \sim 10$), which emit UV photons that reionize the intergalactic medium. This process is called **reionization**. Note that in this stage the gas cloud already has a very low density, thus the universe remains transparent to photons, and the CMB photons can still reach us without being scattered by the gas.

Remark

In fact, there is no direct observation of the dark ages currently, astrophysicists are still trying different methods to probe this era, for example, using 21-cm line observations, which can trace the neutral hydrogen gas before reionization. Overall, the dark ages still remains a large observational gap in our understanding of cosmic history.

7.2 CMB anisotropies

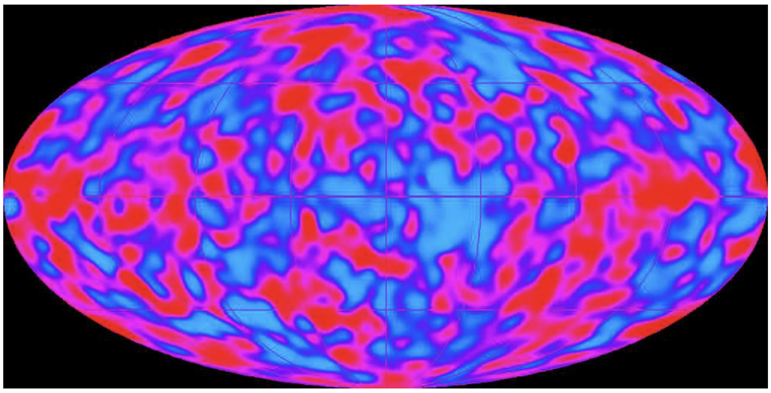

Figure 7.3: The CMB anisotropies measured by the Planck satellite.

Physics Intuition

When the COBE satellite first returned the data of the CMB, Scientists found out that there are tiny fluctuations in the temperature of the CMB across the sky, which is called **CMB anisotropies**. These anisotropies are on the order of 10^{-5} K, which means the temperature of the CMB is not perfectly uniform, but has tiny variations.

The anisotropies provide crucial information about the early universe, as they are the seeds of the large scale structure we see today. The tiny fluctuations in the CMB temperature correspond to the density fluctuations in the early universe, which eventually grow into galaxies and clusters of galaxies due to gravitational instability.

Theory

Define the temperature fluctuation of any point (θ, ϕ) on the sky as:

$$\frac{\Delta T(\theta,\phi)}{T_0} = \frac{T(\theta,\phi) - T_0}{T_0} \sim 10^{-5}$$

where T_0 is the average temperature of the CMB:

$$T_0 = \frac{1}{4\pi}\int T(\theta,\phi) d\Omega \approx 2.725 \text{ K}$$

with $d\Omega = \sin\theta d\theta d\phi$ is the solid angle element.

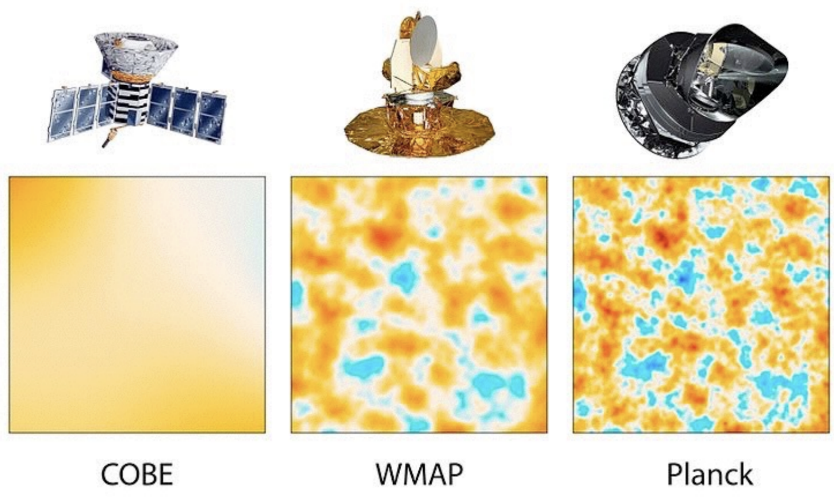

Figure 7.4: The CMB anisotropies measured by the COBE, WMAP and Planck satellites.

EXTRA

The anisotropies were first detected by the COBE satellite in 1992, however, as the angular resolution of COBE was quite low (about 7 degrees), it could only detect the anisotropies on scales larger than that.

To get a higher resolution map of the CMB anisotropies, NASA launched the WMAP satellite in 2001, which had an angular resolution of about 13 arcminutes, and hence could detect the anisotropies on smaller scales. The WMAP data provided a much more detailed map of the CMB anisotropies, and allowed scientists to measure the cosmological parameters with much higher precision.

After that, the Planck satellite was launched in 2009, which had an even higher angular resolution of about 5 arcminutes, and hence could detect the anisotropies on even smaller scales. The Planck data provided the most detailed map of the CMB anisotropies to date, and allowed scientists to measure the cosmological parameters with unprecedented precision.

7.2.1 Spherical Harmonic Decomposition

PHYSICS INTUITION

Recall Fourier decomposition, when we want to analyze a function $y = f(x)$ defined on a line, we can decompose it into a sum of sine and cosine functions with different frequencies, which is called Fourier decomposition.

In the case of CMB anisotropies, we can use a similar method to decompose the temperature fluctuations on the sky, but this time is defined on a sphere, thus we need to use spherical harmonics instead of sine and cosine functions.

FOUNDATION

Just like Fourier decomposition, we can approximate the CMB by a sum of periodic functions, which in this case are the spherical harmonics $Y_{lm}(\theta, \phi)$, where l is the degree of the spherical harmonic, and m is the order of the spherical harmonic.

In details, the l represent how many times the function oscillates, in other words, $2\times$ the number of crossing zeros(nodal lines); while the m represent how many times the function oscillates in the azimuthal direction, in details, the number of nodal lines in the azimuthal direction(meridians) is $|m|$, and the number of nodal lines in the polar direction(parallels) is $l - |m|$.

FOUNDATION

Let $Y_{lm}(\theta, \phi)$ be the spherical harmonic function, we want it have below properties:

- Smooth everywhere
- Single-valued, which means $Y_{lm}(\theta, \phi) = Y_{lm}(\theta, \phi + 2\pi)$
- Orthonormal, which means $\int Y_{lm}(\theta, \phi) Y^*_{l'm'}(\theta, \phi) d\Omega = \delta_{ll'}\delta_{mm'}$

In a spherical coordinates, the Laplacian operator of smooth angular functions can be written as:

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial}{\partial \theta}\right) + \frac{1}{r^2 \sin^2\theta}\frac{\partial^2}{\partial \phi^2}$$

As we only care about the angular part of the function, we can ignore the radial part, and hence we can write the angular part of the Laplacian operator as:

$$\begin{aligned} \nabla^2_{\text{angular}}(Y) &= \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial Y}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial \phi^2} \\ -\lambda Y &= \frac{1}{\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta \frac{\partial Y}{\partial \theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2 Y}{\partial \phi^2} \end{aligned}$$

where λ is the eigenvalue of the Laplacian operator.
And further separation of variables gives us:

$$\begin{aligned} Y(\theta, \phi) &= \Theta(\theta)\Phi(\phi) \\ -\lambda\Theta\Phi &= \frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right)\Phi + \frac{1}{\sin^2\theta}\Theta\frac{d^2\Phi}{d\phi^2} \\ \frac{\sin\theta}{\Theta}\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + \lambda\sin^2\theta &= -\frac{1}{\Phi}\frac{d^2\Phi}{d\phi^2} \end{aligned}$$

As the LHS only depends on θ and the RHS only depends on ϕ, to make the equation hold for all θ and ϕ, both sides must be equal to a constant, which we can denote as m^2. Hence, we can get two separate equations:

$$\begin{aligned} \frac{d^2\Phi}{d\phi^2} + m^2\Phi &= 0 \\ \sin\theta \frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + (\lambda\sin^2\theta - m^2)\Theta &= 0 \end{aligned}$$

First, we can solve the ϕ part of the equation, which gives us:

$$\Phi(\phi) = Ae^{im\phi} + Be^{-im\phi}$$

where A and B are constants. To satisfy the single-valued condition, we need $\Phi(\phi) = \Phi(\phi + 2\pi)$, which gives us:

$$Ae^{im\phi} + Be^{-im\phi} = Ae^{im(\phi+2\pi)} + Be^{-im(\phi+2\pi)} \implies m \in \mathbb{Z}$$

Hence, with the fact that m is integer, we can get the solution for $\Phi(\phi)$ as:

$$\Phi(\phi) = e^{im\phi}$$

Next, we will handle the θ part of the equation, let $x = \cos\theta$, we can rewrite the equation as:

$$\begin{aligned}
\sin\theta \frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + (\lambda \sin^2\theta - m^2)\Theta &= 0 \\
\frac{d}{d\theta}\left(\sin\theta \frac{d\Theta}{d\theta}\right) + \left(\lambda - \frac{m^2}{\sin^2\theta}\right)\Theta &= 0 \\
\frac{d}{dx}\left((1-x^2)\frac{d\Theta}{dx}\right) + \left(\lambda - \frac{m^2}{1-x^2}\right)\Theta &= 0 \\
(1-x^2)\frac{d^2\Theta}{dx^2} - 2x\frac{d\Theta}{dx} + \left(\lambda - \frac{m^2}{1-x^2}\right)\Theta &= 0
\end{aligned}$$

Which is the associated Legendre equation.

We then introduce the constraints of setting the at $theta = 0$ (i.e. poles of sphere), the function should be finite, which means $\Theta(\theta = 0) < \infty$, which only happens when $\lambda = l(l+1)$, where l is a non-negative integer and $|m| \leq l$. Hence, we can get the solution for $\Theta(\theta)$ as:

$$\Theta(\theta) = P_l^m(\cos\theta)$$

where $P_l^m(x)$ is the associated Legendre polynomial, which is defined as:

$$P_l^m(x) = (-1)^m(1-x^2)^{m/2}\frac{d^m}{dx^m}P_l(x)$$

where $P_l(x)$ is the Legendre polynomial of degree l, which is defined as:

$$P_l(x) = \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l$$

Finally, to keep the spherical harmonics orthonormal, we need to introduce a normalization constant N_{lm}, which is given by:

$$N_{lm} = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}$$

Theory

we can decompose the temperature fluctuations into spherical harmonics as:

$$\frac{\Delta T(\theta,\phi)}{T_0} = \sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm} Y_{lm}(\theta,\phi)$$

where a_{lm} are the coefficients of the decomposition.

And the complex form of the general equation for spherical harmonic is:

$$Y_{lm}(\theta,\phi) = N_{lm} P_l^m(\cos\theta) e^{im\phi}$$

where:

- l is the degree of the spherical harmonic, which determines the angular scale of the fluctuation, with $l = 0, 1, 2, \ldots$;
- m is the order of the spherical harmonic, which determines the azimuthal dependence of the fluctuation, with $m = -l, -l+1, \ldots, 0, \ldots, l-1, l$;
- $P_l^m(\cos\theta)$ is the associated Legendre polynomial, which is a function of $\cos\theta$ and depends on l and m:

$$P_l^m(x) = (-1)^m (1-x^2)^{m/2} \frac{d^m}{dx^m} P_l(x)$$

  where $P_l(x)$ is the Legendre polynomial of degree l:

$$P_l(x) = \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l$$

- N_{lm} is the normalization constant, which ensures that the spherical harmonics are orthonormal:

$$N_{lm} = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}$$

Foundation

Recall in Fourier decomposition, the sine and cosine functions form an orthonormal basis for functions defined on a line. Now lets check if the spherical harmonics $Y_{lm}(\theta,\phi)$ also form an orthonormal basis for functions defined on the sphere with below inner product:

$$\int Y_{lm}(\theta,\phi) Y^*_{l'm'}(\theta,\phi) d\Omega = \delta_{ll'}\delta_{mm'}$$

where $\delta_{ll'}$ and $\delta_{mm'}$ are the Kronecker delta functions, and the result should be zero if any 2 different basis functions are multiplied together, and should be one if the same basis function is multiplied together. Hence, by proving the above equation, we can show that the spherical harmonics form an orthonormal basis for functions defined on the sphere.

Starts from L.H.S, we can get:

$$\begin{aligned} I &= \int_0^{2\pi}\int_0^{\pi} Y_{lm}(\theta,\phi)Y^*_{l'm'}(\theta,\phi)\sin\theta d\theta d\phi \\ &= \int_0^{2\pi}\int_0^{\pi} N_{lm}P_l^m(\cos\theta)e^{im\phi}N_{l'm'}P_{l'}^{m'}(\cos\theta)e^{-im'\phi}\sin\theta d\theta d\phi \\ &= N_{lm}N_{l'm'}\int_0^{2\pi} e^{i(m-m')\phi}d\phi\int_0^{\pi} P_l^m(\cos\theta)P_{l'}^{m'}(\cos\theta)\sin\theta d\theta \end{aligned}$$

Now, we can evaluate the first integral:

- If $m = m'$, then the first integral becomes $\int_0^{2\pi} d\phi = 2\pi$;
- If $m \neq m'$, then the first integral becomes:

$$\begin{aligned} \int_0^{2\pi} e^{i(m-m')\phi}d\phi &= \left[\frac{e^{i(m-m')\phi}}{i(m-m')}\right]_0^{2\pi} \\ &= \frac{1}{i(m-m')}(e^{i(m-m')2\pi}-1) \\ &= 0 \end{aligned}$$

And we can write the general result of the first integral as:

$$\int_0^{2\pi} e^{i(m-m')\phi}d\phi = 2\pi\delta_{mm'}$$

Then, for the second integral, we can use the orthogonality of the associated Legendre polynomials, which states that the Legendre polynomials are also orthogonal:

$$\begin{aligned} \int_0^{\pi} P_l^m(\cos\theta)P_{l'}^{m'}(\cos\theta)\sin\theta d\theta &= \int_{-1}^{1} P_l^m(x)P_{l'}^{m'}(x)dx \\ &= \frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}\delta_{mm'} \end{aligned}$$

Combined the above results, we can get:

$$\begin{aligned} I &= N_{lm}N_{l'm'}\int_0^{2\pi} e^{i(m-m')\phi}d\phi\int_0^{\pi} P_l^m(\cos\theta)P_{l'}^{m'}(\cos\theta)\sin\theta d\theta \\ &= N_{lm}N_{l'm'}\cdot 2\pi\cdot\frac{2}{2l+1}\frac{(l+m)!}{(l-m)!}\delta_{ll'}\delta_{mm'} \end{aligned}$$

Because when $l \neq l'$ or $m \neq m'$, the result is zero, we can just focus on the case when $l = l'$ and $m = m'$, which gives us:

$$\begin{aligned} I &= N_{lm}^2 \cdot 2\pi \cdot \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \\ &= \frac{(2l+1)}{4\pi} \frac{(l-m)!}{(l+m)!} \cdot 2\pi \cdot \frac{2}{2l+1} \frac{(l+m)!}{(l-m)!} \\ &= 1 \end{aligned}$$

Hence, we have shown that the spherical harmonics form an orthonormal basis for functions defined on the sphere:

$$\int Y_{lm}(\theta,\phi) Y^*_{l'm'}(\theta,\phi) d\Omega = \delta_{ll'} \delta_{mm'}$$

Further, we can used the general addition theorem for spherical harmonics to get the closure relation:

$$P_l(\cos\gamma) = \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}(\theta_1,\phi_1) Y^*_{lm}(\theta_2,\phi_2)$$

where γ is the angle between the two points (θ_1, ϕ_1) and (θ_2, ϕ_2) on the sphere. Setting $\gamma = 0$, we can get:

$$\begin{aligned} P_l(1) &= \frac{4\pi}{2l+1} \sum_{m=-l}^{l} Y_{lm}(\theta,\phi) Y^*_{lm}(\theta,\phi) \\ 1 &= \frac{4\pi}{2l+1} \sum_{m=-l}^{l} |Y_{lm}(\theta,\phi)|^2 \\ \frac{2l+1}{4\pi} &= \sum_{m=-l}^{l} |Y_{lm}(\theta,\phi)|^2 \end{aligned}$$

This is the **Unsöld's theorem**, which in physical meaning, states that the sum of the squares of the spherical harmonics with the same degree l is a constant, which is independent of the angles θ and ϕ. This theorem is a direct consequence of the orthonormality of the spherical harmonics, and it shows a rotational symmetry in the spherical harmonics, which is a fundamental property of the functions defined on the sphere.

THEORY

The functions $Y_{lm}(\theta,\phi)$ form an orthonormal basis for functions defined on the sphere, which:

$$\int Y_{lm}(\theta,\phi)Y^*_{l'm'}(\theta,\phi)d\Omega = \delta_{ll'}\delta_{mm'}$$

Therefore, summing over m for a fixed l gives us the closure relation:

$$\sum_{m=-l}^{l} |Y_{lm}(\theta,\phi)|^2 = \frac{2l+1}{4\pi}$$

FOUNDATION

Start from the complex form of the spherical harmonics:

$$Y_{lm}(\theta,\phi) = N_{lm}P_l^m(\cos\theta)\left(\cos(m\phi) + i\sin(m\phi)\right)$$

Next, we can get:

$$\begin{aligned}
P_l^{-m}(x) &= \frac{(-1)^{-m}}{2^l l!}(1-x^2)^{-m/2}\frac{d^{l-m}}{dx^{l-m}}(x^2-1)^l \\
&= \frac{(-1)^{-m}}{2^l l!}(1-x^2)^{-m/2}\left((-1)^{-m}\frac{(l-m)!}{(l+m)!}(1-x^2)^m\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l\right) \\
&= \frac{1}{2^l l!}(1-x^2)^{m/2}\frac{(l-m)!}{(l+m)!}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l \\
&= \frac{(l-m)!}{(l+m)!}\left(\frac{1}{2^l l!}(1-x^2)^{m/2}\frac{d^{l+m}}{dx^{l+m}}(x^2-1)^l\right) \\
&= \frac{(l-m)!}{(l+m)!}\left((-1)^m P_l^m(x)\right) \\
&= (-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x)
\end{aligned}$$

And hence:

$$\begin{aligned}
N_{l,-m}P_l^{-m}(x) &= N_{l,-m}(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x) \\
&= \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-(-m))!}{(l+(-m))!}}(-1)^m\frac{(l-m)!}{(l+m)!}P_l^m(x) \\
&= \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}(-1)^m P_l^m(x) \\
&= N_{lm}P_l^m(x)(-1)^m
\end{aligned}$$

Then, we get the $Y_{l,-m}(\theta,\phi)$ in terms of conjugate of $Y_{lm}(\theta,\phi)$:

$$\begin{aligned} Y_{l,-m}(\theta,\phi) &= N_{l,-m}P_l^{-m}(\cos\theta)e^{im\phi} \\ &= (-1)^m N_{lm}P_l^m(\cos\theta)e^{-im\phi} \\ &= (-1)^m Y_{lm}^*(\theta,\phi) \end{aligned}$$

Or we can said that $\forall m \in$, we have:

$$Y_{l,m}(\theta,\phi) = N_{l,|m|}P_l^{|m|}(\cos\theta)e^{-im\phi}$$

However, up to now, we only still get the complex form with:

$$\begin{cases} Y_{lm}(\theta,\phi) = N_{lm}P_l^m(\cos\theta)e^{im\phi}, & m \geq 0 \\ Y_{l,-m}(\theta,\phi) = N_{l,-m}P_l^{-m}(\cos\theta)e^{-im\phi}, & m < 0 \end{cases}$$

with $A_{l,|m|}(\theta) = N_{l,|m|}P_l^{|m|}(\cos\theta)$ is the amplitude of the spherical harmonic.

To get the real form of the spherical harmonics, we can take the linear combination of $Y_{lm}(\theta,\phi)$ and $Y_{l,-m}(\theta,\phi)$:

$$\begin{cases} Y_{lm}(\theta,\phi) + Y_{l,-m}(\theta,\phi) & = A_{l,|m|}(\theta)\left(e^{im\phi} + e^{-im\phi}\right) = 2A_{l,|m|}(\theta)\cos(m\phi) \\ Y_{lm}(\theta,\phi) - Y_{l,-m}(\theta,\phi) & = A_{l,|m|}(\theta)\left(e^{im\phi} - e^{-im\phi}\right) = 2iA_{l,|m|}(\theta)\sin(m\phi) \end{cases}$$

And then we take the magnitude of the above 2 functions and normalize them to get the new basis functions:

$$\begin{cases} 2A_{l,|m|}(\theta)\cos(m\phi) \to \sqrt{2}N_{l,|m|}P_l^{|m|}(\cos\theta)\cos(m\phi), & m > 0 \\ 2iA_{l,|m|}(\theta)\sin(m\phi) \to \sqrt{2}N_{l,|m|}P_l^{|m|}(\cos\theta)\sin(|m|\phi), & m < 0 \end{cases}$$

Some may now ask do we miss the $m = 0$ case? Actually, for $m = 0$, we have:

$$Y_{l0}(\theta,\phi) = N_{l0}P_l^0(\cos\theta)e^{i\cdot 0\cdot\phi} = N_{l0}P_l^0(\cos\theta)$$

Therefore, we get the real form of the spherical harmonics in 3 cases!!

THEORY

As temperature fluctuation $\Delta T(\theta,\phi)/T_0$ is always a real function, to have a easier analysis, we can defined the real spherical harmonics as:

$$Y_{lm}(\theta,\phi) = \begin{cases} \sqrt{2}N_{l|m|}P_l^{|m|}(\cos\theta)\sin(|m|\phi), & m < 0 \\ N_{l0}P_l^0(\cos\theta), & m = 0 \\ \sqrt{2}N_{lm}P_l^m(\cos\theta)\cos(m\phi), & m > 0 \end{cases}$$

where N_{lm} is the normalization factor:

$$N_{lm} = \sqrt{\frac{(2l+1)}{4\pi}\frac{(l-m)!}{(l+m)!}}$$

And the associated Legendre polynomial $P_l^m(x)$ is defined as:

$$P_l^m(x) = (-1)^m(1-x^2)^{m/2}\frac{d^m}{dx^m}P_l(x)$$

which is related to the Legendre polynomial $P_l(x)$ by:

$$P_l(x) = \frac{1}{2^l l!}\frac{d^l}{dx^l}(x^2-1)^l$$

FOUNDATION

Start with the expression and by orthogonality of the spherical harmonics, we can get the coefficients a_{lm} as:

$$\begin{aligned}
\frac{\Delta T(\theta,\phi)}{T_0} &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm}Y_{lm}(\theta,\phi) \\
Y^*_{l'm'}(\theta,\phi)\frac{\Delta T(\theta,\phi)}{T_0} &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm}Y^*_{l'm'}(\theta,\phi)Y_{lm}(\theta,\phi) \\
\int Y^*_{l'm'}(\theta,\phi)\frac{\Delta T(\theta,\phi)}{T_0}d\Omega &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm}\int Y^*_{l'm'}(\theta,\phi)Y_{lm}(\theta,\phi)d\Omega \\
\int Y^*_{l'm'}(\theta,\phi)\frac{\Delta T(\theta,\phi)}{T_0}d\Omega &= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm}\delta_{ll'}\delta_{mm'} \\
\int Y^*_{l'm'}(\theta,\phi)\frac{\Delta T(\theta,\phi)}{T_0}d\Omega &= a_{l'm'}
\end{aligned}$$

THEORY

Besides, we can express the coefficients a_{lm} as:

$$a_{lm} = \int Y^*_{lm}(\theta,\phi)\frac{\Delta T(\theta,\phi)}{T_0}d\Omega$$

The a_{lm} are dimensionless, but sometimes we also defined as $T_0 a_{lm}$, which has the unit of temperature (i.e. in units of μK, with μ is the micro symbol, which means 10^{-6}).

Besides, as Y_{00} is constant and only contribute to the isotropic part of the CMB(the average temperature), we can ignore the $l=0$ term and let $a_{00}=0$ for simplicity.

Extra

The below three figures show the real spherical harmonics $Y_{lm}(\theta, \phi)$ in different representations. The first figure shows the 2D map of the spherical harmonics, where the colour represents the value of the function at each point on the sphere. The second figure shows the polar plot of the spherical harmonics, where the colour represents the value of the function at each point on the sphere. The third figure shows the radial plot of the spherical harmonics, where the radius represents the value of the function at each point on the sphere, and the angle represents the azimuthal direction.

For interested readers, you can check an interactive online tool to visualize the spherical harmonics in different representations: `https://automatoc-ti.github.io/Spherical-Harmonics-Visualizer/`.

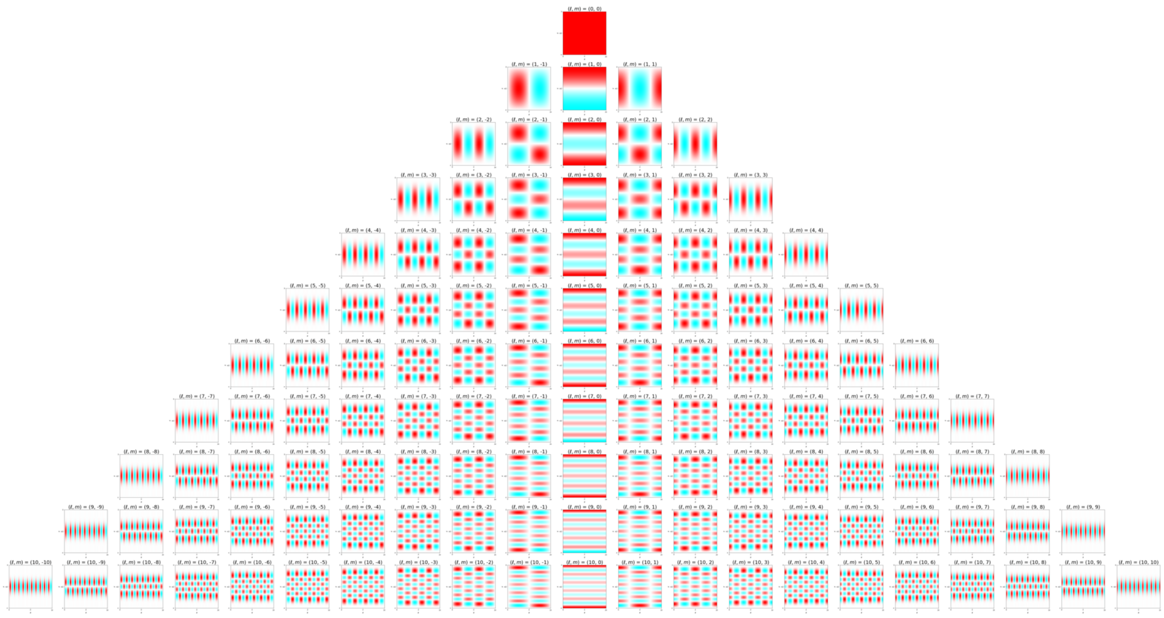

Figure 7.5: The real spherical harmonics $Y_{lm}(\theta, \phi)$ in the 2D map.

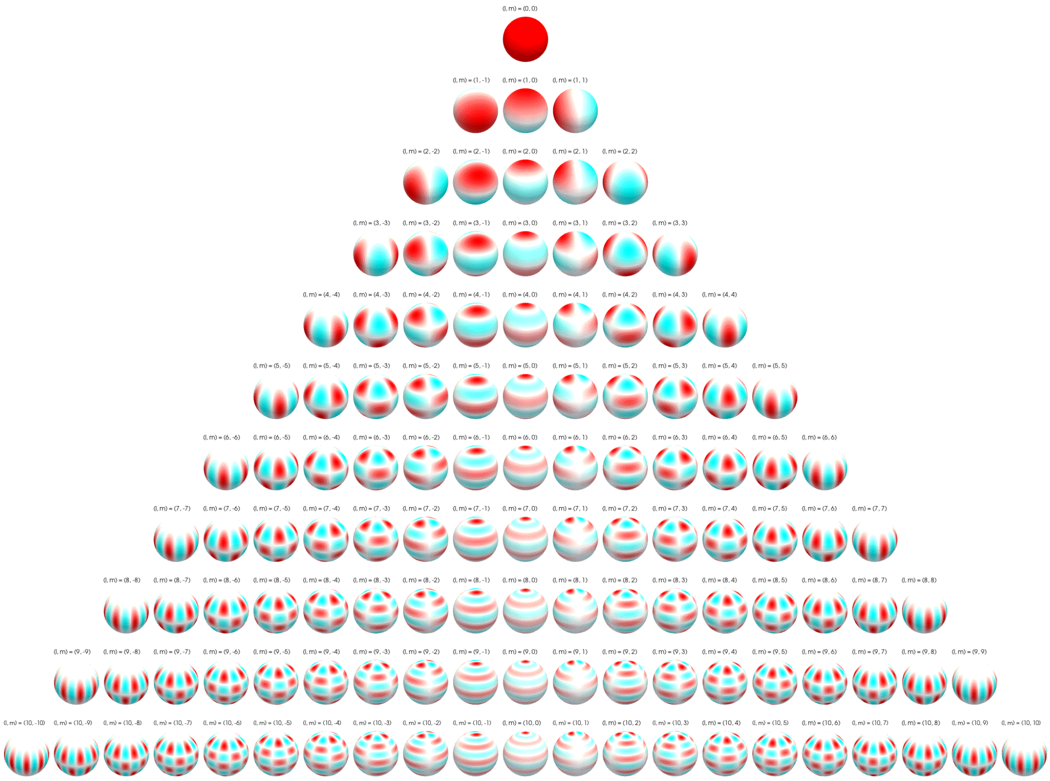

Figure 7.6: The real spherical harmonics $Y_{lm}(\theta, \phi)$ in the polar plot.

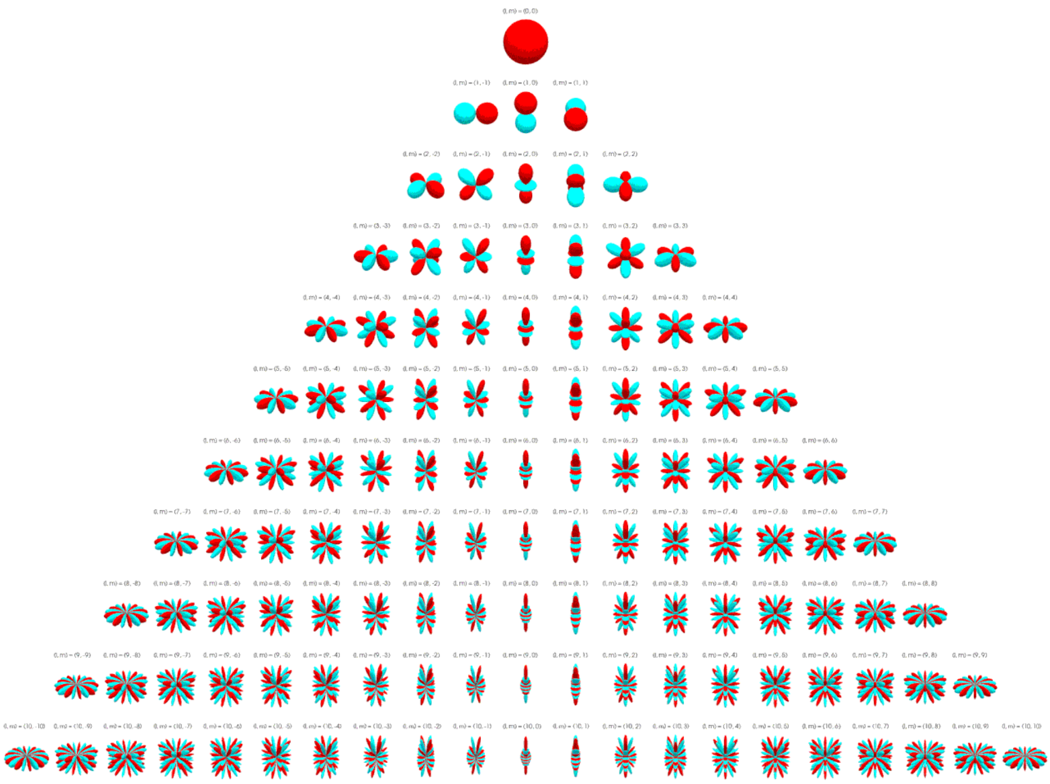

Figure 7.7: The real spherical harmonics $Y_{lm}(\theta, \phi)$ in the radial plot.

7.2.2 Theoretical Power Spectrum

Physics Intuition

In the previous section, we have shown that the temperature fluctuations can be described by the spherical harmonics, now it is time to analyze the statistical properties of the temperature fluctuations, which is the power spectrum.

The **angular power spectrum** represents the variance of the temperature fluctuations exists in a specific angular scale, or in other words, it tells us "how rough" the CMB map when we look at patches of the sky with a specific angular size $\theta \approx 180^\circ/l$. Note that by cosmological principle, the CMB should be statistically isotropic, so we expect all orientations m at the same scale l should have the same average power spectrum C_l, and can be written as:

$$C_l \equiv \frac{1}{2l+1} \sum_{m=-l}^{l} \langle |a_{lm}|^2 \rangle$$

which we sum over all m for a fixed l to get the average power spectrum at scale l.

Further, we can let $\langle |a_{l,-l}|^2 \rangle = \cdots = \langle |a_{l0}|^2 \rangle = \cdots = \langle |a_{l,l}|^2 \rangle$ due to the isotropy of the CMB, and hence we can get:

$$C_l = \langle |a_{lm}|^2 \rangle$$

Remark

The power spectrum C_l we defined here is a **theoretical power spectrum**, which is the average power spectrum we expect to see in the CMB map, thus in the equation, we used the bracket $\langle \cdot \rangle$ to represent the average over all possible realizations of the CMB map.

Foundation

By the cosmological principle, for a different combination of l and m, their coefficients a_{lm} are independent and identically distributed random variables, which we have below properties:

- $\langle a_{lm} \rangle = 0$, which means the average of the coefficients is zero, which is expected as the temperature fluctuations should be around the average temperature T_0;
- if $l \neq l'$ or $m \neq m'$, then $\langle a_{lm} a^*_{l'm'} \rangle = \langle a_{lm} \rangle \langle a^*_{l'm'} \rangle = 0$, which means the coefficients are uncorrelated with each other;
- if $l = l'$ and $m = m'$, then $\langle a_{lm} a^*_{l'm'} \rangle = \langle |a_{lm}|^2 \rangle = C_l$, which means the variance of the coefficients is equal to the power spectrum at scale l.

Therefore, we can get the average temperature variance at scale l as:

$$\begin{aligned}
\langle \left(\frac{\Delta T}{T_0}\right)^2 \rangle &= \langle \left(\sum_{l=0}^{\infty}\sum_{m=-l}^{l} a_{lm}Y_{lm}(\theta,\phi)\right)\left(\sum_{l'=0}^{\infty}\sum_{m'=-l'}^{l'} a^*_{l'm'}Y^*_{l'm'}(\theta,\phi)\right)\rangle \\
&= \sum_{l=0}^{\infty}\sum_{m=-l}^{l}\sum_{l'=0}^{\infty}\sum_{m'=-l'}^{l'} \langle a_{lm}a^*_{l'm'}\rangle Y_{lm}(\theta,\phi)Y^*_{l'm'}(\theta,\phi) \\
&= \sum_{l=0}^{\infty}\sum_{m=-l}^{l} C_l Y_{lm}(\theta,\phi)Y^*_{lm}(\theta,\phi) \\
&= \sum_{l=0}^{\infty} C_l \sum_{m=-l}^{l} |Y_{lm}(\theta,\phi)|^2 \\
&= \sum_{l=0}^{\infty} C_l \cdot \frac{2l+1}{4\pi} \\
&= \frac{1}{4\pi}\sum_{l=0}^{\infty}(2l+1)C_l
\end{aligned}$$

Theory

The average temperature variance at scale l can be expressed as:

$$\langle \left(\frac{\Delta T}{T_0}\right)^2 \rangle = \frac{1}{4\pi}\sum_{l=0}^{\infty}(2l+1)C_l$$

with $C_l = \langle |a_{lm}|^2 \rangle$ is the theoretical power spectrum at scale l.

7.2.3 Observed Power Spectrum

Physics Intuition

However, in reality, we can only observe one realization of the CMB map, which means we can only get one set of coefficients a_{lm}, and hence we can only get one set of power spectrum. Therefore, we need to define the **observed power spectrum** as:

$$\hat{C}_l = \frac{1}{2l+1}\sum_{m=-l}^{l} |a_{lm}|^2$$

which is the power spectrum we can get from the observed CMB map.

FOUNDATION

Again, we can calculate the variance of the temperature fluctuations at scale l using the observed power spectrum:

$$\begin{aligned}
\frac{1}{4\pi}\int\left(\frac{\Delta T}{T_0}\right)^2 d\Omega &= \frac{1}{4\pi}\int\left(\sum_{l=0}^{L}\sum_{m=-l}^{l} a_{lm}Y_{lm}(\theta,\phi)\right)\left(\sum_{l'=0}^{L}\sum_{m'=-l'}^{l'} a^*_{l'm'}Y^*_{l'm'}(\theta,\phi)\right)d\Omega \\
&= \frac{1}{4\pi}\sum_{l=0}^{L}\sum_{m=-l}^{l}\sum_{l'=0}^{L}\sum_{m'=-l'}^{l'} a_{lm}a^*_{l'm'}\int Y_{lm}(\theta,\phi)Y^*_{l'm'}(\theta,\phi)d\Omega \\
&= \frac{1}{4\pi}\sum_{l=0}^{L}\sum_{m=-l}^{l}|a_{lm}|^2 \\
&= \frac{1}{4\pi}\sum_{l=0}^{L}(2l+1)\hat{C}_l
\end{aligned}$$

Note here actually $a_{lm}a^*_{l'm'} \neq 0$ when $l \neq l'$ or $m \neq m'$ in a single realization, the reason why here we can just ignore those terms and simplified the summation into $\sum_{l=0}^{L}\sum_{m=-l}^{l}|a_{lm}|^2$ is because the orthogonality of the spherical harmonic basis. It isolates the diagonal terms and hence we can just focus on the diagonal terms when we calculate the variance of the temperature fluctuations.

THEORY

The variance of the temperature fluctuations at scale l can be expressed as:

$$\frac{1}{4\pi}\int\left(\frac{\Delta T}{T_0}\right)^2 d\Omega = \frac{1}{4\pi}\sum_{l=0}^{L}(2l+1)\hat{C}_l$$

with $\hat{C}_l = \frac{1}{2l+1}\sum_{m=-l}^{l}|a_{lm}|^2$ is the observed power spectrum at scale l. And L is the maximum scale we can observe, which is determined by the resolution of the CMB map.

IMPORTANT NOTE

The **observed power spectrum** $\hat{C}_l$ is an estimator of the **theoretical power spectrum** C_l, which means $\hat{C}_l$ is a random variable that depends on the specific realization of the CMB map we observed, be aware that $\hat{C}_l$ is not equal to C_l, and there is a statistical uncertainty in the estimation of C_l by $\hat{C}_l$, which is called **cosmic variance**.

7.2.4 Cosmic variance

PHYSICS INTUITION

Now lets define the variance of the difference between the observed power spectrum $\hat{C}_l$ and the theoretical power spectrum C_l as:

$$\sigma_{C_l}^2 = \langle (\hat{C}_l - C_l)^2 \rangle$$

which is the variance of the estimation of C_l by $\hat{C}_l$. This variance is called **cosmic variance**, which is the fundamental statistical uncertainty in the estimation of the power spectrum due to the fact that we can only observe one realization of the universe.

By intuition, we can expect that the cosmic variance should be larger at larger scales (smaller l) because we have fewer independent modes to average over, and hence the estimation of C_l is more uncertain. On the other hand, at smaller scales (larger l), we have more independent modes to average over, and hence the estimation of C_l is more accurate, which means the cosmic variance should be smaller at smaller scales.

FOUNDATION

Start with the definition of the cosmic variance:

$$\begin{aligned}
\sigma_{C_l}^2 &= \langle (\hat{C}_l - C_l)^2 \rangle \\
&= \langle \hat{C}_l^2 - 2\hat{C}_l C_l + C_l^2 \rangle \\
&= \langle \hat{C}_l^2 \rangle - 2C_l \langle \hat{C}_l \rangle + C_l^2 \\
&= \langle \hat{C}_l^2 \rangle - C_l^2
\end{aligned}$$

Since:

$$\langle \hat{C}_l \rangle = \langle \frac{1}{2l+1} \sum_{m=-l}^{l} |a_{lm}|^2 \rangle = C_l$$

we can write $\langle \hat{C}_l^2 \rangle$ as:

$$\langle \hat{C}_l^2 \rangle = \frac{1}{(2l+1)^2} \sum_{m=-l}^{l} \sum_{m'=-l}^{l} \langle |a_{lm}|^2 |a_{lm'}|^2 \rangle$$

By assuming the CMB fluctuations are Gaussian random fields, we can get the following properties of the coefficients a_{lm}:

- $m = 0$ case, we have $\langle |a_{l0}|^2 \rangle = \sigma^2 = C_l$, and hence we can get:

$$
\begin{aligned}
\langle |a_{l0}|^4 \rangle &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{\infty} x^4 e^{-\frac{x^2}{2\sigma^2}} dx \\
&= 3\sigma^4 \\
&= 3C_l^2
\end{aligned}
$$

- $m \neq 0$ case, let $a_{lm} = x + iy$, where x and y are independent Gaussian random variables with mean 0 and variance $\sigma^2 = C_l/2$, we can get:

$$
\begin{aligned}
\langle |a_{lm}|^4 \rangle &= \langle (x^2 + y^2)^2 \rangle \\
&= \langle x^4 + 2x^2y^2 + y^4 \rangle \\
&= 3\sigma^4 + 2\sigma^4 + 3\sigma^4 \\
&= 8\sigma^4 \\
&= 2C_l^2
\end{aligned}
$$

And we get:

$$
\begin{aligned}
\langle |a_{lm}|^2 \rangle &= C_l \\
\langle |a_{lm}|^4 \rangle &= \begin{cases} 3C_l^2, & m = 0 \text{ (real case)} \\ 2C_l^2, & m \neq 0 \text{ (complex case)} \end{cases}
\end{aligned}
$$

We can further simplify $\langle \hat{C}_l^2 \rangle$ in 2 cases:

- For $m = m'$,
  - if $m = 0$, then $\langle |a_{l0}|^4 \rangle = 3C_l^2$; The number of zero diagonal term m is 1.
  - if $m \neq 0$, then $\langle |a_{lm}|^4 \rangle = 2C_l^2$; The number of non-zero diagonal terms m is $2l$.
- For $m \neq m'$, we have $\langle |a_{lm}|^2 |a_{lm'}|^2 \rangle = \langle |a_{lm}|^2 \rangle \langle |a_{lm'}|^2 \rangle = C_l^2$; The number of off-diagonal terms is $(2l+1)^2 - 1 - 2l$.

Hence, we can get:

$$
\begin{aligned}
\sum_{m=-l}^{l} \sum_{m'=-l}^{l} \langle |a_{lm}|^2 |a_{lm'}|^2 \rangle &= (3C_l^2) \cdot 1 + (2C_l^2) \cdot 2l + C_l^2 \cdot ((2l+1)^2 - 1 - 2l) \\
&= (3 + 4l + 4l^2 + 4l + 1 - 1 - 2l) C_l^2 \\
&= (4l^2 + 6l + 3) C_l^2 \\
&= (2l+3)(2l+1) C_l^2
\end{aligned}
$$

As a result, we get $\langle \hat{C}_l^2 \rangle$ as:

$$\begin{aligned}\langle \hat{C}_l^2 \rangle &= \frac{1}{(2l+1)^2} \sum_{m=-l}^{l} \sum_{m'=-l}^{l} \langle |a_{lm}|^2 |a_{lm'}|^2 \rangle \\ &= \frac{1}{(2l+1)^2} (2l+3)(2l+1) C_l^2 \\ &= \frac{2l+3}{2l+1} C_l^2\end{aligned}$$

Therefore, the cosmic variance is:

$$\begin{aligned}\sigma_{C_l}^2 &= \langle (\hat{C}_l - C_l)^2 \rangle \\ &= \langle \hat{C}_l^2 \rangle - C_l^2 \\ &= \left(\frac{2l+3}{2l+1} - 1 \right) C_l^2 \\ &= \frac{2}{2l+1} C_l^2\end{aligned}$$

THEORY

The cosmic variance can be expressed as:

$$\sigma_{C_l}^2 = \langle (\hat{C}_l - C_l)^2 \rangle = \frac{2}{2l+1} C_l^2$$

And the equation matches our intuition that the cosmic variance is larger at larger scales (smaller l) and smaller at smaller scales (larger l).

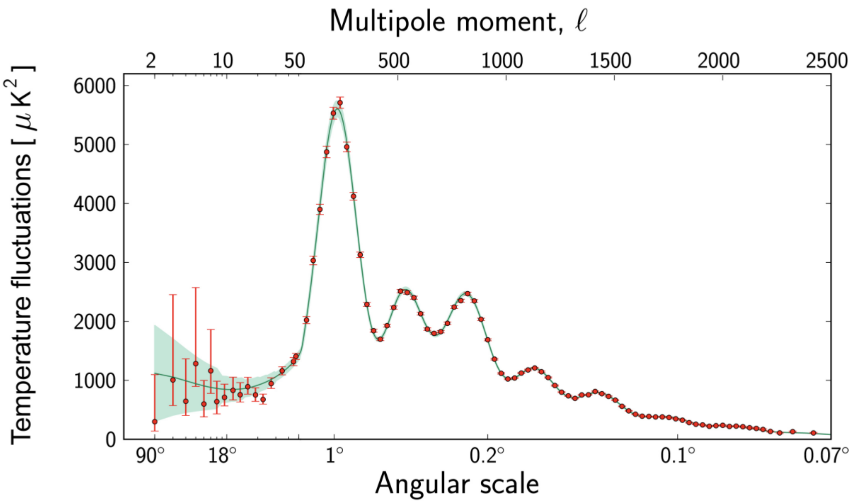

Figure 7.8: The angular power spectrum of the CMB

Extra

The above figure shows the angular power spectrum of the CMB. The x-axis represents the multipole moment l (top) and the corresponding angular scale θ (bottom), related approximately by $\theta \approx 180^\circ / l$. The y-axis represents the scaled power spectrum, typically plotted as $\frac{l(l+1)C_l}{2\pi}$, which quantifies the variance of temperature fluctuations at scale l.

The red dots with error bars represent the observed power spectrum $\hat{C}_l$ derived from CMB maps, while the green solid line represents the theoretical power spectrum predicted by the best-fit ΛCDM model. The green shaded region represents **cosmic variance**, the fundamental statistical uncertainty arising because we can only observe one realization of the universe.

On the far left (low l), the curve is relatively flat. This is the **Sachs-Wolfe plateau**, which corresponds to super-horizon scales (larger than the causal horizon at recombination). These fluctuations are primordial, originating from quantum fluctuations stretched by inflation.

In the middle region ($30 < l < 1000$), we see a series of **acoustic peaks**. These correspond to sound waves (oscillations) in the photon-baryon fluid before recombination.

- The **First Peak** ($l \approx 200$) represents the fundamental mode of oscillation—the maximum compression of the fluid. Its position at 1° indicates that the universe is geometrically **flat**.
- The **Second Peak** represents the first rarefaction (expansion) phase.
- The **Third Peak** represents the second compression phase. The relative heights of these peaks (specifically the ratio of odd to even peaks) provide a precise measurement of the **baryon density** (Ω_b) of the universe.

On the far right ($l > 1000$), the curve drops sharply. This is the **damping tail** (or Silk damping). On these very small scales, photons could diffuse through the plasma before recombination, effectively "smoothing out" the temperature fluctuations and erasing the signal.

Further details about the meaning of above features will be discussed in next few sections, but here we just want to give a brief introduction about the shape of the angular power spectrum and its physical meaning.

7.3 Acoustic Oscillations and Acoustic Peaks

7.3.1 Damped Harmonic Oscillator

FOUNDATION

Recall about the equation of motion for a damped harmonic oscillator, in such a system, the total force is given 2 components: the restoring force $F_r = -kx$ and the damping force by friction $F_d = -b\frac{dx}{dt}$, where k is the stiffness factor and b is the viscous damping coefficient. The equation of motion can be written as:

$$F = m\frac{d^2x}{dt^2} = -kx - b\frac{dx}{dt}$$

And can be rearranged into the standard form:

$$\frac{d^2x}{dt^2} + 2\zeta\omega_0\frac{dx}{dt} + \omega_0^2 x = 0$$

where:

$$\omega_0 = \sqrt{\frac{k}{m}}$$

is the natural angular frequency of the undamped system, and

$$\zeta = \frac{b}{2\sqrt{km}}$$

is the damping ratio.

THEORY

By solving the damped harmonic oscillator equation, we can get the general solution as:

$$x(t) = Ae^{-\zeta\omega_0 t}\sin(\sqrt{1-\zeta^2}\omega_0 t + \phi)$$

where A is the amplitude of the oscillation, and ϕ is the phase constant. The term $e^{-\zeta\omega_0 t}$ represents the exponential decay of the amplitude due to damping, while the term $\sin(\sqrt{1-\zeta^2}\omega_0 t + \phi)$ represents the oscillatory behavior of the system.

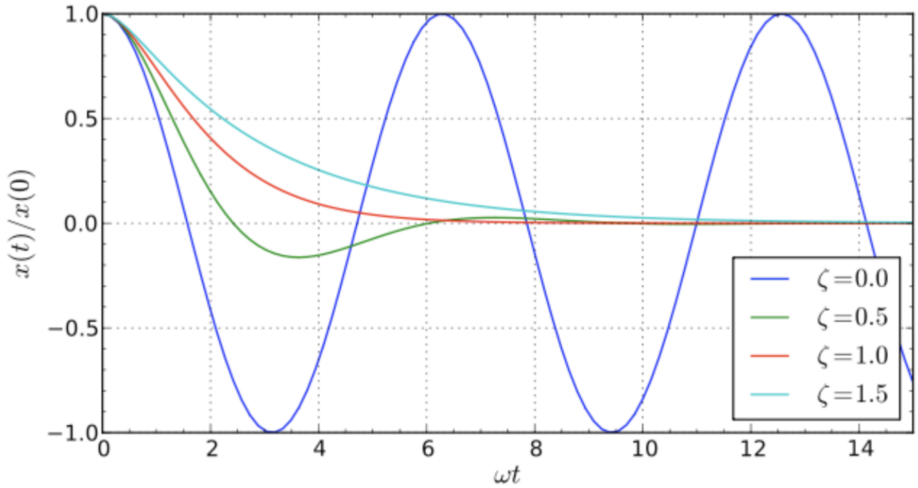

Figure 7.9: The motion of a damped harmonic oscillator for different values of the damping ratio ζ.

Physics Intuition

There is 4 types of damped harmonic oscillator:

- **Undamped** ($\zeta = 0$): the system oscillates indefinitely with a constant amplitude and a natural frequency ω_0. A common example is a mass-spring system without any friction.
- **Underdamped** ($\zeta < 1, \zeta > 0$): the system oscillates with the amplitude gradually decreasing to zero over time. Besides, it oscillates(with a slightly lower frequency than the natural frequency ω_0) because the damping is not strong enough to stop the oscillation immediately:

$$\omega_u = \sqrt{1 - \zeta^2}\omega_0$$

  A common example is a mass-spring system with a small amount of friction.
- **Critically damped** ($\zeta = 1$): the system returns to equilibrium as quickly as possible without oscillating. It does not oscillate because the damping is just strong enough to prevent any oscillation. A common example is a door closer that brings the door to a stop without bouncing.
- **Overdamped** ($\zeta > 1$): the system returns to equilibrium in exponentially decaying without oscillating, but more slowly than in the critically damped case. The larger the damping ratio ζ, the slower the system returns to equilibrium. A common example is a car shock absorber that is too stiff, causing the car to return to its normal position very slowly after hitting a bump.

Remark

We ignore the case of $\zeta < 0$ because it shows a system that gain energy from the environment, which will diverge and hence is not stable.

7.3.2 Acoustic Oscillations in CMB

Physics Intuition

Before recombination, the universe is filled with a hot, dense plasma of photons, electrons, and baryons. The photons are tightly coupled to the electrons through Thomson scattering, and the electrons are coupled to the baryons through Coulomb interactions. This photon-baryon fluid can support sound waves (acoustic oscillations) due to the interplay between the pressure of the photons and the gravitational attraction of the baryons.

The acoustic oscillations in the photon-baryon fluid before recombination can be described by a damped harmonic oscillator equation, where the restoring force is provided by the pressure of the photons, and the damping is caused by the expansion of the universe and the diffusion of photons (Silk damping). The solutions to this equation give rise to the characteristic peaks in the CMB power spectrum, known as acoustic peaks.

Finally, after recombination, the photons decouple and the acoustic oscillations are frozen in, leaving an imprint on the CMB that we can observe today.

Foundation

considering a **fixed volume** bounded by surface, we can get the total number of photons in the volume as:

$$N_\gamma(t) = \int_V n_\gamma(\vec{x},t)dV = \int_V n_\gamma(x,t)d^3x$$

At the same time, the net change of the number of photons in the volume is given by the flux of photons across the surface:

$$\frac{dN_\gamma}{dt} = -\oint_S n_\gamma(\vec{x},t)\vec{v}\cdot d\vec{S}$$

Then we can write the 2 equations together as:

$$\begin{aligned}\frac{d}{dt}\int_V n_\gamma(\vec{x},t)d^3x &= -\oint_S n_\gamma(\vec{x},t)\vec{v}\cdot d\vec{S}\\ \int_V \frac{\partial n_\gamma(\vec{x},t)}{\partial t}d^3x &= -\oint_S n_\gamma(\vec{x},t)\vec{v}\cdot d\vec{S}\end{aligned}$$

By applying the divergence theorem:

$$\oint_S n_\gamma(\vec{x},t)\vec{v}\cdot d\vec{S} = \int_V \nabla\cdot(n_\gamma(\vec{x},t)\vec{v})d^3x$$

we have:

$$\begin{aligned}\int_V \frac{\partial n_\gamma(\vec{x},t)}{\partial t}d^3x &= -\int_V \nabla\cdot(n_\gamma(\vec{x},t)\vec{v})d^3x\\ \int_V\left(\frac{\partial n_\gamma(\vec{x},t)}{\partial t}+\nabla\cdot(n_\gamma(\vec{x},t)\vec{v})\right)d^3x &= 0\end{aligned}$$

Since the above equation holds for any arbitrary volume, we can get the continuity equation for the number density of photons as:

$$\frac{\partial n_\gamma(\vec{x},t)}{\partial t}+\nabla\cdot(n_\gamma(\vec{x},t)\vec{v}) = 0$$

Foundation

The above equation assume a fixed volume, but we know that the **universe is expanding**, so we need to consider the scale factor in the equation:

$$N_\gamma(t) = n_\gamma(t)a^3(t) = \text{constant}$$

Then by conservation of the number of photons, we can get:

$$\begin{aligned}\frac{dN_\gamma}{dt} &= 0\\ \frac{d}{dt}(n_\gamma(t)a^3(t)) &= 0\\ \dot{n}_\gamma(t)a^3(t)+3n_\gamma(t)a^2(t)\dot{a}(t) &= 0\\ \dot{n}_\gamma(t)+3n_\gamma(t)\frac{\dot{a}(t)}{a(t)} &= 0\\ \dot{n}_\gamma(t)+3n_\gamma(t)H(t) &= 0\end{aligned}$$

And substituting the above equation into the continuity equation, we can get:

$$\begin{aligned}\frac{dN_\gamma}{dt} &= -\oint_S n_\gamma(\vec{x},t)\vec{v}\cdot d\vec{S}\\ \int_V\left(\frac{\partial\left(n_\gamma(\vec{x},t)a^3(t)\right)}{\partial t}+\nabla\cdot(n_\gamma(\vec{x},t)\vec{v})\right)d^3x &= 0\\ \int_V\left(\dot{n}_\gamma(t)+3n_\gamma(t)H(t)+\nabla\cdot(n_\gamma(\vec{x},t)\vec{v})\right)d^3x &= 0\\ \dot{n}_\gamma(t)+3n_\gamma(t)H(t)+\nabla\cdot(n_\gamma(\vec{x},t)\vec{v}) &= 0\end{aligned}$$

Theory

The continuity equation for the number density of photons in an expanding universe can be expressed as:

$$\dot{n}_\gamma(t) + 3n_\gamma(t)H(t) + \nabla \cdot (n_\gamma(\vec{x}, t)\vec{v}) = 0$$

where $n_\gamma(t)$ is the number density of photons, $H(t)$ is the Hubble parameter, and $\vec{v}$ is the velocity field of the photons.

In a special case of static universe where $H(t) = 0$, the above equation reduces to the standard continuity equation:

$$\dot{n}_\gamma(t) + \nabla \cdot (n_\gamma(\vec{x}, t)\vec{v}) = 0$$

Foundation

As we only interested in the small scale perturbations of the number density of photons, we can linearize the number desnity of photons as:

$$n_\gamma(\vec{x}, t) \approx \bar{n}_\gamma(t) + \delta n_\gamma(\vec{x}, t)$$

where $\bar{n}_\gamma(t)$ is the average number density of photons, and $\delta n_\gamma(\vec{x}, t)$ is the perturbation of the number density of photons. Substituting the above equation into the continuity equation, we can get:

$$\dot{\bar{n}}_\gamma(t) + \dot{\delta n}_\gamma(\vec{x}, t) + 3(\bar{n}_\gamma(t) + \delta n_\gamma(\vec{x}, t))H(t) + \nabla \cdot ((\bar{n}_\gamma(t) + \delta n_\gamma(\vec{x}, t))\vec{v}) = 0$$
$$\dot{\bar{n}}_\gamma(t) + 3\bar{n}_\gamma(t)H(t) + \dot{\delta n}_\gamma(\vec{x}, t) + 3\delta n_\gamma(\vec{x}, t)H(t) + \nabla \cdot (\bar{n}_\gamma(t)\vec{v}) + \nabla \cdot (\delta n_\gamma(\vec{x}, t)\vec{v}) = 0$$

Since $\bar{n}_\gamma(t)$ is the average number density of photons, it should satisfy the background continuity equation:

$$\dot{\bar{n}}_\gamma(t) + 3\bar{n}_\gamma(t)H(t) = 0$$

we can simplify the above equation into:

$$\dot{\delta n}_\gamma(\vec{x}, t) + 3\delta n_\gamma(\vec{x}, t)H(t) + \nabla \cdot (\bar{n}_\gamma(t)\vec{v}) + \nabla \cdot (\delta n_\gamma(\vec{x}, t)\vec{v}) = 0$$

Furthermore, consider the last 2 terms in the above equation:

- $\nabla \cdot (\bar{n}_\gamma(t)\vec{v})$: since $\bar{n}_\gamma(t)$ is only a function of time, we can take it out of the divergence operator, and we have $\nabla \cdot (\bar{n}_\gamma(t)\vec{v}) = \bar{n}_\gamma(t)\nabla \cdot \vec{v}$. We keep this term because $\nabla \cdot \vec{v}$ is a first order perturbation, and hence it is not negligible in the linear approximation.
- $\nabla \cdot (\delta n_\gamma(\vec{x}, t)\vec{v})$: since both $\delta n_\gamma(\vec{x}, t)$ and $\vec{v}$ are small perturbations, we can ignore the second order term $\nabla \cdot (\delta n_\gamma(\vec{x}, t)\vec{v})$ in the linear approximation.

At the sametime, we compare the magnitude of the remaining 3 terms, we know that $\dot{\delta n_\gamma}(\vec{x},t)$ is the time derivative of the perturbation, which is expected to be of the same order of magnitude as $\delta n_\gamma(\vec{x},t)$;
And $3\delta n_\gamma(\vec{x},t)H(t)$ is the expansion term, which is expected to be smaller than $\dot{\delta n_\gamma}(\vec{x},t)$ because the expansion of the universe is relatively slow compared to the time evolution of the perturbations. Therefore, we can also ignore the expansion term $3\delta n_\gamma(\vec{x},t)H(t)$ in the linear approximation.

As a result, we can simplify the above equation into:

$$
\begin{aligned}
\dot{\delta n_\gamma}(\vec{x},t) + \bar{n}_\gamma(t)\nabla\cdot\vec{v} &\approx 0 \\
\dot{\delta n_\gamma}(\vec{x},t) &\approx -\bar{n}_\gamma(t)\nabla\cdot\vec{v} \\
\frac{\dot{\delta n_\gamma}(\vec{x},t)}{\bar{n}_\gamma(t)} &\approx -\nabla\cdot\vec{v}
\end{aligned}
$$

Recall that $n_\gamma = \frac{2\zeta(3)}{\pi^2}T^3$, we can get $\bar{n}_\gamma(t) = CT^3(t)$, and hence we can write the above equation as:

$$
\frac{\delta n_\gamma(\vec{x},t)}{\bar{n}_\gamma(t)} = \frac{3CT_0^2(t)\delta T(\vec{x},t)}{CT_0^3(t)} = 3\frac{\delta T(\vec{x},t)}{T_0(t)} \equiv 3\Theta(\vec{x},t)
$$

Substituting the above equation into the previous equation, we can get:

$$
\dot{\Theta}(\vec{x},t) \approx -\frac{1}{3}\nabla\cdot\vec{v}
$$

Theory

By considering the small scale perturbations of the number density of photons in an expanding universe, we can derive the following equation:

$$
\dot{\Theta}(\vec{x},t) = -\frac{1}{3}\nabla\cdot\vec{v}
$$

where:

$$
\Theta(\vec{x},t) = \frac{\delta T(\vec{x},t)}{T_0(t)} = \frac{1}{3}\frac{\delta n_\gamma(\vec{x},t)}{\bar{n}_\gamma(t)}
$$

is the temperature perturbation, and $\vec{v}$ is the velocity field of the photons.

FOUNDATION

Fourier transforming the temperature perturbation equation:

$$\dot{\Theta}(\vec{k},t) \approx -\frac{1}{3}\nabla \cdot \vec{v}(\vec{k},t)$$
$$\dot{\Theta}(\vec{k},t) \approx -\frac{1}{3}i\vec{k} \cdot \vec{v}(\vec{k},t)$$
$$\dot{\Theta}(\vec{k},t) \approx -\frac{1}{3}ikv(\vec{k},t)$$

At same time, we can also derive the Euler equation for the velocity field of the photons. Ignoring gravity and expansion of universe first, so that we can focus on the small scale oscillatory behavior of the system:

$$\dot{v} = -\frac{\nabla \delta P_\gamma}{\rho_\gamma + P_\gamma}$$
$$\dot{v} = -\frac{\nabla \delta P_\gamma}{\frac{4}{3}\rho_\gamma}$$

While the pressure perturbation δP_γ is in terms of Θ as:

$$\frac{\delta P_\gamma}{\bar{P}_\gamma} = 4\frac{\delta T}{T_0} = 4\Theta$$
$$\delta P_\gamma = 4\Theta \bar{P}_\gamma = 4\Theta \frac{1}{3}\bar{\rho}_\gamma = \frac{4}{3}\Theta \bar{\rho}_\gamma$$

Substituting the above equation into the Euler equation, we can get:

$$\dot{v} = -\frac{\nabla \delta P_\gamma}{\frac{4}{3}\rho_\gamma}$$
$$\dot{v} = -\frac{\nabla \left(\frac{4}{3}\Theta \bar{\rho}_\gamma\right)}{\frac{4}{3}\rho_\gamma}$$
$$\dot{v} = -\Theta$$
$$\dot{v} = -ik\Theta$$

Finally, by taking the time derivative of the temperature perturbation equation, we can get:

$$\ddot{\Theta}(\vec{k},t) \approx -\frac{1}{3}ik\dot{v}(\vec{k},t)$$
$$\ddot{\Theta}(\vec{k},t) + \frac{1}{3}k^2\Theta(\vec{k},t) \approx 0$$

Theory

We can derive the equation of temperature perturbation into a simple harmonic oscillator equation as:

$$\ddot{\Theta}(\vec{k},t) + c_s^2 k^2 \Theta(\vec{k},t) = 0$$

with the sound speed $c_s = \frac{1}{\sqrt{3}}$, and hence $\omega_0 = c_s k = \frac{k}{\sqrt{3}}$.

By comparing the equation and the harmonic oscillator equation, we can see that it miss the damping term(friction term) in the equation, we will discuss more about this in next few sections.

Remark

Although it is oscillation of photon-baryon fluid, we can still call it "acoustic oscillation" because the sound waves in the photon-baryon fluid are analogous to the sound waves in air, and they both involve the propagation of pressure perturbations through a medium. The term "acoustic" is used to emphasize the wave-like nature of these oscillations and their role in shaping the CMB power spectrum.

Important Note

The above model is an oversimplified model that ignores the damping effect and effect of baryons, but it can still capture the essential physics and provide a good intuition about the acoustic oscillations in the photon-baryon fluid before recombination. Therefore, we will still use the above model to discuss about the acoustic peaks in the power spectrum, and we will discuss more about the damping effect and effect of baryons in next few sections.

7.3.3 Acoustic Peaks in the Power Spectrum

Physics Intuition

After making the model of acoustic oscillation in the photon-baryon fluid, we can now try to understand how the acoustic peaks in the CMB power spectrum are formed. The acoustic peaks correspond to the modes of oscillation that have reached either maximum compression or maximum rarefaction at the time of recombination. These modes will leave a strong imprint on the CMB power spectrum, resulting in the characteristic peaks that we observe.

Foundation

Recall the relationship between physical wavelength λ and the comoving wavelength λ_c:

$$\lambda = a(t)\lambda_c$$

as $k \propto \frac{1}{\lambda}$, we can get the relationship between physical angular wave number k and comoving angular wave number k^c as:

$$k^c = a(t)k$$

Let ω_0 be the angular frequency, and c_s be the sound speed, we get:

$$\begin{aligned}
\omega_0 &= c_s k \\
\omega_0 &= c_s \frac{k^c}{a} \\
\omega_0 &= \frac{c_s}{a} k^c \\
\omega_0 &= c_s^c k^c
\end{aligned}$$

where $c_s^c = \frac{c_s}{a}$ is the comoving sound speed.

Then, recall the relationship between agnular scale θ and multipole moment l:

$$\theta = \frac{2\pi}{l}$$

And with long enough distance assumption, we let $\frac{\lambda}{d_A} = \tan\theta \approx \theta$, which d_A is the angular diameter distance, so we can get:

$$\begin{aligned}
\frac{\lambda}{d_A} &= \frac{2\pi}{l} \\
\frac{2\pi}{kd_A} &= \frac{2\pi}{l} \\
k &= \frac{l}{d_A} \\
k^c &= l\frac{a}{d_A} \\
k^c &= \frac{l}{d_A^c}
\end{aligned}$$

where $d_A^c = \frac{d_A}{a}$ is the comoving angular diameter distance.

Substituting the above equation back to the ω_0 equation, and considering the i-th mode of the oscillation, we can get:

$$\begin{aligned}
\omega_0^i &= c_s^c k_i^c \\
\omega_0^i &= c_s^c \frac{l_i}{d_A^c} \\
\omega_0^i &= \frac{c_s^c}{d_A^c} l(k_i)
\end{aligned}$$

where $l(k_i) = l_i$ is the mode number of the oscillation.

THEORY

The i-th mode of angular wave number k_i in the acoustic oscillation corresponds to a multipole moment l_i in the CMB power spectrum, and the relationship can be expressed as:

$$k_i^c = \frac{l_i}{d_A^c}$$

where d_A^c is the comoving angular diameter distance.

And hence, the angular frequency of the $i-th$ mode can be expressed as:

$$\omega_0^i = \frac{c_s^c}{d_A^c} l(k_i)$$

where c_s^c is the comoving sound speed, and $l(k_i)$ is the mode number of the oscillation.

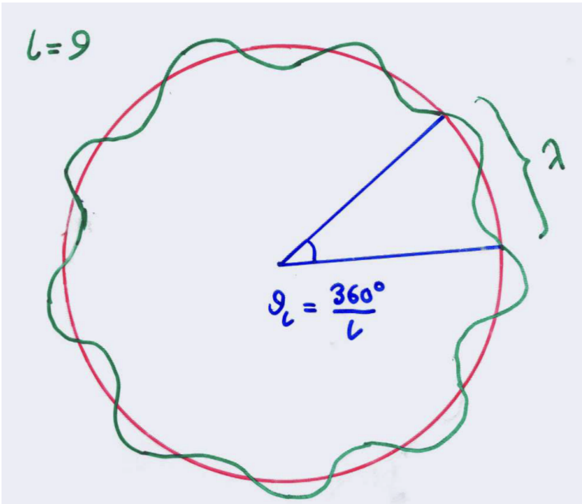

Figure 7.10: The rough relationship between the angular scale θ and the multipole moment l.

FOUNDATION

Now lets define the sound horizon r_s^c as the comoving distance that a sound wave can travel from the beginning of the universe to the time of recombination t_{rec}:

$$r_s^c(t_{rec}) = \int_0^{t_{rec}} c_s^c dt = \int_0^{t_{rec}} \frac{c_s}{a} dt$$

Further, assume that for a specific mode i, it achieves the maximum compression or maximum rarefaction at the time of recombination t_{rec}, its angular frequency

ω_0^i must be an integer multiple of π:

$$\omega_0^i t_{rec} = n\pi \quad , n = 1, 2, 3, \dots$$

And hence:

$$\begin{aligned}
\frac{c_s^c}{d_A^c} l(k_i) t_{rec} &= n\pi \\
l(k_i) &= n\pi \frac{d_A^c}{c_s^c t_{rec}} \\
l(k_i) &\approx n\pi \frac{d_A^c}{r_s^c(t_{rec})} \\
l(k_i) &= n l_A
\end{aligned}$$

where we used the approximation that $r_s^c(t_{rec}) \approx c_s^c t_{rec}$, and we defined the acoustic scale l_A as:

$$l_A = \pi \frac{d_A^c}{r_s^c(t_{rec})} = \frac{\pi}{\theta_s} \sim 220$$

with θ_s is sound horizon angle, which is the angular size of the sound horizon at the time of recombination, and it is defined as:

$$\theta_s = \frac{r_s^c(t_{rec})}{d_A^c}$$

THEORY

The multipole moments of the acoustic peaks in the CMB power spectrum can be expressed as integer multiples of the acoustic scale l_A:

$$l(k_i) = n l_A$$

where n is a positive integer, and l_A is defined as:

$$l_A = \pi \frac{d_A^c}{r_s^c(t_{rec})} = \frac{\pi}{\theta_s} \sim 220$$

with d_A^c is the comoving angular diameter distance, and $r_s^c(t_{rec})$ is the comoving sound horizon at the time of recombination.

PHYSICS INTUITION

The above equation shows that the positions of the acoustic peaks in the CMB power spectrum are determined by 2 important factors: the comoving angular diameter distance d_A^c and the comoving sound horizon $r_s^c(t_{rec})$.

The comoving angular diameter distance d_A^c depends on the geometry of the universe and the expansion history, while the comoving sound horizon $r_s^c(t_{rec})$ depends on the physical conditions of the early universe, such as the baryon density and the radiation density.

Therefore, by measuring the positions of the acoustic peaks in the CMB power spectrum, we can infer important information about the geometry and composition of the universe.

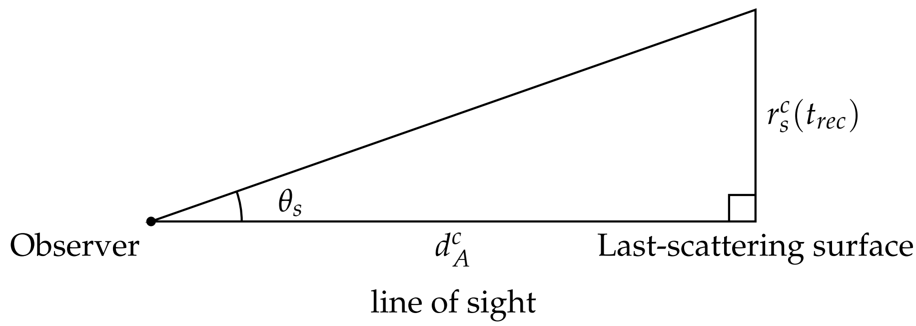

Figure 7.11: Analogy of right-triangle geometry for the sound-horizon angle.

7.3.4 Comoving Diameter Distance

PHYSICS INTUITION

The comoving angular diameter distance $d_A^c(t_{rec})$ shows the comoving distance that light travels from the last-scattering surface to us. Note it is dependent on spacetime evolution **after** recombination.

FOUNDATION

In general case where space can be flat or non-flat, we defined:

$$\Omega_0 = \Omega_{m,0} + \Omega_{\Lambda,0} + \Omega_{r,0} + \Omega_{k,0}$$

As radiation-dominated era end much earlier than recombination (around 50,000 years after the big bang), we can ignore the contribution of radiation to the total energy density of the universe at the time of recombination, and hence we can write Ω_0 as:

$$\Omega_0 = \Omega_{m,0} + \Omega_{\Lambda,0} + \Omega_{k,0}$$

with $\Omega_{k,0} = 1 - \Omega_{m,0} - \Omega_{\Lambda,0}$ is the curvature density parameter.

Using above definition, we can write the comoving angular diameter distance as:

$$\begin{aligned}
d_A^c(t_{rec}) &= \int_{t_{rec}}^{t_0} \frac{dt}{a(t)} \\
d_A^c(a_{rec}) &= \int_{a_{rec}}^{a_0} \frac{da}{a^2 H(a)} \\
&= \int_{a_{rec}}^{a_0} \frac{da}{a^2 H_0 \sqrt{\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0} + \Omega_{k,0} a^{-2}}} \\
&= \frac{1}{H_0} \int_{a_{rec}}^{a_0} \frac{da}{a^2 \sqrt{\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0} + (-\Omega_{m,0} - \Omega_{\Lambda,0}) a^{-2}}} \\
&= \frac{1}{H_0} \int_{a_{rec}}^{a_0} \frac{da}{a^2 \sqrt{\Omega_{m,0}(a^{-3} - a^{-2}) + \Omega_{\Lambda,0}(1 - a^{-2}) + a^{-2}}} \\
&= \frac{1}{H_0} \int_{a_{rec}}^{a_0} \frac{da}{\sqrt{\Omega_{m,0}(a - a^2) - \Omega_{\Lambda,0}(a^2 - a^4) + a^2}} \\
&= \frac{1}{H_0} \int_{a_{rec}}^{a_0} \frac{da}{\sqrt{\Omega_0'(a - a^2) - \Omega_{\Lambda,0}(a - a^4) + a^2}}
\end{aligned}$$

if we define $\Omega_0' = 1 - \Omega_{m,0} - \Omega_{\Lambda,0}$ as the curvature density parameter.

Theory

The comoving angular diameter distance from the last-scattering surface to us can be expressed as:

$$d_A^c(t_{rec}) = \frac{1}{H_0} \int_{a_{rec}}^{a_0} \frac{da}{\sqrt{\Omega_{m,0}(a - a^2) - \Omega_{\Lambda,0}(a^2 - a^4) + a^2}}$$

where H_0 is the Hubble constant, $\Omega_{m,0}$ is the matter density parameter, $\Omega_{\Lambda,0}$ is the dark energy density parameter, and a_{rec} and a_0 are the scale factors at the time of recombination and today, respectively.

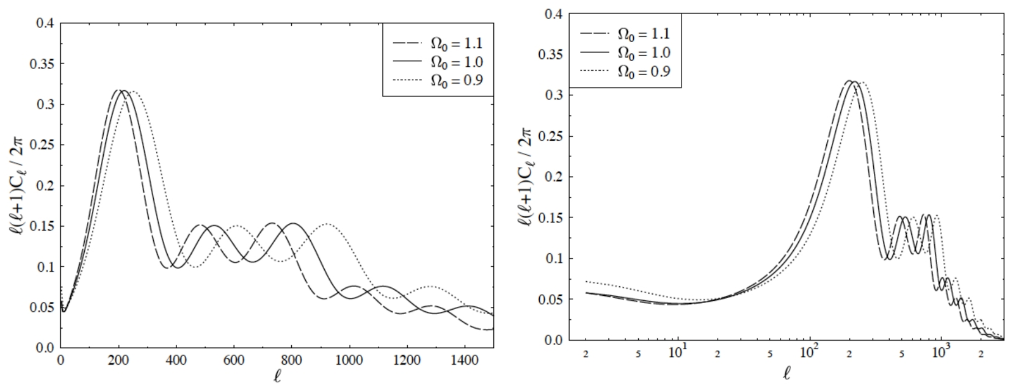

Figure 7.12: The effect of different values of Ω_0 on the CMB power spectrum.

Physics Intuition

The above figure show the results of the CMB power spectrum for different values of Ω_0' (the figure used Ω_0 to represent Ω_0' as different notations are used).

With a fixed value of $\Omega_{\Lambda,0}$, we know the increasing value of Ω_0' will cause a decrease of d_A^c, it make sense as it implies increasing of $\Omega_{m,0}$, which means more matter in the universe, and stronger deceleration of the expansion of the universe, and hence less distance and time that light can travel from the last-scattering surface to us.

On the other hand, with a fixed value of Ω_0', we know the increasing value of $\Omega_{\Lambda,0}$ will cause an increase of d_A^c, it also make sense as it implies more dark energy in the universe, which means stronger acceleration of the expansion of the universe, and hence more distance and time that light can travel from the last-scattering surface to us.

Therefore, by measuring the positions of the acoustic peaks in the CMB power spectrum, we can infer important information about the curvature of the universe and the ratio of matter and dark energy in the universe.

EXTRA

you may wonder how we can measure the ratio of matter $\Omega_{m,0}$ and dark energy $\Omega_{\Lambda,0}$ and curvature $\Omega_{k,0}$ in the universe by just looking at the positions of the acoustic peaks in the CMB power spectrum, as there is 2 degree of freedom in the parameters, but we only have 1 degree of freedom in the positions of the acoustic peaks.

The answer is we also measure other features in the CMB power spectrum, such as the relative heights of the acoustic peaks, damping tail at small scales, and the Sachs-Wolfe effect at large scales, which can provide us more information about the universe, and help us to break the degeneracy between the parameters. We will discuss more about these features in next few sections.

7.3.5 Comoving Sound Horizon

Physics Intuition

The comoving sound horizon $r_s(t_{rec})$ shows the comoving distance that a sound wave can travel from the beginning of the universe to the time of recombination. Note it is dependent on spacetime evolution **before** recombination.

Foundation

The definition of the comoving sound horizon can be written as:

$$
\begin{aligned}
r_s^c(t_{rec}) &= \int_0^{t_{rec}} c_s^c(t)dt \\
&= \int_0^{t_{rec}} \frac{c_s(t)}{a(t)}dt \\
r_s^c(a_{rec}) &= \int_0^{a_{rec}} \frac{c_s(a)}{a^2 H(a)}da \\
&= \int_0^{a_{rec}} \frac{c_s(a)}{a^2 H_0 \sqrt{\Omega_{m,0}a^{-3} + \Omega_{\Lambda,0} + \Omega_{k,0}a^{-2} + \Omega_{r,0}a^{-4}}}da \\
&= \frac{1}{H_0} \int_0^{a_{rec}} \frac{c_s(a)}{\sqrt{\Omega_{m,0}a + \Omega_{\Lambda,0}a^4 + \Omega_{k,0}a^2 + \Omega_{r,0}}}da \\
&= \frac{1}{H_0} \int_0^{a_{rec}} \frac{c_s(a)}{\sqrt{\Omega_{\Lambda,0}a^4 + (1-\Omega_0')a^2 + \Omega_{m,0}a + \Omega_{r,0}}}da
\end{aligned}
$$

where $c_s(a)$ is the sound speed as a function of scale factor, and $H(a)$ is the Hubble parameter as a function of scale factor, and we used the definition of $\Omega_0' = 1 - \Omega_{m,0} - \Omega_{\Lambda,0} - \Omega_{r,0}$ as the curvature density parameter.

We can further take the assumption that in early stage of the universe $a \ll 1$, we can ignore the contribution of dark energy and curvature to the total energy density of the universe, and hence we can write $r_s^c(t_{rec})$ as:

$$
r_s^c(a_{rec}) = \frac{1}{H_0} \int_0^{a_{rec}} \frac{c_s(a)}{\sqrt{\Omega_{m,0}a + \Omega_{r,0}}}da
$$

Foundation

From the previous sections, we know the $c_s = \frac{1}{\sqrt{3}}$, but this is only true for the pure photon fluid, while in the real universe, we have baryons as well, which will affect the sound speed. The sound speed in the photon-baryon fluid can be expressed as:

$$\begin{aligned}
c_s^2(a) &= \frac{\dot{P}}{\dot{\rho}} \\
c_s^2(a) &= \frac{\dot{P}_\gamma + \dot{P}_b}{\dot{\rho}_\gamma + \dot{\rho}_b} \\
c_s^2(a) &= \frac{\dot{P}_\gamma + 0}{\dot{\rho}_\gamma + \dot{\rho}_b} \\
c_s^2(a) &= \frac{\dot{\rho}_\gamma / 3}{\dot{\rho}_\gamma + \dot{\rho}_b}
\end{aligned}$$

Using the fluid continuity $\dot{\rho} = -3H(\rho + P)$, we can get:

$$\begin{cases} \dot{\rho}_\gamma = -4H\rho_\gamma \\ \dot{\rho}_b = -3H\rho_b \end{cases}$$

Substituting the above equation back to the sound speed equation, we can get:

$$\begin{aligned}
c_s^2(a) &= \frac{\dot{\rho}_\gamma / 3}{\dot{\rho}_\gamma + \dot{\rho}_b} \\
c_s^2(a) &= \frac{\frac{-4H\rho_\gamma}{3}}{-4H\rho_\gamma - 3H\rho_b} \\
c_s^2(a) &= \frac{1}{3} \frac{4\rho_\gamma}{4\rho_\gamma + 3\rho_b} \\
c_s^2(a) &= \frac{1}{3} \frac{1}{1 + \frac{3\rho_b}{4\rho_\gamma}} \\
c_s^2(a) &= \frac{1}{3} \frac{1}{1 + \frac{3}{4} \frac{\Omega_{b,0} a^{-3}}{\Omega_{\gamma,0} a^{-4}}} \\
c_s^2(a) &= \frac{1}{3} \frac{1}{1 + \frac{3}{4} \frac{\Omega_{b,0}}{\Omega_{\gamma,0}} a}
\end{aligned}$$

Therefore, we have the sound speed in the photon-baryon fluid as:

$$c_s^2(a) = \begin{cases} \frac{1}{3} & \text{, for pure photon fluid} \\ \frac{1}{3} \frac{1}{1 + \frac{3}{4} \frac{\Omega_{b,0}}{\Omega_{\gamma,0}} a} & \text{, for photon-baryon fluid} \end{cases}$$

FOUNDATION

We can actually writing above equations in terms of physical density parameters $\omega_i = \Omega_{i,0}h^2$, where h is the dimensionless Hubble parameter:

$$h = \frac{H_0}{100\,\mathrm{km/s/Mpc}}$$

and we can get:

$$c_s^2(a) = \frac{1}{3}\frac{1}{1+\frac{3}{4}\frac{\Omega_{b,0}}{\Omega_{\gamma,0}}a}$$
$$c_s^2(a) = \frac{1}{3}\frac{1}{1+\frac{3}{4}\frac{\omega_b/h^2}{\omega_\gamma/h^2}a}$$
$$c_s^2(a) = \frac{1}{3}\frac{1}{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}$$

substituting the above equation back to the $r_s^c(a_{rec})$ equation, we can get:

$$\begin{aligned}
r_s^c(a_{rec}) &= \frac{1}{H_0}\int_0^{a_{rec}} \frac{c_s(a)}{\sqrt{\Omega_{m,0}a + \Omega_{r,0}}}da \\
&= \frac{1}{100h}\int_0^{a_{rec}} \frac{\frac{1}{\sqrt{3}}\frac{1}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}}}{\sqrt{\omega_m\frac{a}{h^2}+\omega_r\frac{1}{h^2}}}da \\
&= \frac{1}{100}\int_0^{a_{rec}} \frac{\frac{1}{\sqrt{3}}\frac{1}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}}}{\sqrt{\omega_m a+\omega_r}}da \\
&= \frac{1}{100}\int_0^{a_{rec}} \frac{1}{\sqrt{3}}\frac{1}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}}\frac{1}{\sqrt{\omega_m a+\omega_r}}da \\
&\approx 3000\mathrm{Mpc}\frac{1}{\sqrt{3\omega_r}}\int_0^{a_{rec}} \frac{1}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}}\frac{1}{\sqrt{1+\frac{\omega_m}{\omega_r}a}}da
\end{aligned}$$

THEORY

The comoving sound horizon at the time of recombination can be expressed as:

$$r_s^c(t_{rec}) = r_s^c(\omega_b,\omega_d) = \frac{3000\mathrm{Mpc}}{\sqrt{3\omega_r}}\int_0^{a_{rec}} \frac{1}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}}\frac{1}{\sqrt{1+\frac{\omega_m}{\omega_r}a}}da$$

where ω_b is the physical baryon density parameter, ω_γ is the physical photon density parameter ($\approx 2.4702\times10^{-5}$ by CMB); ω_m is the physical matter density parameter, and ω_r is the physical radiation density parameter ($\approx 4.1756\times10^{-5}$ by CMB).

Physics Intuition

From the above equation, by given ω_γ and ω_r we can see that the roles of baryons and photons in the acoustic peaks; when ω_b increases, the baryon slow down the sound speed, it can be think as the more baryons in the photon-baryon fluid, the more inertia the fluid has, and hence the slower the sound speed.

On the other hand, when ω_γ increases, the photons speed up the sound speed, it can be think as the more photons in the photon-baryon fluid, the more pressure the fluid has, and hence the faster the sound speed.

Meanwhile, when ω_m increases, by Friedmann equation we can see that the expansion rate of the universe will increase, and hence the recombination time t_{rec} happens earlier, hence a shorter time for the sound wave to travel, and hence a smaller sound horizon $r_s^c(t_{rec})$. And same for ω_r as well.

Question Example 7.1: free parameters

y $\theta_s = \frac{r_s^c(t_{rec})}{d_A^c}$, and the equation of $r_s^c(t_{rec})$ and d_A^c, we can see that the acoustic scale l_A is dependent on the parameters $\Omega_{k,0}$, $\Omega_{\Lambda,0}$, ω_b, and ω_d. But why ignore the dependence on ω_γ and ω_r and Hubble constant H_0?

Solution.
The reason is that ω_γ are well constrained by the CMB temperature, while the radiation density ω_r is depends by photon density ω_γ and neutrino density ω_ν, and the neutrino density is also well constrained by the CMB (by $(\frac{4}{11})^{\frac{1}{3}}$), hence we can treat ω_γ and ω_r as fixed parameters.

On the other hand, the Hubble constant H_0 is depends on other energy density parameters through the Friedmann equation, and hence we can also treat it as a derived parameter, and ignore its dependence in the acoustic scale l_A.

Question Example 7.2: Numerical Estimate of the Acoustic Scale l_A

Use representative values near recombination and Planck-like parameters:

$$z_{rec} \approx 1090, \quad \Omega_{m,0} \approx 0.315, \quad \Omega_{\Lambda,0} \approx 0.685, \quad h \approx 0.674, \quad \omega_b \approx 0.0224.$$

Break the estimate into three steps:
(a). compute d_A^c;
(b). compute r_s^c;
(c). compute the acoustic peak scale.

Solution.
(a). **Comoving angular diameter distance**
Use

$$d_A^c(z_{rec}) = \int_0^{z_{rec}} \frac{dz}{H(z)}, \qquad H(z) = H_0\sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}}$$

(radiation gives only a small correction for this distance integral near late times).
Numerically:

$$d_A^c(z_{rec}) \approx 13.9\,\text{Gpc} = 1.39 \times 10^4\,\text{Mpc}.$$

(b). **Comoving sound horizon**
Use

$$r_s^c(t_{rec}) = \int_0^{a_{rec}} \frac{c_s(a)}{a^2 H(a)}\,da, \qquad c_s(a) = \frac{1}{\sqrt{3\left(1 + \frac{3}{4}\frac{\omega_b}{\omega_\gamma}a\right)}}.$$

Numerically (with the above Planck-like parameters):

$$r_s^c(t_{rec}) \approx 144\,\text{Mpc}.$$

(c). **Acoustic peak scale**
First compute

$$\theta_s = \frac{r_s^c}{d_A^c} \approx \frac{144}{1.39 \times 10^4} \approx 1.04 \times 10^{-2}\,\text{rad} \approx 0.60^\circ.$$

Then

$$l_A = \frac{\pi}{\theta_s} = \pi\frac{d_A^c}{r_s^c} \approx \frac{3.1416}{1.04 \times 10^{-2}} \approx 302 \approx 300.$$

Therefore, the fundamental acoustic spacing scale is $l_A \sim 300$, while the first temperature peak is observed near $l \sim 220$ due to phase-shift/projection effects.

Question Example 7.3: More related questions

For below questions, we used the **rec** subscript to represent the time of decoupling, but some textbooks may use **dec** subscript to representing it.
(a). Calculate the physical sound horizon at decoupling moment:

$$r_s(t_{rec}) = a_{rec} r_s^c(t_{rec}) = a_{rec} \int_0^{t_{rec}} c_s^c(t) dt$$

in terms of z_{rec}, H_0, h, $\omega_x = \Omega_{x,0} h^2$. Include the effect of radiation and dust components in expansion law.
(b). What is the seperation between the two neighbored acoustic peaks in the CMB angular power spectrum for the case $\Omega_{\Lambda,0} = 0$ and $\Omega_{\Lambda,0} = 1 - \Omega_{d,0}$ respectively?
(c). Calculate the numerical value of r_s and l_A for $z_{rec} = 1090$, $\Omega_{d,0} = 0.3$, $\Omega_{\Lambda,0} = 0.0$, $h = 0.7$, and $\omega_b = 0.02$.

Solution.
(a). First, recall the equation of comoving sound horizon:

$$r_s^c(t_{rec}) = \frac{3000\text{Mpc}}{\sqrt{3\omega_r}} \int_0^{a_{rec}} \frac{1}{\sqrt{1 + \frac{3}{4}\frac{\omega_b}{\omega_\gamma} a}} \frac{1}{\sqrt{1 + \frac{\omega_m}{\omega_r} a}} da$$

The using the relation of $a = \frac{1}{1+z}$, and $3000\text{Mpc} \approx \frac{hc}{H_0} = \frac{h}{H_0}$, we can get:

$$\begin{aligned}
r_s(t_{rec}) &= a_{rec} r_s^c(t_{rec}) \\
&= \frac{1}{1+z_{rec}} \frac{h}{H_0\sqrt{3\omega_r}} \int_\infty^{z_{rec}} \frac{1}{\sqrt{1 + \frac{3}{4}\frac{\omega_b}{\omega_\gamma}\frac{1}{1+z}}} \frac{1}{\sqrt{1 + \frac{\omega_m}{\omega_r}\frac{1}{1+z}}} \frac{-dz}{(1+z)^2} \\
&= \frac{1}{1+z_{rec}} \frac{h}{H_0\sqrt{3\omega_r}} \int_{z_{rec}}^\infty \frac{1}{\sqrt{1 + z + \frac{3}{4}\frac{\omega_b}{\omega_\gamma}}} \frac{1}{\sqrt{1 + z + \frac{\omega_m}{\omega_r}}} \frac{dz}{(1+z)} \\
&= \frac{1}{1+z_{rec}} \frac{h}{H_0\sqrt{3\omega_r}} \int_{z_{rec}}^\infty \frac{\sqrt{4\omega_\gamma}\sqrt{\omega_r}}{\sqrt{4(1+z)\omega_\gamma + 3\omega_b}\sqrt{(1+z)\omega_r + \omega_m}} \frac{dz}{(1+z)} \\
&= \frac{2h\sqrt{\omega_\gamma}}{H_0\sqrt{3}(1+z_{rec})} \int_{z_{rec}}^\infty \frac{1}{\sqrt{4(1+z)\omega_\gamma + 3\omega_b}\sqrt{(1+z)\omega_r + \omega_m}} \frac{dz}{(1+z)}
\end{aligned}$$

Hence we get the integral expression that the question asked for. However, in case you are a stubborn mathematician that want to ge tthe analytical expression of the above integral, i will show you the detailed steps also: Let's define $A = \omega_r$, $B = \omega_m$, $C = 4\omega_\gamma$, and $D = 3\omega_b$, and $u = 1 + z$, then we can write the above integral as:

$$\begin{aligned}
I &= \int_{u_{rec}}^{\infty} \frac{1}{\sqrt{Cu+D}\sqrt{Au+B}} \frac{du}{u} \\
&\xrightarrow{v=1/u} \int_{1/u_{rec}}^{0} \frac{-dv/v^2}{(1/v)\sqrt{\frac{C+Dv}{v}}\sqrt{\frac{A+Bv}{v}}} \\
&= \int_{1/u_{rec}}^{0} \frac{-dv}{\sqrt{(A+Bv)(C+Dv)}} \\
&= \int_{0}^{1/u_{rec}} \frac{dv}{\sqrt{(A+Bv)(C+Dv)}} \\
&= \int_{0}^{1/u_{rec}} \frac{dv}{\sqrt{BDv^2+(AD+BC)v+AC}}
\end{aligned}$$

Given $BD > 0$, we can use the standard primitive

$$\int \frac{dv}{\sqrt{(A+Bv)(C+Dv)}} = \frac{1}{\sqrt{BD}} \ln\left|2\sqrt{BD}\sqrt{(A+Bv)(C+Dv)} + 2BDv + AD + BC\right| + \text{const}$$

and hence:

$$\begin{aligned}
I &= \frac{1}{\sqrt{BD}} \ln\left|2\sqrt{BD}\sqrt{(A+Bv)(C+Dv)} + 2BDv + AD + BC\right| \Bigg|_0^{1/u_{rec}} \\
&= \frac{1}{\sqrt{BD}} \ln\left[\frac{2\sqrt{BD}\sqrt{\left(A+\frac{B}{u_{rec}}\right)\left(C+\frac{D}{u_{rec}}\right)} + \frac{2BD}{u_{rec}} + AD + BC}{2\sqrt{ABCD} + AD + BC}\right]
\end{aligned}$$

Writing it back in terms of u_{rec} inside radicals:

$$I = \frac{1}{\sqrt{BD}} \ln\left[\frac{2\sqrt{BD}\,\sqrt{(Au_{rec}+B)(Cu_{rec}+D)} + 2BD + (AD+BC)u_{rec}}{u_{rec}\,(2\sqrt{ABCD} + AD + BC)}\right]$$

where $u_{rec} = 1 + z_{rec}$, $A = \omega_r$, $B = \omega_m$, $C = 4\omega_\gamma$, $D = 3\omega_b$.
Therefore, substituting this I back to the physical sound horizon expression gives:

$$\begin{aligned}
r_s(t_{rec}) &= \frac{2h\sqrt{\omega_\gamma}}{H_0\sqrt{3}(1+z_{rec})} I \\
&= \frac{2h\sqrt{\omega_\gamma}}{H_0\sqrt{3}(1+z_{rec})\sqrt{BD}} \ln\left[\frac{2\sqrt{BD}\,\sqrt{(Au_{rec}+B)(Cu_{rec}+D)}}{u_{rec}\,(2\sqrt{ABCD}+AD+BC)}\right. \\
&\quad \left. + \frac{2BD+(AD+BC)u_{rec}}{u_{rec}\,(2\sqrt{ABCD}+AD+BC)}\right]
\end{aligned}$$

In terms of the cosmological parameters only ($u_{rec} = 1 + z_{rec}$):

$$\begin{aligned}
r_s(t_{rec}) = \frac{2h\sqrt{\omega_\gamma}}{H_0\sqrt{3}(1+z_{rec})\sqrt{3\omega_b\omega_m}} \ln&\left[\frac{2\sqrt{3\omega_b\omega_m}\,\sqrt{(\omega_r u_{rec}+\omega_m)(4\omega_\gamma u_{rec}+3\omega_b)}}{u_{rec}\,(4\sqrt{3\omega_r\omega_m\omega_\gamma\omega_b}+3\omega_r\omega_b+4\omega_m\omega_\gamma)}\right. \\
&\left. + \frac{6\omega_b\omega_m + (3\omega_r\omega_b+4\omega_m\omega_\gamma)u_{rec}}{u_{rec}\,(4\sqrt{3\omega_r\omega_m\omega_\gamma\omega_b}+3\omega_r\omega_b+4\omega_m\omega_\gamma)}\right]
\end{aligned}$$

(b). **Separation between neighboring acoustic peaks**
Assumptions.
1. Use leading-order peak locations $l_n \simeq n l_A$ (ignore phase-shift corrections from driving/projection):

$$\Delta l \equiv l_{n+1} - l_n \simeq (n+1) l_A - n l_A = l_A.$$

2. For comparing the two cosmologies, keep pre-recombination physics fixed (z_{rec}, ω_b, ω_γ, ω_r, ω_m), so the same $r_s^c(t_{rec})$ is used in both cases. 3. Geometry difference enters through d_A^c only:

$$l_A = \pi \frac{d_A^c}{r_s^c}, \qquad d_A^c = \frac{3000}{h} \int_{a_{rec}}^{1} \frac{da}{a^2 E(a)}, \qquad E(a) = \sqrt{\Omega_{d,0} a^{-3} + \Omega_{\Lambda,0} + \Omega_{k,0} a^{-2}}.$$

For this question, $\Omega_{d,0} = 0.3$, $h = 0.7$, $a_{rec} = 1/(1+1090) = 1/1091$, and

$$\Omega_{k,0} = 1 - \Omega_{d,0} - \Omega_{\Lambda,0}.$$

Define

$$I_d(\Omega_{\Lambda,0}) \equiv \int_{a_{rec}}^{1} \frac{da}{a^2 E(a)}, \qquad d_A^c = \frac{3000}{h} I_d.$$

Then numerically:

$$\begin{aligned}
\Omega_{\Lambda,0} = 0 \ (\Omega_{k,0} = 0.7): \quad & I_d \approx 2.7818, \\
& d_A^c \approx \frac{3000}{0.7} \times 2.7818 \approx 1.192 \times 10^4 \,\text{Mpc}. \\
\Omega_{\Lambda,0} = 1 - \Omega_{d,0} = 0.7 \ (\Omega_{k,0} = 0): \quad & I_d \approx 3.1945, \\
& d_A^c \approx \frac{3000}{0.7} \times 3.1945 \approx 1.369 \times 10^4 \,\text{Mpc}.
\end{aligned}$$

Using $r_s^c \approx 144.93\,\text{Mpc}$ (computed in part (c)):

$$\begin{aligned}
\Delta l\big|_{\Omega_{\Lambda,0}=0} \simeq l_A = \pi \frac{1.192 \times 10^4}{144.93} \approx 258.4, \\
\Delta l\big|_{\Omega_{\Lambda,0}=0.7} \simeq l_A = \pi \frac{1.369 \times 10^4}{144.93} \approx 296.8.
\end{aligned}$$

Therefore,

$$\Delta l\big|_{\Omega_{\Lambda,0}=0} \approx 258, \qquad \Delta l\big|_{\Omega_{\Lambda,0}=0.7} \approx 297,$$

and the fractional increase is

$$\frac{296.8 - 258.4}{258.4} \approx 14.9\% \approx 15\%.$$

(c). **Numerical values of r_s and l_A for the given set**
Assumptions.
1. Use the problem parameters:

$$z_{rec} = 1090, \quad \Omega_{d,0} = 0.3, \quad \Omega_{\Lambda,0} = 0, \quad h = 0.7, \quad \omega_b = 0.02.$$

2. Adopt chapter constants:

$$\omega_\gamma = 2.4702 \times 10^{-5}, \qquad \omega_r = 4.1756 \times 10^{-5}.$$

3. Use $\omega_m = \Omega_{d,0}h^2 = 0.147$ and the chapter approximation formula

$$r_s^c(t_{rec}) = \frac{3000\,\mathrm{Mpc}}{\sqrt{3\omega_r}} \int_0^{a_{rec}} \frac{da}{\sqrt{1+\frac{3}{4}\frac{\omega_b}{\omega_\gamma}a}\,\sqrt{1+\frac{\omega_m}{\omega_r}a}}, \qquad a_{rec} = \frac{1}{1+z_{rec}}.$$

First evaluate the dimensionless factors in the integrand:

$$a_{rec} = \frac{1}{1+1090} = \frac{1}{1091} \approx 9.166 \times 10^{-4},$$
$$R_b \equiv \frac{3}{4}\frac{\omega_b}{\omega_\gamma} = \frac{3}{4}\frac{0.02}{2.4702 \times 10^{-5}} \approx 607.24,$$
$$R_m \equiv \frac{\omega_m}{\omega_r} = \frac{0.147}{4.1756 \times 10^{-5}} \approx 3520.45.$$

So

$$r_s^c = \frac{3000}{\sqrt{3\omega_r}} \underbrace{\int_0^{a_{rec}} \frac{da}{\sqrt{1+R_b a}\sqrt{1+R_m a}}}_{I_s}.$$

Numerically,

$$I_s \approx 5.407 \times 10^{-4},$$
$$\frac{3000}{\sqrt{3\omega_r}} \approx 2.680 \times 10^5\,\mathrm{Mpc},$$
$$r_s^c \approx \left(2.680 \times 10^5\right)\left(5.407 \times 10^{-4}\right) \approx 144.93\,\mathrm{Mpc}.$$

Then physical sound horizon:

$$r_s(t_{rec}) = a_{rec} r_s^c = \frac{1}{1091} \times 144.93\,\mathrm{Mpc} \approx 0.13285\,\mathrm{Mpc} \approx 133\,\mathrm{kpc}.$$

Next compute l_A for the same parameter set ($\Omega_{\Lambda,0} = 0$):

$$\Omega_{k,0} = 1 - \Omega_{d,0} - \Omega_{\Lambda,0} = 1 - 0.3 - 0 = 0.7,$$
$$d_A^c = \frac{3000}{0.7} \int_{a_{rec}}^1 \frac{da}{a^2\sqrt{0.3a^{-3} + 0.7a^{-2}}} \approx 1.192 \times 10^4\,\mathrm{Mpc},$$
$$l_A = \pi \frac{d_A^c}{r_s^c} \approx \pi \frac{1.192 \times 10^4}{144.93} \approx 258.4.$$

Therefore, for the given parameter set,

$$r_s(t_{rec}) \approx 0.133\,\mathrm{Mpc}\ (\approx 133\,\mathrm{kpc}), \qquad l_A \approx 258.$$

(A verification script is provided in `code/verify_cmb_acoustic_bc.py`.)

7.4 More Features of the Power Spectrum

From the previous section, we have discussed about the acoustic peaks in the power spectrum, which is the most prominent feature in the power spectrum. However, you may noticed we have ignored the damping term in the equation of temperature perturbation, which is not our real case. In fact, during the oscillation of the photon-baryon fluid, the oscillation is not perfect, but is damped by the expansion of the universe and the diffusion of photons (Silk damping). Therefore, we need to consider the damping term in the equation of temperature perturbation, which will give us more features in the power spectrum, such as the damping tail at small scales.

Besides, there are also other features in the power spectrum, such as the Sachs-Wolfe effect at large scales, and the relative heights of the acoustic peaks, which can provide us more information about the universe, such as the geometry of the universe, we will discuss more about these features in below subsections.

7.4.1 Damping Tail

PHYSICS INTUITION

If we further consider the damping term, we can write the oscillation equation of the photon-baryon fluid as:

$$\ddot{\Theta} + c_s^2 k^2 f(c_i)\dot{\Theta} + c_s^2 k^2 \Theta = g(c_i)$$

where $f(c_i)$ is the damping term by the expansion of the universe and the diffusion of photons, and $g(c_i)$ is the force driven term. The c_i are the parameters that affect the damping and source terms, such as the baryon density, the matter density, the dark matter density, etc.

We can simply write the damping coeffcient as:

$$c \propto k^2 \propto l^2$$

Therefore, for the small scales (large l), the damping term will be more significant, and hence the amplitude of the oscillation will be smaller, which will give us a damping tail in the power spectrum at small scales.

FOUNDATION

To connect the above oscillator picture with the observed power spectrum, we can separate damping into two intuitive effects:

- **Expansion damping (Hubble friction):** the expansion of the universe re-moves energy from oscillations, similar to friction in a damped oscillator.

- **Diffusion damping (Silk damping):** before decoupling, photons still scatter many times and perform a random walk, so they can leak from hot regions to cold regions and smooth out temperature contrast.

The second effect is the dominant reason for the sharp small-scale suppression. A useful way to state it is: modes with wavelength smaller than the photon diffusion scale are exponentially erased. In k-space, this is often written schematically as:

$$\Theta(k, \eta_*) \simeq \Theta_{\text{osc}}(k, \eta_*) \, \exp\left[-\left(\frac{k}{k_D} \right)^2 \right]$$

where k_D is the diffusion-damping scale (evaluated near recombination), and η_* is recombination time.

Since larger k means smaller physical scale, the exponential factor suppresses high-k modes much more strongly. This is exactly the same trend as your earlier statement $c \propto k^2 \propto l^2$: higher-$k$ (thus higher-l) modes damp faster.

THEORY

A compact way to summarize the damping-tail physics is:

$$\Theta(k) \text{ is multiplied by } \exp\left[-\left(\frac{k}{k_D} \right)^2 \right] \quad \Longrightarrow \quad C_l \text{ is suppressed at large } l.$$

Using the projection relation $l \sim k \, d_A^c$ (with d_A^c the comoving angular-diameter distance to last scattering), we define a damping multipole $l_D \sim k_D d_A^c$. Then the observed spectrum is often approximated by:

$$C_l \simeq C_l^{(\text{no damping})} \, \exp\left[-2 \left(\frac{l}{l_D} \right)^2 \right]$$

up to order-one modeling details.
Therefore, the high-l side of the CMB spectrum is not another acoustic peak sequence, but a progressively damped tail: the smaller the angular scale, the stronger the suppression by photon diffusion (plus expansion damping).

7.4.2 Sachs-Wolfe Effect

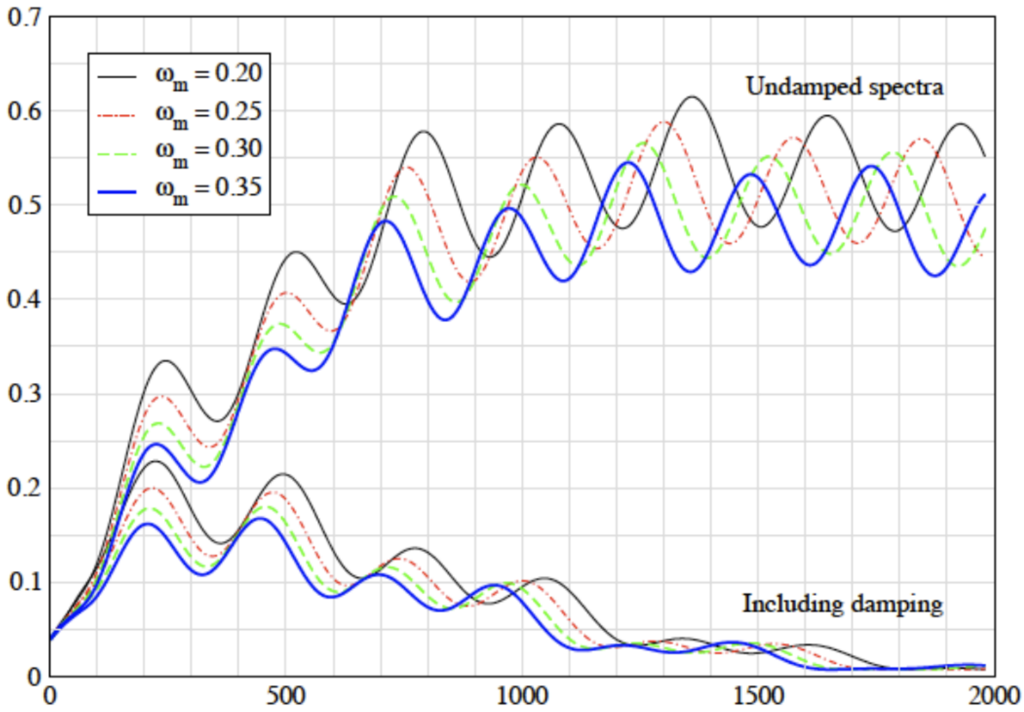

Figure 7.13: The effect of damping on the CMB power spectrum.

Physics Intuition

The Sachs-Wolfe (SW) effect describes how gravity changes the observed CMB temperature when photons travel from the last-scattering surface to us.

A simple picture is: imagine photons are emitted from different "gravitational landscapes" (potential wells and hills). If a photon climbs out from a potential well, it loses energy (gravitational redshift), so we observe it colder; if it is emitted from a potential hill, it gains relative energy and looks hotter. This converts gravitational potential perturbations directly into temperature anisotropies.

There are two related contributions:

- **Ordinary Sachs-Wolfe (OSW):** set at recombination, mainly important on very large angular scales.
- **Integrated Sachs-Wolfe (ISW):** accumulated during photon propagation when gravitational potentials evolve with time.

Therefore, the SW effect is the key physical reason why the low-l (large-angle) part of the CMB power spectrum carries direct information about primordial metric perturbations and late-time expansion history.

Foundation

Let the line-of-sight direction be $\hat{n}$ and conformal time be η. The observed temperature perturbation can be written schematically as:

$$\Theta(\hat{n}) \equiv \frac{\Delta T}{T_0}(\hat{n}) = [\Theta_0 + \Psi]_* + \int_{\eta_*}^{\eta_0} (\dot{\Psi} + \dot{\Phi})\, d\eta$$

where:

- Θ_0 is the intrinsic photon temperature perturbation at last scattering;
- Ψ and Φ are metric (gravitational potential) perturbations;
- subscript $*$ means evaluated at recombination;
- the integral term is along the photon path from recombination to today.

The first bracket term is the **ordinary SW** contribution, while the line-of-sight integral is the **ISW** contribution.

For super-horizon modes at recombination in matter domination (and negligible anisotropic stress, so $\Phi \simeq \Psi$), the ordinary SW term reduces to the well-known approximation:

$$\left(\frac{\Delta T}{T_0}\right)_{\text{OSW}} \simeq \frac{1}{3}\Psi_*$$

(up to sign convention of the potential).

Physically, this means large-angle CMB anisotropy is roughly one-third of the primordial gravitational potential perturbation at last scattering.

For the ISW term:

- During strict matter domination, potentials are nearly constant, so $\dot{\Psi} + \dot{\Phi} \approx 0$, and ISW is small.
- Near radiation-matter transition (early ISW) and in dark-energy domination (late ISW), potentials evolve, so ISW becomes non-zero.

Theory

The Sachs-Wolfe contribution to CMB anisotropy can be summarized as:

$$\Theta(\hat{n}) = \underbrace{[\Theta_0 + \Psi]_*}_{\text{ordinary SW}} + \underbrace{\int_{\eta_*}^{\eta_0} (\dot{\Psi} + \dot{\Phi}) d\eta}_{\text{integrated SW}}$$

and for large scales at recombination:

$$\left(\frac{\Delta T}{T_0}\right)_{\text{OSW}} \simeq \frac{1}{3}\Psi_*$$

In the angular power spectrum, this physics appears as the low-l Sachs-Wolfe plateau (roughly $l \lesssim 30$), where:

$$\frac{l(l+1)C_l}{2\pi} \approx \text{constant}$$

while late-time ISW slightly modifies the largest scales and provides sensitivity to dark energy and spatial curvature.

7.4.3 Relative Heights of the Acoustic Peaks

PHYSICS INTUITION

To understand why different peaks have different heights, we only need three physical ideas that we already introduced in previous sections: **compression vs rarefaction**, **baryon inertia**, and **gravitational potential wells**.

Before recombination, each Fourier mode of the photon-baryon fluid oscillates like a driven oscillator. At decoupling, we take a "snapshot" of all modes:

- modes caught at **maximum compression** (odd peaks: 1st, 3rd, 5th, ...) produce one set of peak heights;
- modes caught at **maximum rarefaction** (even peaks: 2nd, 4th, ...) produce another set.

If we add more baryons, the fluid has more mass (inertia). This shifts the oscillation zero-point toward compression, so compressions become deeper while rarefactions become shallower. Therefore, increasing baryon density enhances **odd peaks relative to even peaks**.

Dark matter contributes mainly through gravity: it deepens potential wells and changes how strongly modes are driven around horizon entry. This affects the overall peak envelope and, in particular, the height of higher peaks (for example the third peak) relative to lower ones.

So the pattern of peak heights is not random: odd-even contrast mainly traces baryons, while the broader peak-height pattern across multiple peaks is sensitive to total matter and dark matter.

FOUNDATION

A useful approximate form of the photon temperature perturbation for one mode is:

$$\Theta_0(k,\eta_*) + \Psi(k,\eta_*) \approx A(k)\cos\big(k\,r_s(\eta_*)\big) - R_*\,\Psi(k,\eta_*)$$

where:

- $r_s(\eta_*)$ is the sound horizon at recombination;
- $R_* \equiv \frac{3\rho_b}{4\rho_\gamma}\big|_*$ measures baryon loading;
- $A(k)$ encodes driving effects from the evolving gravitational potentials.

The cosine term gives the alternating compression/rarefaction phases, while the $-R_*\Psi$ term is the baryon-loading shift. As R_* increases:

- compression extrema (odd peaks) are boosted;
- rarefaction extrema (even peaks) are suppressed.

Meanwhile, matter content controls potential evolution around horizon entry:

- more matter (especially non-baryonic dark matter) makes potentials decay less during the radiation-to-matter transition;
- this changes the driving efficiency and therefore the relative heights among the first few peaks, especially the third peak compared with the first two.

THEORY

The relative peak heights provide complementary parameter sensitivity:

$$\begin{aligned} \text{odd/even contrast} &\longrightarrow \omega_b\ (= \Omega_{b,0}h^2), \\ \text{higher-peak pattern (e.g. 3rd vs 1st/2nd)} &\longrightarrow \omega_m\ (= \Omega_{m,0}h^2), \\ &\text{with } \omega_m - \omega_b \text{ isolating dark matter density } \omega_c. \end{aligned}$$

In practice, CMB fits use the full spectrum shape, but an intuitive summary is:

- larger $\omega_b \Rightarrow$ stronger odd peaks relative to even peaks;
- larger ω_c (or total ω_m at fixed ω_b) $\Rightarrow$ modified driving and a characteristic change in the third and higher peaks.

Therefore, combining peak positions (geometry) with peak heights (dynamics) lets us separately constrain baryons, dark matter, and total matter in the standard cosmological model.

7.5 CMB and ΛCDM-Standard Model

Extra

The discovery of the Cosmic Microwave Background (CMB) in 1964 provided decisive evidence for the Hot Big Bang model, but the composition and energy budget of the universe remained uncertain for decades.

In the 1970s, cosmological models were often constrained to baryonic matter alone, motivated by Occam's razor and early estimates of the cosmic density. However, these pure-baryon models faced severe theoretical and observational tensions. First, Big Bang Nucleosynthesis (BBN) strictly limited the baryon density to $\Omega_b \lesssim 0.05$, far below the critical density suggested by inflation. Second, because baryons were tightly coupled to photons until recombination ($z \sim 1100$), radiation pressure suppressed gravitational collapse, making it impossible to form the observed large-scale structure within the age of the universe. Additionally, early estimates of the Hubble constant implied a universe younger than the oldest globular clusters, creating the age problem.

In the 1980s, the Cold Dark Matter (CDM) paradigm emerged, proposing that the universe is dominated by non-baryonic, weakly interacting particles that decoupled from radiation early on. Because CDM is unaffected by photon pressure, its density perturbations could begin growing immediately after inflation, providing the gravitational scaffolding for baryonic structure formation. While CDM successfully resolved the structure formation problem, a flat, matter-dominated universe still required a low Hubble constant ($H_0 \lesssim 50$ km/s/Mpc) to match stellar ages, which increasingly conflicted with direct distance-ladder measurements.

In the 1990s, two major extensions were explored: the mixed Hot+Cold Dark Matter (HCDM) model and the ΛCDM model. The 1992 COBE detection of CMB anisotropies, combined with large-scale structure surveys, began to favor models with a cosmological constant Λ. The discovery of cosmic acceleration in 1998 via Type Ia supernovae provided direct evidence for Λ (or dark energy), which naturally resolved the age problem by altering the expansion history and increasing the cosmic age for a given H_0. By the late 1990s and early 2000s, ΛCDM emerged as the consensus model, consistently fitting CMB power spectra, baryon acoustic oscillations, supernova distances, and the growth of structure, eventually becoming the standard model of cosmology.

CHAPTER 8

Dark Matter (Lecture 20-22)

8.1 Evidence for Dark Matter

Physics Intuition

Dark matter is a form of matter that does not emit, absorb, or reflect light, making it invisible to electromagnetic observations. Its existence is inferred from its gravitational effects on visible matter, radiation, and the large-scale structure of the universe.

There are multiple lines of evidence for dark matter, including: gravitational lensing, expansion rate of the universe, the CMB anisotropies, and other measurements. We will focus on the most direct evidence for dark matter: galaxy rotation curves, in below sections. We will also briefly discuss an alternative explanation for the observed phenomena: Modified Newtonian Dynamics (MOND).

Extra

The name of dark matter was first named d by Jacobus Kapteyn in 1922, he claimed that "dark matter" was not required in galactic Solar neighbourhood. In 1932, Jan Oort suggested that there was twice as much dark matter as luminous matter in the Solar neighbourhood. However, latter observations showed that he was wrong, therefore the discovery of dark matter was credited to Fritz Zwicky who made the first correct claim for the existence of dark matter in 1933. Zwicky concluded from measurements of the redshifts of galaxies in the Coma cluster that their velocities are much larger than the escape velocity due to the visible mass of the cluster.

Between 1950s and 1970s, Vera Rubin and Kent Ford built upon earlier galactic rotation studies and conducted a systematic survey of spiral galaxies. While their pioneering 1970 paper on the Andromeda galaxy hinted at anomalous outer velocities, it was their work in the mid-to-late 1970s that definitively demonstrated flat rotation curves across many galaxies. These curves showed that stars and gas orbit at nearly constant speeds far beyond the visible disk, rather than declining as expected from luminous matter alone. This systematic evidence firmly established that galaxies are embedded in massive, extended halos of dark matter.

8.1.1 Galaxy Rotation Curves

FOUNDATION

Start from the Newtonian gravity, the rotational velocity v of a star orbiting at a distance r from the center of a galaxy can be derived from the balance of gravitational force and centripetal force:

$$\frac{GM(r)}{r^2} = \frac{v^2}{r}$$
$$v^2 = \frac{GM(r)}{r}$$

Then assuming that the mass distribution of the galaxy is concentrated in the center, and density decreases with distance:

$$\rho \propto r^{-n}$$

where $n > 0$, but $n < 3$ to ensure that the total mass is finite. For typical values, $n \approx 2 \sim 3$.
Now we can get the mass distribution of the galaxy:

$$\begin{aligned} M(r) &= \int_0^r 4\pi r'^2 \rho(r') dr' \\ &\propto \int_0^r r'^{2-n} dr' \\ &\propto r^{3-n} \end{aligned}$$

Therefore, the rotational velocity can be derived as:

$$v^2 \propto \frac{r^{3-n}}{r}$$

THEORY

In newtonian gravity, the rotational velocity v of a star orbiting at a distance r from the center of a galaxy can be expressed as:

$$v = r^{\frac{2-n}{2}}$$

where n is the power-law index of the mass density distribution of the galaxy.

Physics Intuition

For $n \approx 2$, we have $v \propto r^0$, which means that the rotational velocity v is constant at large distances from the galaxy center, which is consistent with the observed flat rotation curves.

However, from the visible matter distribution, we expect $n > 2$ as stellar density falls rapidly at edges of galaxies, which would predict a declining rotation curve. The conflict between the expected declining rotation curve and the observed flat rotation curve is one of the key pieces of evidence for the existence of dark matter, which provides additional mass that extends beyond the visible matter, leading to a flat rotation curve.

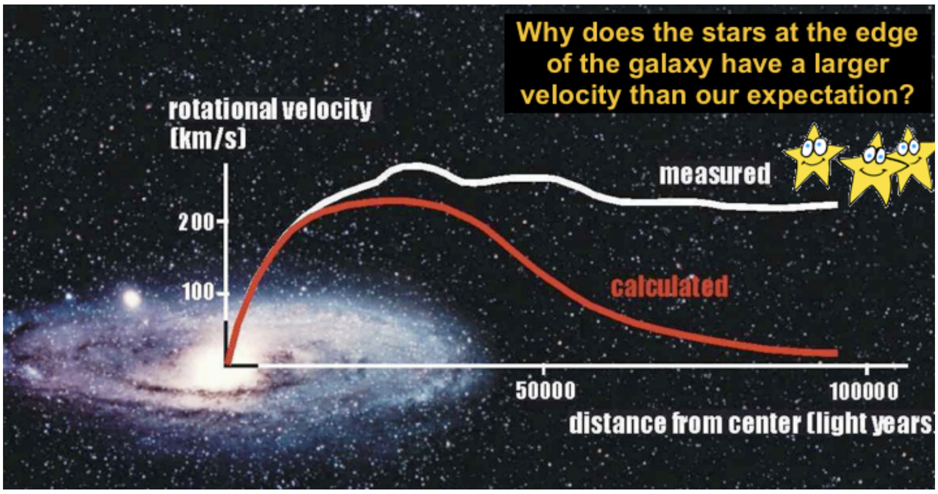

Figure 8.1: The rotation curve of a galaxy: The red curves represent the expected rotation curve based on the visible matter distribution, which shows a declining trend at large distances. The white curve represents the observed rotation curve, which remains flat at large distances, indicating the presence of dark matter.

Extra

To describe the observed rotation curves like above figure, the Navarro-Frenk-White (NFW) profile is commonly used to model the dark matter halo of galaxies. The NFW profile is given by:

$$\rho(r) = \frac{\rho_0}{\frac{r}{r_s}\left(1 + \frac{r}{r_s}\right)^2}$$

where ρ_0 is a characteristic density and r_s is a scale radius. The NFW profile predicts a density in 3 parts of the galaxy: (1) Inner region ($r \ll r_s$): $\rho(r) \propto r^{-1}$, which means that density drops but velocity increases as we slightly move away from the center of the galaxy. It matches the rapid increase of the rotation curve in the inner region of the galaxy.

(2) Intermediate region ($r \approx r_s$): $\rho(r) \propto r^{-2}$, which means that the density decreases and the velocity remains constant as we move further away from the center of the galaxy. It matches the flat rotation curve in the intermediate region of the galaxy.

(3) Outer region ($r \gg r_s$): $\rho(r) \propto r^{-3}$, which means that the density decreases more rapidly and the velocity decreases as we move further away from the center of the galaxy. It matches the declining rotation curve in the outer region of the galaxy.

8.1.2 Modified Newtonian Dynamics (MOND)

PHYSICS INTUITION

Note that existence of dark matter is just a hypothesis to explain the observed phenomena, and there are alternative explanations. One of the most popular alternative explanation is Modified Newtonian Dynamics (MOND), which was proposed by Mordehai Milgrom in 1983. MOND modifies Newton's second law of motion at very low accelerations, which are typically found in the outer regions of galaxies. According to MOND, the effective gravitational force becomes stronger than predicted by Newtonian gravity at these low accelerations, which can explain the flat rotation curves of galaxies without invoking dark matter.

FOUNDATION

In MOND, an interpolation function $\mu(a/a_0)$ is introduced too bridge the gap between the Newtonian regime (where $a \gg a_0$) and the MOND regime (where $a \ll a_0$). The modified equation of motion can be written as:

$$a\mu\left(\frac{a}{a_0}\right) = a_N = \frac{GM}{r^2}$$

where a is the actual acceleration, a_N is the Newtonian acceleration, and we required $\mu(x)$ that:

$$\mu(x) = \begin{cases} 1 & \text{for } x \gg 1 \quad \text{(Newtonian regime)} \\ x & \text{for } x \ll 1 \quad \text{(MOND regime)} \end{cases}$$

A common choice for the interpolation function is:

$$\mu(x) = \frac{x}{\sqrt{1+x^2}}$$

Here we only consider the Deep-MOND regime, where $a \ll a_0$, the equation of motion can be simplified to:

$$\begin{aligned} a\mu\left(\frac{a}{a_0}\right) &= a_N \\ a\left(\frac{a}{a_0}\right) &= a_N \\ a^2 &= a_0 a_N \\ a &= \sqrt{a_0 a_N} \\ \frac{v^2}{r} &= \sqrt{a_0 \frac{GM}{r^2}} \\ v^4 &= GMa_0 \end{aligned}$$

This result shows that the rotational velocity v becomes constant at large distances from the galaxy center, which is consistent with the observed flat rotation curves.

Theory

MOND suggest an modified theory of Newton's laws, with:

$$F = ma\mu\left(\frac{a}{a_0}\right)$$

where $\mu(x)$ is an interpolation function that transitions between the Newtonian regime and the MOND regime, and a_0 is a characteristic acceleration scale below which MOND effects become significant, and by galaxy-scale observations, a_0 is found to be approximately $1.2 \times 10^{-10}\,\mathrm{m/s^2}$. And we can see that as a_0 is extremely small, our labatory experiments are not sensitive enough to detect the MOND effects.

Meanwhile, if scale up to galaxy scale, where the acceleration is much smaller than a_0:

$$v = (GMa_0)^{1/4}$$

the rotational velocity v becomes constant at large distances from the galaxy center, which is consistent with the observed flat rotation curves.

Physics Intuition

While MOND can successfully explain the flat rotation curves of galaxies, it faces 2 main challenges:

(1) Evidence of DM can be found in different physical systems (e.g. galaxy clusters, CMB anisotropies, large scale structure), and if we want to use MOND to explain all these phenomena, gravity must be modified in a more complex way, while the dark matter hypothesis can explain all these phenomena simply.

(It is more related to the principle of Occam's Razor, which states that among competing hypotheses, the one with the fewest assumptions should be selected, another similar case is the geocentric model and the heliocentric model, where the former can explain the observed phenomena by introducing epicycles, while the latter can explain all the phenomena without introducing any epicycle.)

(2) MOND cannot explain the Bullet Cluster, which is a collision of two galaxy clusters. The gravitational lensing observations of the Bullet Cluster show that most of the mass is located in the regions where there is little visible matter, which is consistent with the dark matter hypothesis but cannot be explained by MOND.

Figure 8.2: The Bullet Cluster(NASA, Aug 2006): The blue regions indicate the distribution of dark matter, while the pink regions indicate the distribution of hot gas (which is the dominant form of visible matter in galaxy clusters). The gravitational lensing observations show that most of the mass is located in the blue regions, which is consistent with the dark matter hypothesis but cannot be explained by MOND.

Extra

When 2 galaxy clusters collide, if dark matter exist, we will expect to see the hot gas (which is the dominant form of visible matter in galaxy clusters) interacts with each other and slows down, while the dark matter (which does not interact with itself or with visible matter) passes through without slowing down. Therefore, we will expect to see the distribution of dark matter and visible matter are separated. On the other hand, if MOND is correct, we will expect to see the distribution of dark matter and visible matter are the same, because there is no dark matter in MOND.

The observations of the Bullet Cluster show that the distribution of dark matter and visible matter are separated, which is consistent with the dark matter hypothesis but cannot be explained by MOND.

8.2 Properties of Dark Matter

In the below sub-sections, we will discuss the properties of dark matter in a introduction level.

8.2.1 Baryonic and Non-baryonic Dark Matter

Physics Intuition

We can classify dark matter into two categories: baryonic and non-baryonic. Baryonic dark matter consists of ordinary matter (protons, neutrons, electrons) that does not emit or absorb significant light, such as MACHOs (Massive Astrophysical Compact Halo Objects), which include black holes, neutron stars, brown dwarfs, and rogue planets. The primary method to detect MACHOs is gravitational microlensing, which occurs when a MACHO passes in front of a background star and temporarily magnifies its light. However, microlensing surveys have detected too few events to account for the total dark matter in galactic halos, indicating that MACHOs can only make up a small fraction of dark matter.

Non-baryonic dark matter candidates, such as WIMPs (Weakly Interacting Massive Particles), axions, and sterile neutrinos, are strongly favored by current observations. Cosmological data from the cosmic microwave background and Big Bang nucleosynthesis show that ordinary baryonic matter accounts for only about 5% of the universe's total mass-energy budget, while dark matter makes up roughly 27%, proving that the vast majority of dark matter must be non-baryonic. These candidates are also well-motivated theoretically, as they naturally arise in extensions of the Standard Model of particle physics. Unlike baryonic dark matter in the form of compact objects, non-baryonic dark matter does not produce discrete microlensing events that temporarily brighten background stars. Instead, it exerts a smooth, diffuse gravitational influence, which is fully consistent with observed galactic rotation curves, large-scale structure, and cosmological lensing.

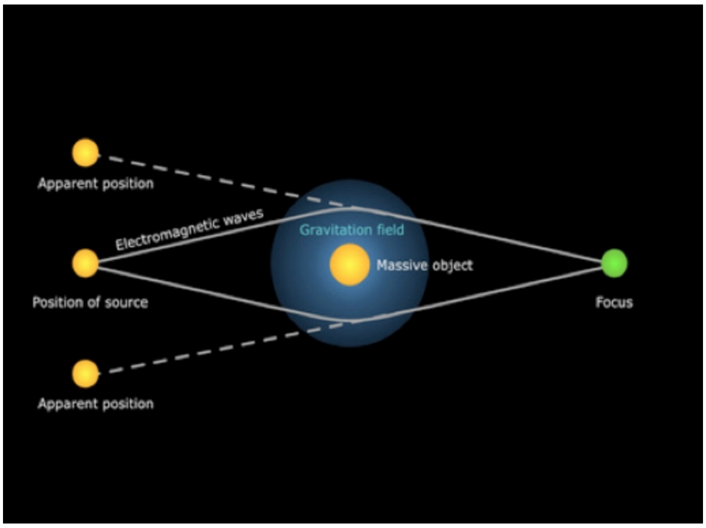

Figure 8.3: Gravitational microlensing: When a MACHO passes in front of a background star, it can temporarily magnify the light from the star, creating a characteristic light curve.

Extra

In case you have zero knowledge about star evolution, below is a brief introduction of the formation of some objects that can be MACHOs:

(1) Brown Dwarfs: Stars are formed from the gravitational collapse of molecular clouds, and the mass of the collapsing cloud determines the type of star that is formed. If the mass of the collapsing cloud is less than about 0.08 times the mass of the Sun, it will not be able to sustain nuclear fusion in its core, and it will become a brown dwarf. Brown dwarfs are sub-stellar objects that are too massive to be planets but not massive enough to be stars. They emit mainly in the infrared and are very faint, making them difficult to detect.

(2) White Dwarf: when the star reach its end of life, it will collapse under its own gravity, and if the mass of the collapsing part of star is below the Chadrasekhar limit (about 1.4 times the mass of the Sun), it will become a white dwarf. White dwarfs become fade and cool over time, and finally become black dwarfs, which are cold and dark objects that do not emit any light.

(3) Neutron Star: While if the mass of the collapsing part of star is above the Chadrasekhar limit but below the Tolman–Oppenheimer–Volkoff limit (about 2-3 times the mass of the Sun), it will become a neutron star. It is an extremely dense object composed mostly of neutrons, and is super small with only about 10-20 kilometers in diameter.

(4) Black Hole: If the mass of the collapsing part of star is above the Tolman–Oppenheimer–Volkoff limit, it will become a black hole. Black holes are regions of spacetime where gravity is so strong that nothing, not even light, can

escape from it. They can be formed from the collapse of massive stars, and they can also be formed from the merger of smaller black holes or from the collapse of dense regions in the early universe.

8.2.2 Cold and Hot Dark Matter

Physics Intuition

Dark matter is classified based on its velocity distribution (or free-streaming length) at the time of decoupling: cold dark matter (CDM), warm dark matter (WDM), and hot dark matter (HDM). HDM consists of particles that were relativistic at decoupling, typically with very small masses (eV scale or less, like active neutrinos). Their high thermal velocities cause significant free-streaming, which erases small-scale density fluctuations. This would lead to a "top-down" structure formation scenario, where the largest structures form first and later fragment, which is inconsistent with the observed hierarchical, "bottom-up" formation of galaxies and clusters. Consequently, HDM can only constitute a minor fraction of the total dark matter budget. Active neutrinos are the primary example of HDM.

On the other hand, CDM consists of particles that were non-relativistic at decoupling. This category spans a wide mass range: it includes heavy thermal relics like WIMPs (GeV–TeV scale) and extremely light non-thermally produced particles like axions (μeV–meV scale). CDM's low velocity dispersion allows gravitational collapse on small scales first, driving the bottom-up hierarchical structure formation that precisely matches cosmological observations. The total dark matter density is approximately 27% of the critical density of the universe, with CDM comprising the dominant component. Current cosmological and astrophysical data strongly favor a CDM-dominated universe.

8.2.3 Stableness and Lifetime

Physics Intuition

What makes a particle stable (or sufficiently long-lived)? There are two primary conditions: (1) The particle must carry a conserved quantum number (e.g., electric charge, baryon number, or a discrete symmetry charge like a Z_2 parity). (2) It must be the lightest particle carrying that conserved quantity. If a lighter state exists with the same conserved charge, the heavier particle can decay into it while respecting the conservation law.

For example, the electron is absolutely stable because it is the lightest particle with non-zero electric charge. Similarly, the proton is effectively stable because it is the lightest baryon, and baryon number is conserved at the perturbative level in the Standard Model. Conversely, the muon and tau leptons are unstable because they are heavier than the electron and can decay via the weak interaction into lighter leptons while conserving total lepton number.

Therefore, if dark matter consists of particles, they must be stable or have lifetimes significantly exceeding the age of the universe ($\tau_{\text{DM}} \gtrsim 10^{17}$ s). This typically requires that dark matter particles carry an exactly or approximately conserved quantum number (often imposed via a discrete symmetry) and be the lightest state charged under it, or that their decay channels are highly suppressed by symmetry, large mass scales, or tiny couplings.

Extra

The lifetime of other objects as reference:

- Electron: $> 4.6 \times 10^{26}$ years
- Muon: 2.2×10^{-6} seconds
- Tau: 2.9×10^{-13} seconds
- Proton: $> 10^{34}$ years (current experimental lower limit)
- Age of the universe: $\approx 1.38 \times 10^{10}$ years

8.2.4 Decoupling and Interactions

Physics Intuition

So far, we have known the properties of dark matter as:
(1) It is non-luminous and electrically neutral.
(2) It is non-baryonic and very likely to be a new type of particle that does not exist in the Standard Model.
(3) It is cold (non-relativistic) at the time of decoupling, (i.e. $m_\chi \gg T_{\text{CMB}}$), which means that it has a very small velocity dispersion and can form structures on small scales.
(4) It is stable or has a lifetime much longer than the age of the universe.

Now lets discuss about the interactions and relic abundance of dark matter.

Physics Intuition

If we assume the DM particles remain in thermal equilibrium and never decoupled, then the number density of DM particles would be given by the Maxwell-Boltzmann distribution:

$$n_{\chi,\text{eq}} = g_\chi \left(\frac{m_\chi T}{2\pi} \right)^{3/2} e^{-m_\chi/T}$$

And we will see that the number density of DM particles would be exponentially suppressed at low temperatures, which means that the relic abundance of DM would be negligible. Therefore, we need to consider the case where DM particles decoupled from the thermal bath at some point in the early universe, which allows them to have a non-negligible relic abundance today.

Generally, assumed the stable DM particle χ is in thermal equilibrium with the Standard Model particles Y in the early universe, and the dominant interaction is $\chi\bar{\chi} \leftrightarrow Y\bar{Y}$.

Physics Intuition

At the early stage of the universe $T \gg m_\chi$, the process of $\chi\bar{\chi}$ creation and annihilation are in equilibrium, leading χ to be present in large quantities alongside the various Standard Model particles.

As the universe expands and cools down, the temperature drops below the mass of the DM particle ($T < m_\chi$), and the process of $\chi\bar{\chi}$ creation becomes exponentially suppressed due to the Boltzmann factor, while the annihilation process continues to deplete the number of χ particles.

Eventually, when the annihilation rate falls below the expansion rate of the universe (i.e., $\Gamma_{\text{ann}} < H$), the DM particles "freeze out" and their comoving number density becomes constant, leading to a relic abundance that can account for the observed dark matter density today.

Question Example 8.1: Equation of DM interactions

Why the equation is $\chi\bar{\chi} \leftrightarrow Y\bar{Y}$, but not $\chi \leftrightarrow Y\bar{Y}$?

Solution.
If the process is $\chi \leftrightarrow Y\bar{Y}$, then the DM particle χ can decay into Standard Model particles Y and $\bar{Y}$, which would violate the stability requirement of dark matter. Since we know that dark matter must be stable or have a lifetime much longer than the age of the universe, the process $\chi \leftrightarrow Y\bar{Y}$ is not allowed.

Further, if the process is $\chi \leftrightarrow Y\bar{Y}$, the quantum numbers of χ must be the same as the combined quantum numbers of Y and $\bar{Y}$, which would require χ to carry 0 value of the quantum numbers. This also violates the requirement that dark matter must carry a conserved quantum number to ensure its stability.

8.2.5 Boltzmann Equation for DM Relic Abundance

Foundation

Recall the general form of the Boltzmann equation for the number density n_χ of DM particles:

$$\frac{dn_\chi}{dt} + 3Hn_\chi = C[n_\chi]$$

Where LHS actually is the number density conversation equation (you can check chapter 7.3.2 to see how the $3Hn_\chi$ term arises), and RHS is the collision term that describes the net effect of annihilation of DM particles. For the process $\chi\bar{\chi} \leftrightarrow Y\bar{Y}$, the collision term can be written as:

$$C[n_\chi] = -\langle\sigma_{\chi\bar{\chi}}v\rangle(n_\chi^2 - n_{\chi,\text{eq}}^2)$$

where $\langle\sigma_{\chi\bar{\chi}}v\rangle$ is the thermally averaged annihilation cross section times velocity, n_χ is the actual number density of DM particles, and $n_{\chi,\text{eq}}$ is the equilibrium number density of DM particles. The term n_χ^2 represents the annihilation of DM particles, while the term $n_{\chi,\text{eq}}^2$ represents the creation of DM particles from the thermal bath.

On the other words:

- Forward process $\chi\bar{\chi} \to Y\bar{Y}$: the rate is proportional to n_χ^2 because it requires two DM particles to annihilate.
- Reverse process $Y\bar{Y} \to \chi\bar{\chi}$: the rate is proportional to $n_{\chi,\mathrm{eq}}^2$ because it depends on the equilibrium abundance of DM particles that can be produced from the thermal bath.

Now we define the decay rate as:

$$\Gamma_\chi = \langle \sigma_{\chi\bar{\chi}} v \rangle n_{\chi,\mathrm{eq}}$$

Substituting the collision term into the Boltzmann equation, we get:

$$\begin{aligned}
\frac{dn_\chi}{dt} + 3Hn_\chi &= -\langle \sigma_{\chi\bar{\chi}} v \rangle n_{\chi,\mathrm{eq}}^2 \left(\frac{n_\chi^2}{n_{\chi,\mathrm{eq}}^2} - 1 \right) \\
\frac{dn_\chi}{dt} + 3Hn_\chi &= n_{\chi,\mathrm{eq}} \Gamma_\chi \left(1 - \frac{n_\chi^2}{n_{\chi,\mathrm{eq}}^2} \right)
\end{aligned}$$

THEORY

The Boltzmann equation for the number density of DM particles can be written as:

$$\begin{aligned}
\frac{dn_\chi}{dt} + 3Hn_\chi &= n_{\chi,\mathrm{eq}} \Gamma_\chi \left(1 - \frac{n_\chi^2}{n_{\chi,\mathrm{eq}}^2} \right) \\
&= -\langle \sigma_{\chi\bar{\chi}} v \rangle (n_\chi^2 - n_{\chi,\mathrm{eq}}^2)
\end{aligned}$$

where Γ_χ is the decay rate of DM particles, n_χ is the actual number density of DM particles, and $n_{\chi,\mathrm{eq}}$ is the equilibrium number density of DM particles.

IMPORTANT NOTE

Some may confused with the 2 notations n_χ and $n_{\chi,\mathrm{eq}}$, the former is the actual number density of DM particles, which means the real number density of DM particles we are having in the universe at the given time, and it affects the rate of DM annihilation, because the more DM particles we have, the more likely they will annihilate with each other.
While the latter is the equilibrium number density of DM particles, which means the number density of DM particles we would have if they were in thermal equilibrium with the thermal bath at the given time. It depends only on current temperature T via statistical mechanics:

$$n_{\chi,\mathrm{eq}} = g_\chi \left(\frac{m_\chi T}{2\pi} \right)^{3/2} e^{-m_\chi/T}$$

and it affects the rate of DM creation, because it shows the number density of DM particles that can be produced from the thermal bath at the given time.

Physics Intuition

Therefore, at early universe $T \gg m_\chi$, we have $n_{\chi,\text{eq}} \approx n_\chi$, which means rate of DM creation and annihilation are balanced, and DM particles are in thermal equilibrium.

As the universe expands and cools down, we have $T < m_\chi$, which means $n_{\chi,\text{eq}} < n_\chi$, the DM particles are no longer in thermal equilibrium, and the annihilation process dominates over the creation process.

And at the freeze-out point, we have $\Gamma_\chi = H$, which means the annihilation rate equals the expansion rate of the universe, and $n_{\chi,\text{eq}} \ll n_\chi$, but the expansion of the universe is so fast that the DM particles cannot find each other to annihilate, so the term $n_\chi^2 - n_{\chi,\text{eq}}^2$ is cancel out, which freeze out the number density of DM particles.

Remark

In real case, the stage of $n_{\chi,\text{eq}} < n_\chi$ is just short as when $m_\chi < T$, the decay rate $\Gamma_\chi \propto n_{\chi,\text{eq}}$ also drops as $e^{-m_\chi/T}$, which means the 2 conditions nearly happen at the same time, and the freeze-out happens immediately after the DM particles are no longer in thermal equilibrium. So we can treat the intermediate stage and freeze-out stage as the same stage.

8.2.6 Freeze-out Temperature

Physics Intuition

Through the above Boltzmann equation, we can see that the term $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is the key parameter that determines the relic abundance of DM particles. If $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is too large, the DM particles will annihilate too efficiently during the $n_{\chi,\text{eq}} < n_\chi$ stage, leading to a very small relic abundance.

On the other hand, if $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is too small, the DM particles will not annihilate efficiently, and they will freeze out with a large number density, leading to a relic abundance that exceeds the observed dark matter density.

Now lets find out the freeze-out temperature and the relic abundance of DM particles by solving the Boltzmann equation. We will see that if $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is around the weak scale, we can get the correct relic abundance that matches the observed dark matter density, which is known as the WIMP miracle.

Foundation

Theory

The freeze-out temperature T_{FO} can be expressed as:

$$x_{\mathrm{FO}} \equiv \frac{m_\chi}{T_{\mathrm{FO}}} \approx \ln\left[c(c+2)\cdot\sqrt{\frac{45}{8}}\frac{g_\chi}{2\pi^3}\cdot\frac{m_\chi M_{\mathrm{Pl}}\left(a+6b/x_{\mathrm{FO}}\right)}{\sqrt{g_*}\cdot\sqrt{x_{\mathrm{FO}}}}\right]$$

with $c \approx 0.5$ is a numerical determined quantity, g_χ is the number of internal degrees of freedom of the DM particle, M_{Pl} is the Planck mass, g_* is the effective number of relativistic degrees of freedom at freeze-out, and a and b are the coefficients in the expansion of the thermally averaged annihilation cross section:

$$\langle\sigma_{\chi\bar{\chi}}v\rangle = a + b\langle v^2\rangle + \mathcal{O}(\langle v^4\rangle)$$

8.2.7 Relic Abundance

FOUNDATION

First, define the comoving abundance of DM particles as:

$$Y_\chi \equiv \frac{n_\chi}{s} = \frac{n_{\text{dec}} a^{-3}}{s_{\text{dec}} a^{-3}} = \frac{n_{\text{dec}}}{s_{\text{dec}}}$$

The Y_χ is the number of DM particles per unit entropy, which is a convenient quantity to use because it remains constant after freeze-out, even as the universe continues to expand and cool down.

and we can rewrite the LHS of the Boltzmann equation as:

$$\frac{dn_\chi}{dt} + 3Hn_\chi = s\frac{dY_\chi}{dt} + Y_\chi \frac{ds}{dt} + 3HsY_\chi$$

Note the entropy density s is $proptoa^{-3}$, which means $\frac{ds}{dt} = -3Hs$, so we have:

$$\frac{dn_\chi}{dt} + 3Hn_\chi = s\frac{dY_\chi}{dt} + Y_\chi(-3Hs) + 3HsY_\chi = s\frac{dY_\chi}{dt}$$

Similarly, we can rewrite the RHS of the Boltzmann equation as:

$$-\langle \sigma_{\chi\bar{\chi}} v \rangle (n_\chi^2 - n_{\chi,\text{eq}}^2) = -\langle \sigma_{\chi\bar{\chi}} v \rangle s^2 (Y_\chi^2 - Y_{\chi,\text{eq}}^2)$$

Dividing both sides by s, we get:

$$\frac{dY_\chi}{dt} = -\langle \sigma_{\chi\bar{\chi}} v \rangle s (Y_\chi^2 - Y_{\chi,\text{eq}}^2)$$

Now, lets define the inverse temperature variable $x = \frac{m_\chi}{T}$, and:

$$\begin{aligned}\frac{dx}{dt} &= \frac{d}{dt}\left(\frac{m_\chi}{T}\right) \\ &= -\frac{m_\chi}{T^2}\frac{dT}{dt} \\ &= -\frac{m_\chi}{T^2}(-HT) \\ &= Hx\end{aligned}$$

which we using the fact that the temperature of the universe scales as $T \propto a^{-1}$, so:

$$\begin{aligned}\frac{dT}{dt} &= \frac{d}{dt}(Ca(t)) \\ &= \frac{C}{a^2}\frac{da}{dt} \\ &= -\frac{Ta}{a^2}\dot{a} \\ &= -HT\end{aligned}$$

Therefore, we can rewrite the Boltzmann equation as:

$$\begin{aligned}
\frac{dY_\chi}{dx}\frac{dx}{dt} &= -\langle\sigma_{\chi\bar{\chi}}v\rangle s(Y_\chi^2 - Y_{\chi,\mathrm{eq}}^2) \\
\frac{dY_\chi}{dx} &= \frac{1}{Hx}\left(-\langle\sigma_{\chi\bar{\chi}}v\rangle s(Y_\chi^2 - Y_{\chi,\mathrm{eq}}^2)\right) \\
\frac{dY_\chi}{dx} &= -\frac{\lambda}{x^2}(Y_\chi^2 - Y_{\chi,\mathrm{eq}}^2)
\end{aligned}$$

with $\lambda = \frac{s\langle\sigma_{\chi\bar{\chi}}v\rangle}{H}x$.

In fact, we can get details value of λ by substituting the expressions of s and H:

$$\begin{cases} s &= \frac{2\pi^2}{45}g_{*s}T^3 \\ H &= \sqrt{\frac{8\pi G}{3}\rho} = \sqrt{\frac{8\pi G}{3}\frac{\pi^2}{30}g_*T^4} \end{cases}$$

Note here we assume the decoupling happens during the radiation-dominated era, so the energy density ρ and the entropy density s are dominated by the relativistic particles, and we use the relativistic in chpater 5.2 and 5.4 to get the above expressions.

Defining the Planck mass as $M_{\mathrm{Pl}} = \frac{\hbar c}{\sqrt{G}} = \frac{1}{\sqrt{G}} \approx 2.4\times 10^{18}\,\mathrm{GeV}$, we can rewrite the expression of λ as:

$$\begin{aligned}
\lambda &= \langle\sigma_{\chi\bar{\chi}}v\rangle\frac{s}{H}x \\
&= \langle\sigma_{\chi\bar{\chi}}v\rangle\frac{\frac{2\pi^2}{45}g_{*s}T^3}{\sqrt{\frac{8\pi^3}{90}}\frac{\sqrt{g_*}}{M_{\mathrm{Pl}}}T^2}x \\
&= \langle\sigma_{\chi\bar{\chi}}v\rangle\left(\frac{2\pi^2}{45}\cdot\sqrt{\frac{90}{8\pi^3}}\right)\cdot\frac{g_{*s}}{\sqrt{g_*}}\cdot M_{\mathrm{Pl}}\cdot Tx \\
&= \langle\sigma_{\chi\bar{\chi}}v\rangle\left(\sqrt{\frac{\pi}{45}}\right)\cdot\sqrt{g_*}\cdot M_{\mathrm{Pl}}\cdot m_\chi
\end{aligned}$$

where we take the assumption that $g_{*s}\approx g_*$ in last step, which is valid when there is no significant change in the number of relativistic degrees of freedom around the freeze-out temperature (i.e. before the neutrino decoupling).

THEORY

The Boltzmann equation for the comoving abundance can be written as:

$$\frac{dY_\chi}{dx} = -\frac{\lambda}{x^2}(Y_\chi^2 - Y_{\chi,\mathrm{eq}}^2)$$

with $\lambda = \langle\sigma_{\chi\bar{\chi}}v\rangle\left(\sqrt{\frac{\pi}{45}}\right)\cdot\sqrt{g_*}\cdot M_{\mathrm{Pl}}\cdot m_\chi$.

FOUNDATION

THEORY

The relic abundance of DM particles can be expressed as:

$$\Omega_\chi h^2 \approx 0.1 \left(\frac{x_{\rm FO}}{20}\right) \left(\frac{g_*}{80}\right)^{-1/2} \left(\frac{a + 3b/x_{\rm FO}}{3 \times 10^{-26}\,{\rm cm}^3/{\rm s}}\right)^{-1}$$

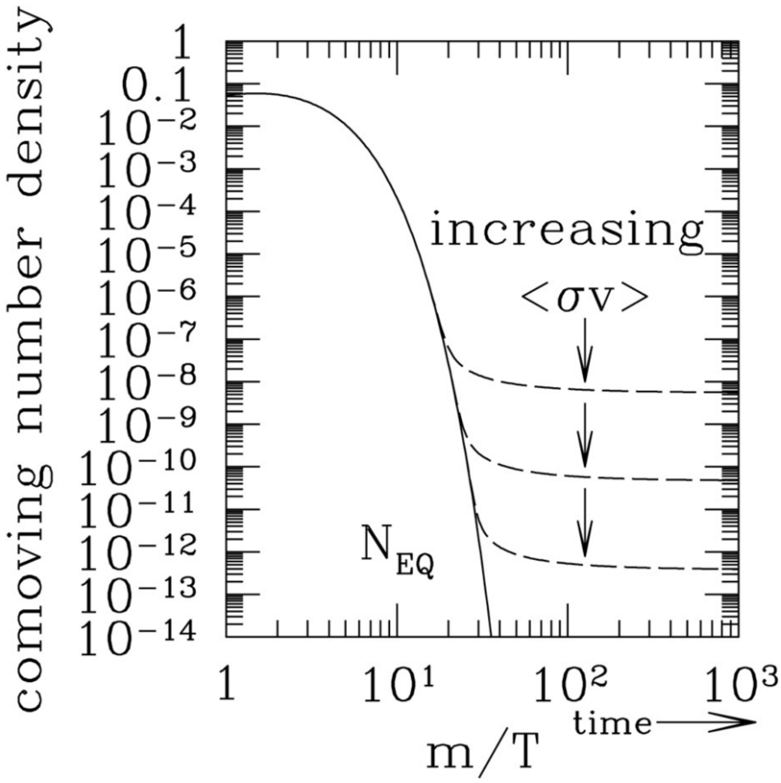

Figure 8.4: The evolution of the comoving number density Y_χ of DM in the freeze-out scenario.

EXTRA

The above figure shows the evolution of the comoving number density Y_χ of DM particles as a function of the inverse temperature variable $x = m_\chi/T$. The solid line in the figure represents the **Equilibrium Number Density** $Y_{\chi,\rm eq}$, while the dashed line represents the **Actual Number Density** Y_χ.

In the early universe, when x is small, the dashed line closely follows the solid line, indicating that the DM particles are in thermal equilibrium with the thermal bath. As the universe expands and cools down, $T \ll m$ and x increases, and the solid drops exponentially and tends to zero, when new particles cannot be created but remain particles can still annihilate.

However, the dashed line does not drop to zero, and tends to a constant value(i.e. the relic density), the 2 lines separate at the freeze-out point ($m_c hi/T \approx 20$), as the free-out stops the annihilation process continues, and the number density of DM particles is frozen at the value at freeze-out, which is much larger than the equilibrium number density at that time.

By the equation we have derived in previous subsection, we know that the cross section $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is also the key parameter that determines the relic abundance of DM particles, it can also be explained by the figure: if $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is small, the particles interact weakly, and they will stop annihilating and freeze out earlier, which means the dashed line will separate from the solid line at a smaller value of x, and the relic abundance will be larger. On the other hand, if $\langle \sigma_{\chi\bar{\chi}} v \rangle$ is large, the particles interact strongly, and they will continue to annihilate until a later time, which means the dashed line will separate from the solid line at a larger value of x, and the relic abundance will be smaller.

8.2.8 WIMP Miracle

PHYSICS INTUITION

To get the relic abundance that we observed today, we need $\langle \sigma_{\chi\bar{\chi}} v \rangle \approx 3 \times 10^{-26}\,\mathrm{cm}^3/\mathrm{s}$. Interestingly, if we assume the particle is around the weak scale (i.e. $m_\chi \approx 100$ GeV, like W, Z bosons), and it interacts with the weak-force strength (i.e. $\alpha^2/(100\,\mathrm{GeV})^2 \approx 10^{-26}\,\mathrm{cm}^3/\mathrm{s}$). This coincidence of alignment between the cosmologic observations and the particle physics scale is known as the WIMP miracle, which makes Weakly Interacting Massive Particles (WIMPs) a very attractive candidate for dark matter. And if the WIMP miracle is true, we should be able to detect WIMPs through their interactions with ordinary matter, which motivates the experimental searches for WIMPs in non-gravitational ways, such as direct detection, indirect detection, and collider detection.

8.3 Non-gravitational Detections of Dark Matter

8.3.1 Direct Detection

8.3.2 Indirect Detection

8.3.3 Collider Detection

8.4 Other Candidates for Dark Matter

As mentioned before, we still don not know what really dark matter is, so except for the WIMP candidates, there are still many hypothetical candidates for dark matter, and we will briefly introduce some of them below. Note that the candidates we will introduce are not exhaustive, and there are many other candidates that are proposed in the literature.

EXTRA

() Axions: Axions are hypothetical particles that were originally proposed to solve the strong CP problem in quantum chromodynamics (QCD). They are very light, neutral, and weakly interacting particles that could be produced in the early universe and could account for dark matter. Axions can be detected through their coupling to photons, which allows them to convert into photons in the presence of a strong magnetic field. Experiments such as ADMX (Axion Dark Matter Experiment) are currently searching for axions by looking for this conversion process. While axions are a well-motivated dark matter candidate, they have not yet been detected, and their existence remains speculative.

EXTRA

() Sterile Neutrinos: Sterile neutrinos are hypothetical particles that do not interact via the weak force, unlike the three known active neutrino flavors (electron, muon, and tau neutrinos). They are called "sterile" because they do not participate in standard weak interactions, making them difficult to detect. However, they can mix with active neutrinos and could be produced in the early universe through this mixing. If sterile neutrinos have a mass in the keV range, they could be a viable candidate for warm dark matter. They can be detected indirectly through their decay into active neutrinos and photons, which would produce a characteristic X-ray signal. While there have been some tentative hints of sterile neutrinos from X-ray observations, their existence has not been confirmed, and they remain a speculative dark matter candidate.

EXTRA

() Primordial Black Holes: Primordial black holes (PBHs) are hypothetical black holes that could have formed in the early universe due to high-density fluctuations. They are considered potential candidates for dark matter. While PBHs with masses around 10^{15} g would be evaporating today via Hawking radiation (and are thus constrained by gamma-ray observations), there is a viable mass window roughly between 10^{17} and 10^{23} g (asteroid masses) where they could constitute all or a significant fraction of dark matter while evading current observational constraints. However, their existence remains speculative and unconfirmed. Unlike classical MACHOs, which are baryonic objects formed through stellar evolution, PBHs formed in the early universe and are typically treated as a distinct dark matter candidate category.

CHAPTER A

Declaration of AI Assistance

A.1 Use of Artificial Intelligence in This Document

This course guide was developed with assistance from artificial intelligence (AI) in the following capacities:

- **Template Design:** AI provided guidance on structuring the document layout, choosing appropriate color schemes.
- **LaTeX Code Generation:** AI assisted in writing and refactoring LaTeX code, including package configurations, box styling (tcolorbox), TikZ decorations, and custom command definitions.
- **LaTeX Structure Refactoring:** AI helped reorganize and optimize the document structure for maintainability, readability, and professional presentation.
- **Mathematical Formula Formatting:** AI provided assistance in formatting and presenting mathematical equations in clear, readable notation aligned with conventions.

Author Verification: All content, including scientific accuracy, theoretical derivations, example problems, and pedagogical choices, has been carefully reviewed and verified by the author. The AI assistance was limited to technical formatting and structural optimization, not to the generation or validation of course content itself.

CHAPTER B

Mathematical Tools & Derivations

B.1 Useful Identities

Trigonometric

$$\sin^2\theta + \cos^2\theta = 1, \quad \tan\theta = \sin\theta/\cos\theta, \quad \sin(A \pm B) = \sin A\cos B \pm \cos A \sin B.$$

Hyperbolic

$$\cosh^2 x - \sinh^2 x = 1, \quad \sinh(x \pm y) = \sinh x \cosh y \pm \cosh x \sinh y.$$

$$\sinh x = \frac{e^x - e^{-x}}{2}, \qquad \cosh x = \frac{e^x + e^{-x}}{2}, \qquad \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}.$$

Common Integrals

$$\begin{aligned} \int x^n\,dx &= \tfrac{x^{n+1}}{n+1} + C \quad (n \neq -1), & \int e^{ax}\,dx &= \tfrac{1}{a}e^{ax} + C, \\ \int \frac{dx}{x} &= \ln|x| + C, & \int \sin x\,dx &= -\cos x + C. \end{aligned}$$

Gamma–Zeta Integrals

$$\begin{aligned} \int_0^\infty \frac{x^{s-1}}{e^x - 1}\,dx &= \Gamma(s)\zeta(s), && \Re(s) > 1, \\ \int_0^\infty \frac{x^{s-1}}{e^x + 1}\,dx &= (1 - 2^{1-s})\Gamma(s)\zeta(s), && \Re(s) > 0. \end{aligned}$$

$$\begin{aligned} \int_0^\infty \frac{x^2}{e^x - 1}\,dx &= 2\zeta(3), & \int_0^\infty \frac{x^2}{e^x + 1}\,dx &= \tfrac{3}{2}\zeta(3), \\ \int_0^\infty \frac{x^3}{e^x - 1}\,dx &= \frac{\pi^4}{15} = 6\zeta(4), & \int_0^\infty \frac{x^3}{e^x + 1}\,dx &= \frac{7\pi^4}{120} = \tfrac{7}{8}\,6\zeta(4). \end{aligned}$$

Taylor Expansions

$$f(x) = f(a) + f'(a)(x-a) + \tfrac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$\begin{aligned}
\sin x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \tfrac{x^3}{3!} + \tfrac{x^5}{5!} - \cdots \\
\cos x &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \tfrac{x^2}{2!} + \tfrac{x^4}{4!} - \cdots \\
e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \tfrac{x^2}{2!} + \cdots \\
\sqrt{1+x} &= \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \tfrac{1}{2}x - \tfrac{1}{8}x^2 + \cdots
\end{aligned}$$

Common Approximations

Small-argument limits ($|x| \ll 1$):

$$\sin x \approx x, \qquad \cos x \approx 1 - \frac{x^2}{2}, \qquad \tan x \approx x,$$

$$\sinh x \approx x, \qquad \cosh x \approx 1 + \frac{x^2}{2}, \qquad \tanh x \approx x.$$

Large positive argument ($x \gg 1$):

$$\sinh x \approx \frac{1}{2}e^x, \qquad \cosh x \approx \frac{1}{2}e^x, \qquad \tanh x \approx 1.$$

Large negative argument ($x \ll -1$):

$$\sinh x \approx -\frac{1}{2}e^{-x}, \qquad \cosh x \approx \frac{1}{2}e^{-x}, \qquad \tanh x \approx -1.$$

B.2 Differential Geometry Primer

The **metric tensor** $g_{\mu\nu}$ encodes the geometry of spacetime: $ds^2 = g_{\mu\nu}\, dx^\mu\, dx^\nu$. The **Christoffel symbols** (not tensors) are defined as

$$\Gamma^\lambda_{\mu\nu} = \tfrac{1}{2} g^{\lambda\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \tag{B.1}$$

The Riemann curvature tensor is defined as

$$R^\rho{}_{\sigma\mu\nu} = \partial_\mu \Gamma^\rho_{\nu\sigma} - \partial_\nu \Gamma^\rho_{\mu\sigma} + \Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma} - \Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}. \tag{B.2}$$

The **Ricci tensor** is the contraction $R_{\mu\nu} = R^\lambda{}_{\mu\lambda\nu}$ of the Riemann curvature tensor, and the **Ricci scalar** is $R = g^{\mu\nu} R_{\mu\nu}$.

These enter the Einstein field equations as:

$$R_{\mu\nu} - \tfrac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \tag{B.3}$$

Mostly-Plus vs. Mostly-Minus Signatures

The choice of metric signature affects the signs of above tensors in below ways:

Tensor	**Symbol**	**Sign Change**
Metric	$g_{\mu\nu}$	Flips $(-)$
Inverse metric	$g^{\mu\nu}$	Flips $(-)$
Christoffel symbols	$\Gamma^\lambda_{\mu\nu}$	No change
Riemann(1,3)	$R^\rho{}_{\sigma\mu\nu}$	No change
Riemann(0,4)	$R_{\rho\sigma\mu\nu}$	Flips $(-)$
Ricci tensor	$R_{\mu\nu}$	No change
Ricci scalar	R	Flips $(-)$
Einstein tensor	$G_{\mu\nu}$	No change
Stress-energy tensor	$T_{\mu\nu}$	No change

B.3 Relativistic Limits

The energy-momentum relation is $E^2 = p^2c^2 + m^2c^4$. So in a relativistic limit, we can have:

$$\{E, m, \rho\} \ll \{p, T\}$$

which any one of the values in left-hand set is much smaller than any one of the values in the right-hand set.

B.4 Spherical Harmonics & Associated Legendre Polynomials

We adopt the Condon–Shortley phase convention:

$$Y_l^m(\theta,\phi) = N_{lm}\, P_l^m(\cos\theta)\, e^{im\phi},$$
$$N_{lm} = \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}},$$
$$Y_l^{-m}(\theta,\phi) = (-1)^m N_{lm}\, P_l^m(\cos\theta)\, e^{-im\phi}, \qquad (m>0).$$

The real spherical harmonics are constructed as

$$Y_{l,m}^{(c)}(\theta,\phi) = \frac{1}{\sqrt{2}}\Big(Y_l^{-m} + (-1)^m Y_l^m\Big) = \sqrt{2}\, N_{lm}\, P_l^m(\cos\theta)\cos(m\phi), \qquad m>0,$$
$$Y_{l,m}^{(s)}(\theta,\phi) = \frac{1}{i\sqrt{2}}\Big(Y_l^{-m} - (-1)^m Y_l^m\Big) = \sqrt{2}\, N_{lm}\, P_l^m(\cos\theta)\sin(m\phi), \qquad m<0,$$
$$Y_{l,0}(\theta,\phi) = N_{l0}\, P_l(\cos\theta).$$

Associated Legendre Polynomials $P_l^m(x)$ for $0 \le l \le 4$

Let $x \equiv \cos\theta$.

$l=0:$ $P_0^0(x) = 1.$

$l=1:$ $P_1^0(x) = x,$ $\quad P_1^1(x) = -(1-x^2)^{1/2}.$

$l=2:$ $P_2^0(x) = \frac{1}{2}(3x^2-1),$ $\quad P_2^1(x) = -3x(1-x^2)^{1/2},$ $\quad P_2^2(x) = 3(1-x^2).$

$l=3:$ $P_3^0(x) = \frac{1}{2}(5x^3-3x),$ $\quad P_3^1(x) = -\frac{3}{2}(5x^2-1)(1-x^2)^{1/2},$
$P_3^2(x) = 15x(1-x^2),$ $\quad P_3^3(x) = -15(1-x^2)^{3/2}.$

$l=4:$ $P_4^0(x) = \frac{1}{8}(35x^4-30x^2+3),$ $\quad P_4^1(x) = -\frac{5}{2}(7x^3-3x)(1-x^2)^{1/2},$
$P_4^2(x) = \frac{15}{2}(7x^2-1)(1-x^2),$ $\quad P_4^3(x) = -105x(1-x^2)^{3/2},$
$P_4^4(x) = 105(1-x^2)^2.$

Real Spherical Harmonics Table for $0 \le l \le 4$

$$
\begin{aligned}
l = 0 :\quad & Y_{0,0} = \sqrt{\frac{1}{4\pi}}. \\
l = 1 :\quad & Y_{1,0} = \sqrt{\frac{3}{4\pi}}\, P_1^0(\cos\theta), \\
& Y_{1,1}^{(c)} = \sqrt{\frac{3}{4\pi}}\, P_1^1(\cos\theta)\cos\phi, \qquad Y_{1,1}^{(s)} = \sqrt{\frac{3}{4\pi}}\, P_1^1(\cos\theta)\sin\phi. \\
l = 2 :\quad & Y_{2,0} = \sqrt{\frac{5}{4\pi}}\, P_2^0(\cos\theta), \\
& Y_{2,1}^{(c)} = \sqrt{\frac{5}{12\pi}}\, P_2^1(\cos\theta)\cos\phi, \qquad Y_{2,1}^{(s)} = \sqrt{\frac{5}{12\pi}}\, P_2^1(\cos\theta)\sin\phi, \\
& Y_{2,2}^{(c)} = \sqrt{\frac{5}{48\pi}}\, P_2^2(\cos\theta)\cos(2\phi), \qquad Y_{2,2}^{(s)} = \sqrt{\frac{5}{48\pi}}\, P_2^2(\cos\theta)\sin(2\phi). \\
l = 3 :\quad & Y_{3,0} = \sqrt{\frac{7}{4\pi}}\, P_3^0(\cos\theta), \\
& Y_{3,1}^{(c)} = \sqrt{\frac{7}{24\pi}}\, P_3^1(\cos\theta)\cos\phi, \qquad Y_{3,1}^{(s)} = \sqrt{\frac{7}{24\pi}}\, P_3^1(\cos\theta)\sin\phi, \\
& Y_{3,2}^{(c)} = \sqrt{\frac{7}{240\pi}}\, P_3^2(\cos\theta)\cos(2\phi), \qquad Y_{3,2}^{(s)} = \sqrt{\frac{7}{240\pi}}\, P_3^2(\cos\theta)\sin(2\phi), \\
& Y_{3,3}^{(c)} = \sqrt{\frac{7}{2880\pi}}\, P_3^3(\cos\theta)\cos(3\phi), \qquad Y_{3,3}^{(s)} = \sqrt{\frac{7}{2880\pi}}\, P_3^3(\cos\theta)\sin(3\phi). \\
l = 4 :\quad & Y_{4,0} = \sqrt{\frac{9}{4\pi}}\, P_4^0(\cos\theta), \\
& Y_{4,1}^{(c)} = \sqrt{\frac{9}{40\pi}}\, P_4^1(\cos\theta)\cos\phi, \qquad Y_{4,1}^{(s)} = \sqrt{\frac{9}{40\pi}}\, P_4^1(\cos\theta)\sin\phi, \\
& Y_{4,2}^{(c)} = \sqrt{\frac{1}{80\pi}}\, P_4^2(\cos\theta)\cos(2\phi), \qquad Y_{4,2}^{(s)} = \sqrt{\frac{1}{80\pi}}\, P_4^2(\cos\theta)\sin(2\phi), \\
& Y_{4,3}^{(c)} = \sqrt{\frac{1}{1120\pi}}\, P_4^3(\cos\theta)\cos(3\phi), \qquad Y_{4,3}^{(s)} = \sqrt{\frac{1}{1120\pi}}\, P_4^3(\cos\theta)\sin(3\phi), \\
& Y_{4,4}^{(c)} = \sqrt{\frac{1}{8960\pi}}\, P_4^4(\cos\theta)\cos(4\phi), \qquad Y_{4,4}^{(s)} = \sqrt{\frac{1}{8960\pi}}\, P_4^4(\cos\theta)\sin(4\phi).
\end{aligned}
$$

The above keeps the exact $\sqrt{\cdot}$ normalization and explicit angular dependence (θ, ϕ), while using a real-valued basis.

CHAPTER C

Constants & Conversion Factors

C.1 Physical Constants

Quantity	Symbol	Value (SI)
Speed of light	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$
Hubble constant today	H_0	$67.8 \pm 0.77\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$
Cosmological constant	Λ	$1.11 \times 10^{-52}\,\mathrm{m^{-2}}$
Cosmological constant	Λ	$2.036 \times 10^{-35}\,\mathrm{s^{-2}}$
Planck constant	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s}$
Boltzmann constant	k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Fermi constant	G_F	$1.166 \times 10^{-5}\,\mathrm{GeV^{-2}}$
Fine-structure constant today	α	$1/137.036$
Planck mass	M_{Pl}	$2.176 \times 10^{-8}\,\mathrm{kg}$

C.2 Conversion Factors

Astronomical Units

Conversion	Value
1 astronomical unit (AU)	$1.496 \times 10^{11}\,\mathrm{m}$
1 light-year (ly)	$9.461 \times 10^{15}\,\mathrm{m}$
1 parsec (pc)	$3.086 \times 10^{16}\,\mathrm{m}$
1 parsec (pc)	$3.262\,\mathrm{ly}$
1 year (yr)	$3.156 \times 10^{7}\,\mathrm{s}$
1 Solar mass ($M_\odot$)	$1.989 \times 10^{30}\,\mathrm{kg}$
1 solar radius ($R_\odot$)	$6.957 \times 10^{8}\,\mathrm{m}$
1 Earth mass ($M_\oplus$)	$5.972 \times 10^{24}\,\mathrm{kg}$
1 Earth radius ($R_\oplus$)	$6.371 \times 10^{6}\,\mathrm{m}$
CMB temperature (T_{CMB})	$2.725\,\mathrm{K}$
CνB temperature ($T_{\mathrm{C}\nu\mathrm{B}}$)	$1.95\,\mathrm{K}$

Energy Units

Conversion	Value
1 eV	1.602×10^{-19} J
1 electron mass (m_e)	$0.511\,\mathrm{MeV}/c^2$
1 electron mass (m_e)	9.109×10^{-31} kg
1 proton mass (m_p)	$938.3\,\mathrm{MeV}/c^2$
1 proton mass (m_p)	1.673×10^{-27} kg
1 neutron mass (m_n)	$939.6\,\mathrm{MeV}/c^2$
1 neutron mass (m_n)	1.675×10^{-27} kg

C.3 SI Unit Prefixes

Prefix	Name	Factor
10^{-12}	pico (p)	10^{-12}
10^{-9}	nano (n)	10^{-9}
10^{-6}	micro (μ)	10^{-6}
10^{-3}	milli (m)	10^{-3}
10^{3}	kilo (k)	10^{3}
10^{6}	mega (M)	10^{6}
10^{9}	giga (G)	10^{9}
10^{12}	tera (T)	10^{12}

C.4 Main Events in Universe

Event	Time t	Redshift z	Temperature T (eV)
Planck epoch	10^{-43} s	–	–
Grand unification	10^{-36} s	–	–
Inflation	10^{-34} s (?)	–	–
Baryogenesis	?	?	?
EW phase transition	2×10^{-11} s	10^{15}	10^{11}
QCD phase transition	2×10^{-5} s	10^{12}	1.5×10^{8}
Dark matter freeze-out	?	?	?
Neutrino decoupling	1 s	6×10^{9}	10^{6}
Electron-positron annihilation	6 s	2×10^{9}	5×10^{5}
Big Bang nucleosynthesis	3 min	4×10^{8}	10^{5}
Matter-radiation equality	6×10^{4} yr	3400	0.75
Recombination	$(2.6–3.8) \times 10^{5}$ yr	1100–1400	0.26–0.33
Photon decoupling	3.8×10^{5} yr	1000–1200	0.23–0.28
Reionization	$(1–4) \times 10^{8}$ yr	11–30	$(2.6–7.0) \times 10^{-3}$
Dark energy-matter equality	9×10^{9} yr	0.4	3.3×10^{-4}
Present	1.38×10^{10} yr	0	2.4×10^{-4}

C.5 Nuclei Binding Energies & Atomic Masses

Nucleus	Atomic Mass (u)	B(MeV)	B/A (MeV/u)
e^-	0.00054858	–	–
p	1.0072765	–	–
n	1.0086649	–	–
^{1}H	1.0078250	0.000000	0.000000
^{2}H	2.0141018	2.224515	1.112258
^{3}H	3.0160493	8.481773	2.827258
^{3}He	3.0160293	7.718036	2.572679
^{4}He	4.0026032	28.295803	7.073951
^{6}Li	6.0151223	31.994603	5.332434
^{7}Li	7.0160040	39.244654	5.606379
^{8}Be	8.0053051	56.499667	7.062458
^{9}Be	9.0121821	58.165096	6.462788
^{12}C	12.0000000	92.162051	7.680171
^{13}C	13.0033550	97.108223	7.469863
^{14}N	14.0030740	104.658962	7.475640
^{15}N	15.0001090	115.492214	7.699481
^{16}O	15.9949150	127.619412	7.976213
^{17}O	16.9991320	131.762631	7.750743
^{18}O	17.9991610	139.806972	7.767054
^{20}Ne	19.9924400	160.645559	8.032278
^{24}Mg	23.9850420	198.257479	8.260728
^{28}Si	27.9769270	236.537285	8.447760
^{32}S	31.9720710	271.781333	8.493167
^{56}Fe	55.9349420	492.255833	8.790283

Where **B** is the total binding energy of the nucleus, and **B/A** is the average binding energy per nucleon.

Note: $u = 1.661 \times 10^{-27}\,\mathrm{kg} = 931.5\,\mathrm{MeV}/c^2$.

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